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(54) **PORTABLE COOLER WITH ACTIVE TEMPERATURE CONTROL**

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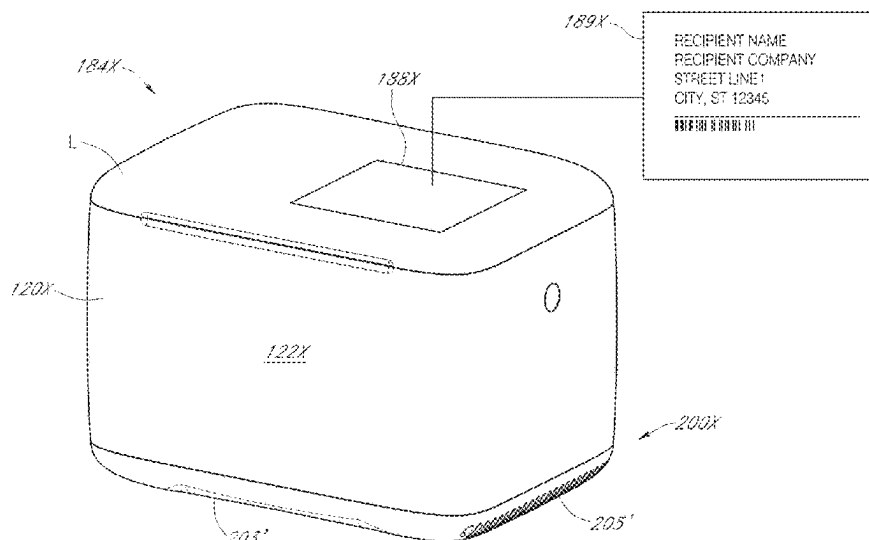
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(57) **ABSTRACT**

A portable cooler container with active temperature control includes a container body with an insulated chamber to receive medicines, vaccines, biological samples or other medical goods. A control system is at least partially disposed between an outer wall and an inner wall of the container body. An electronic display screen on one of the lid and the container body configured to display shipping address information for the portable cooler container.

18 Claims, 78 Drawing Sheets



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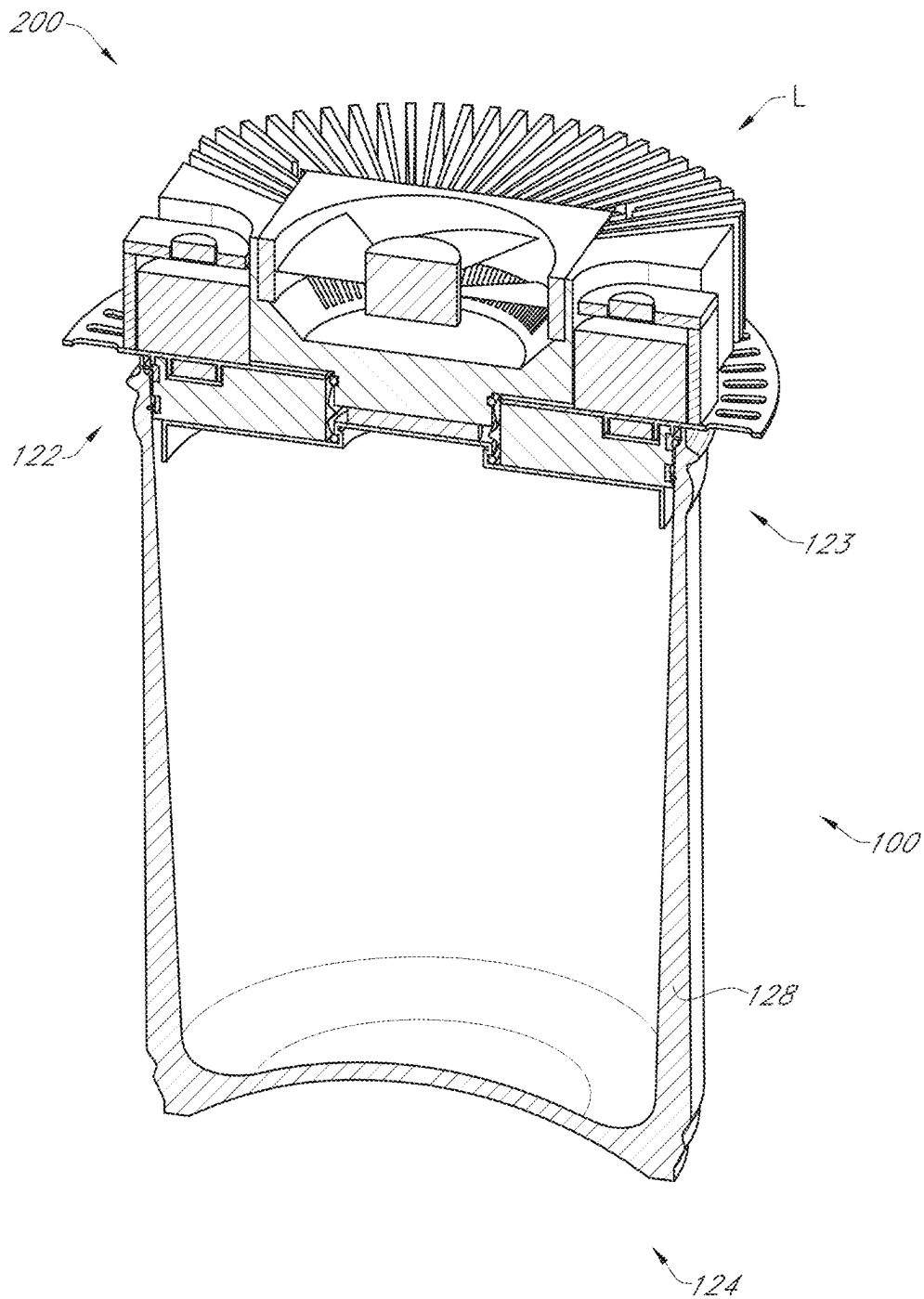


FIG. 1A

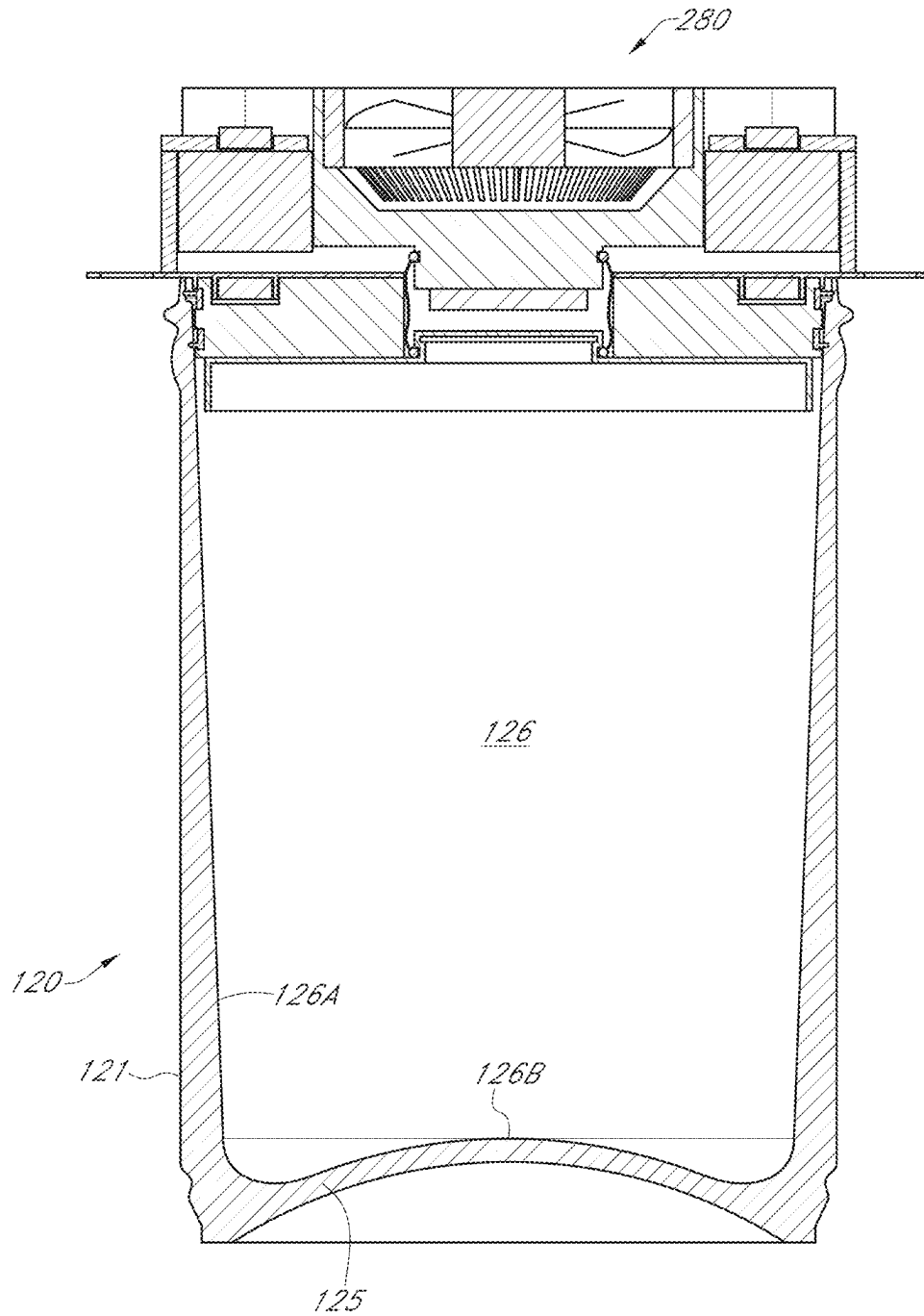


FIG. 1B

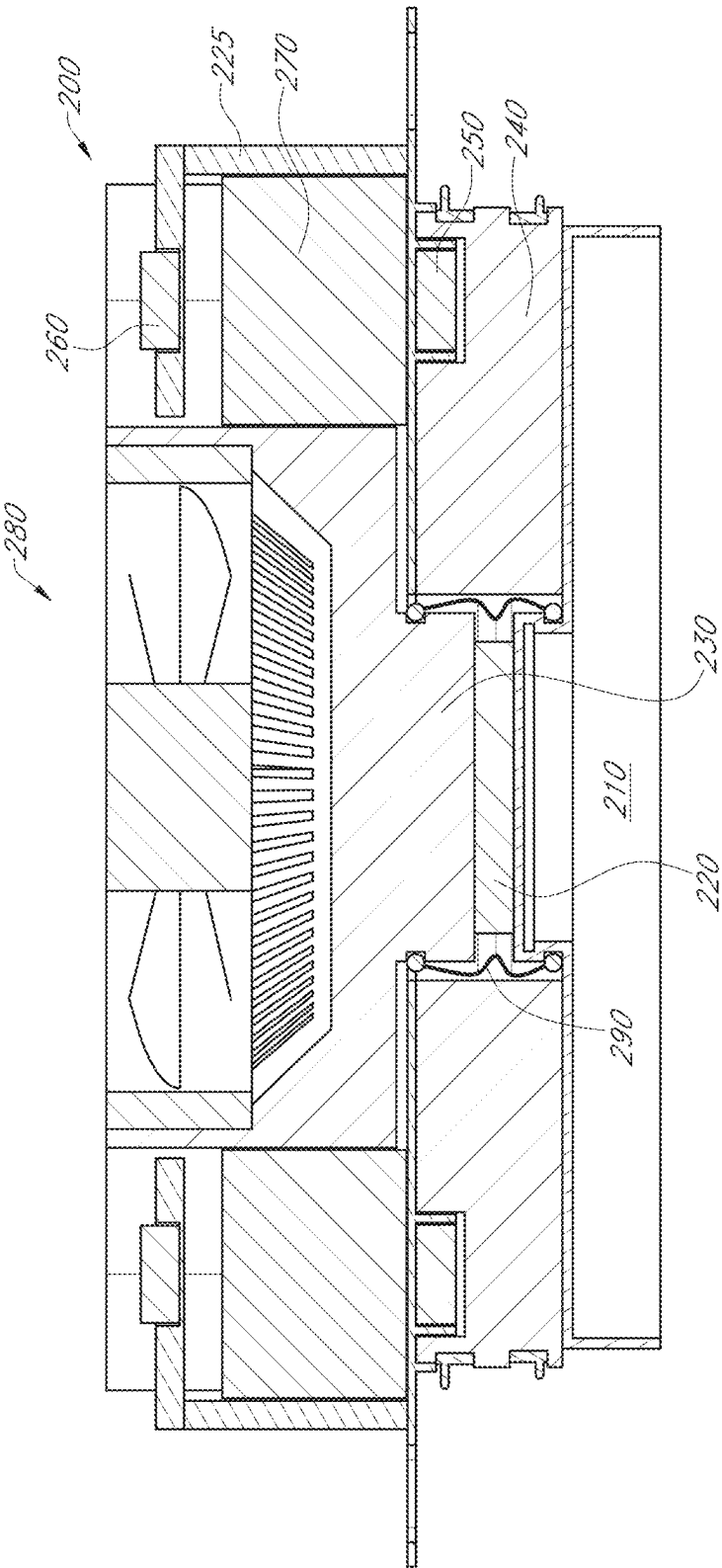


FIG. 1C

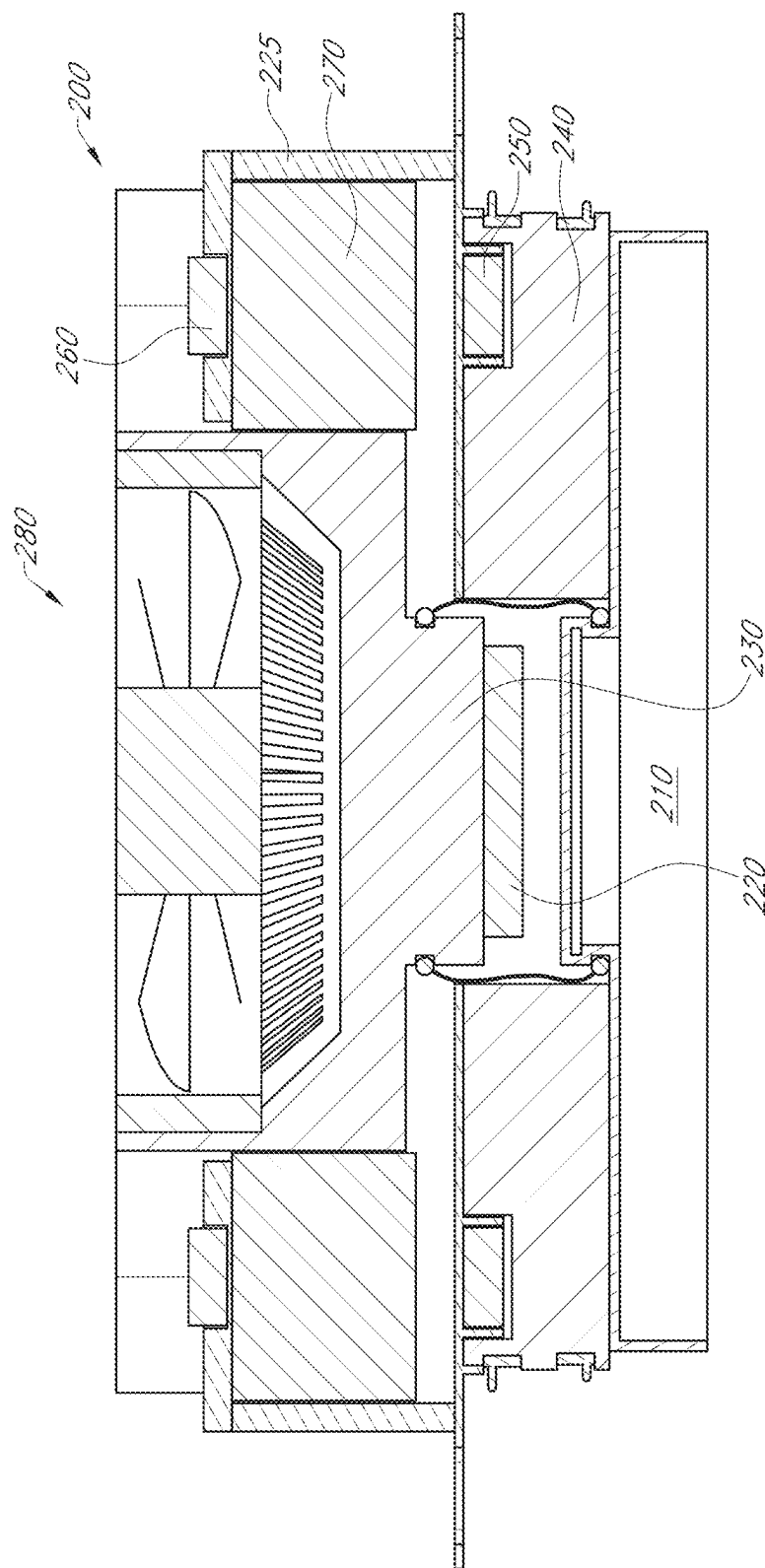


FIG. 1D

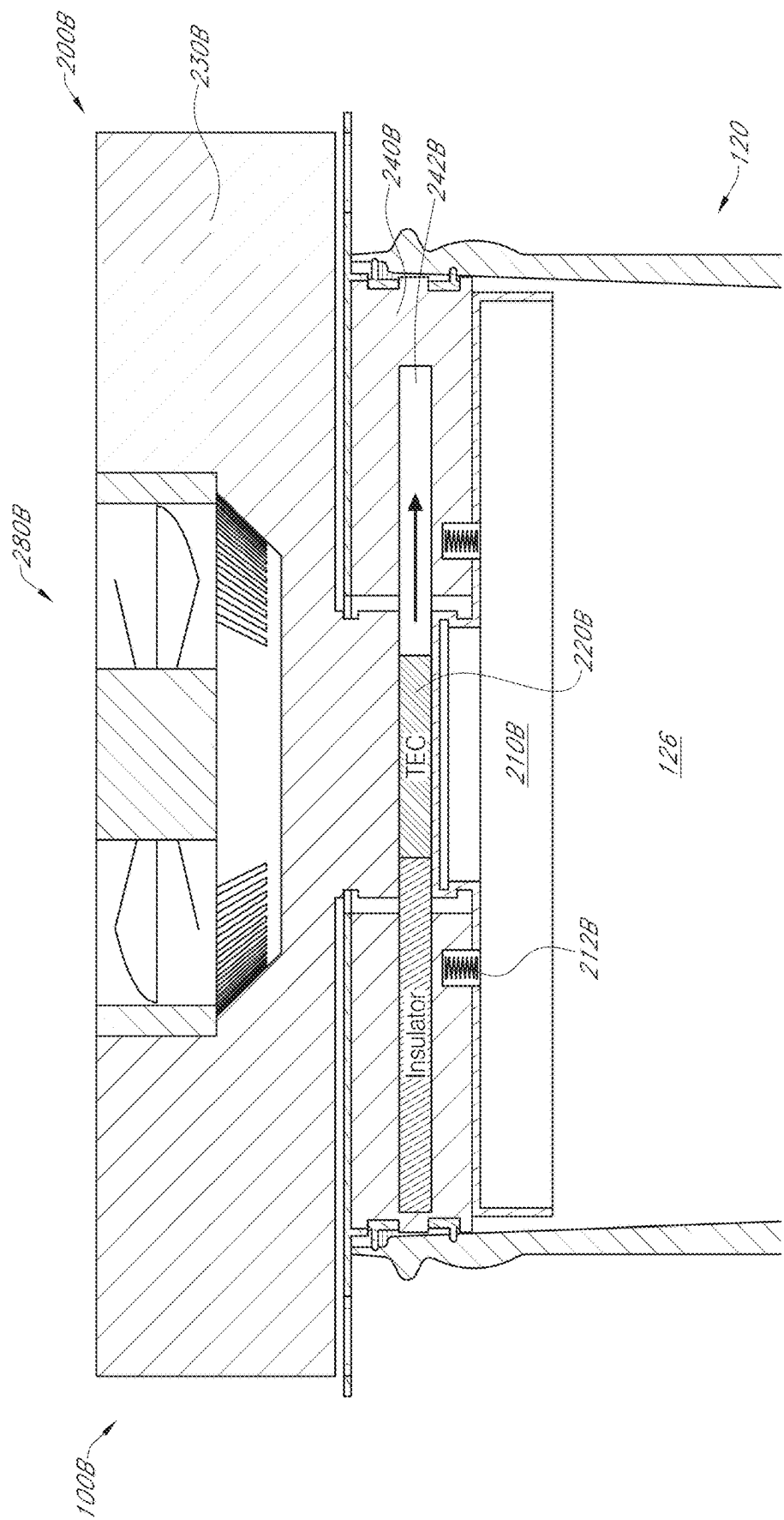


FIG. 2A

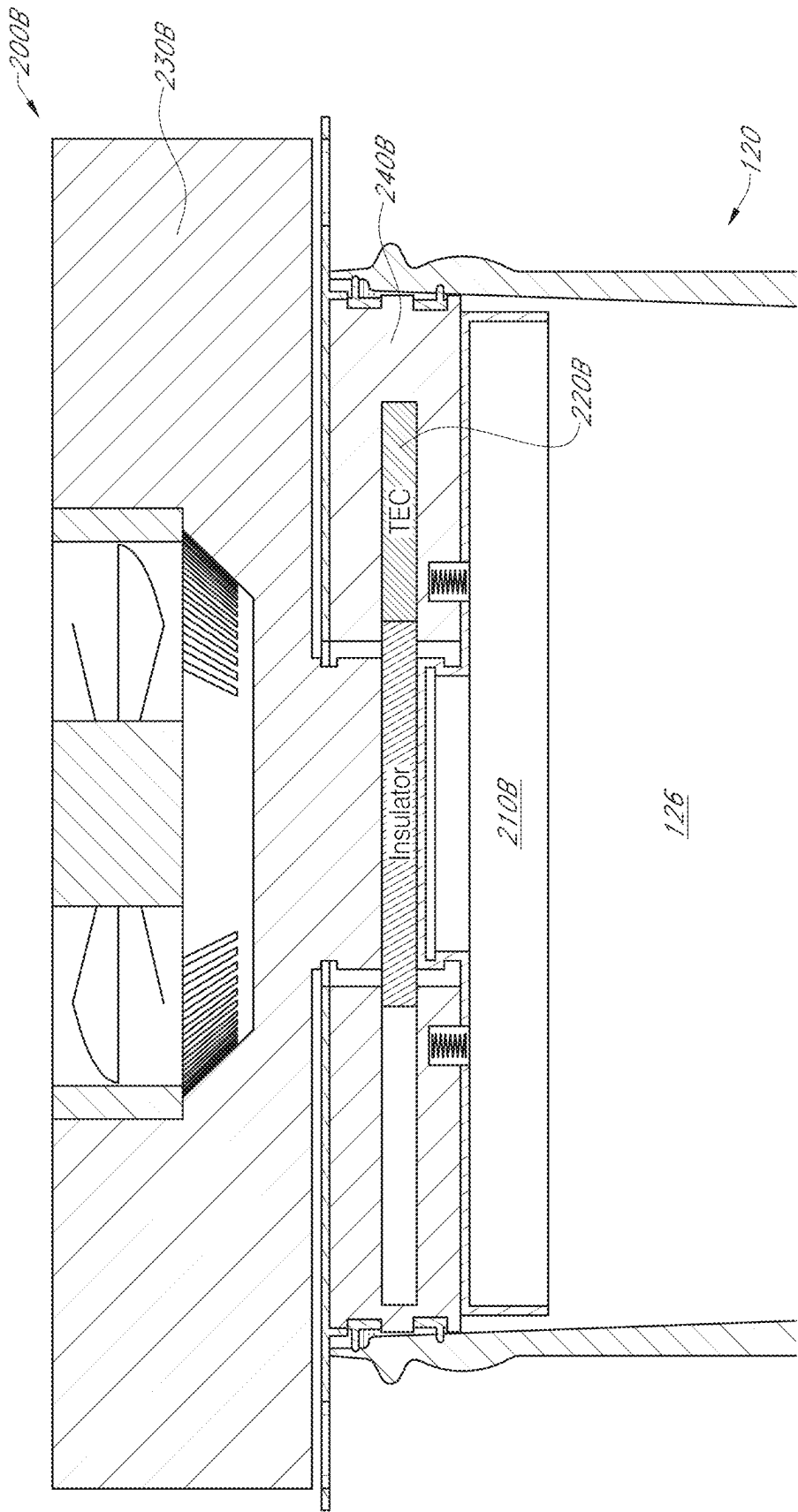


FIG. 2B

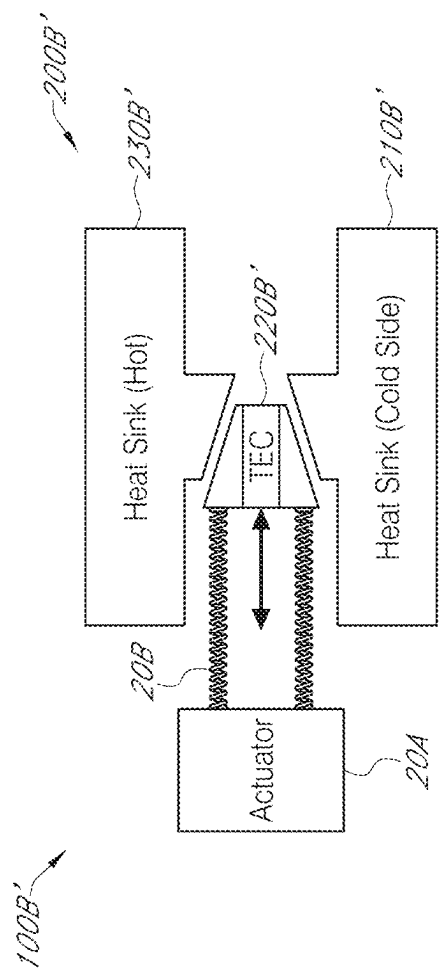


FIG. 2C

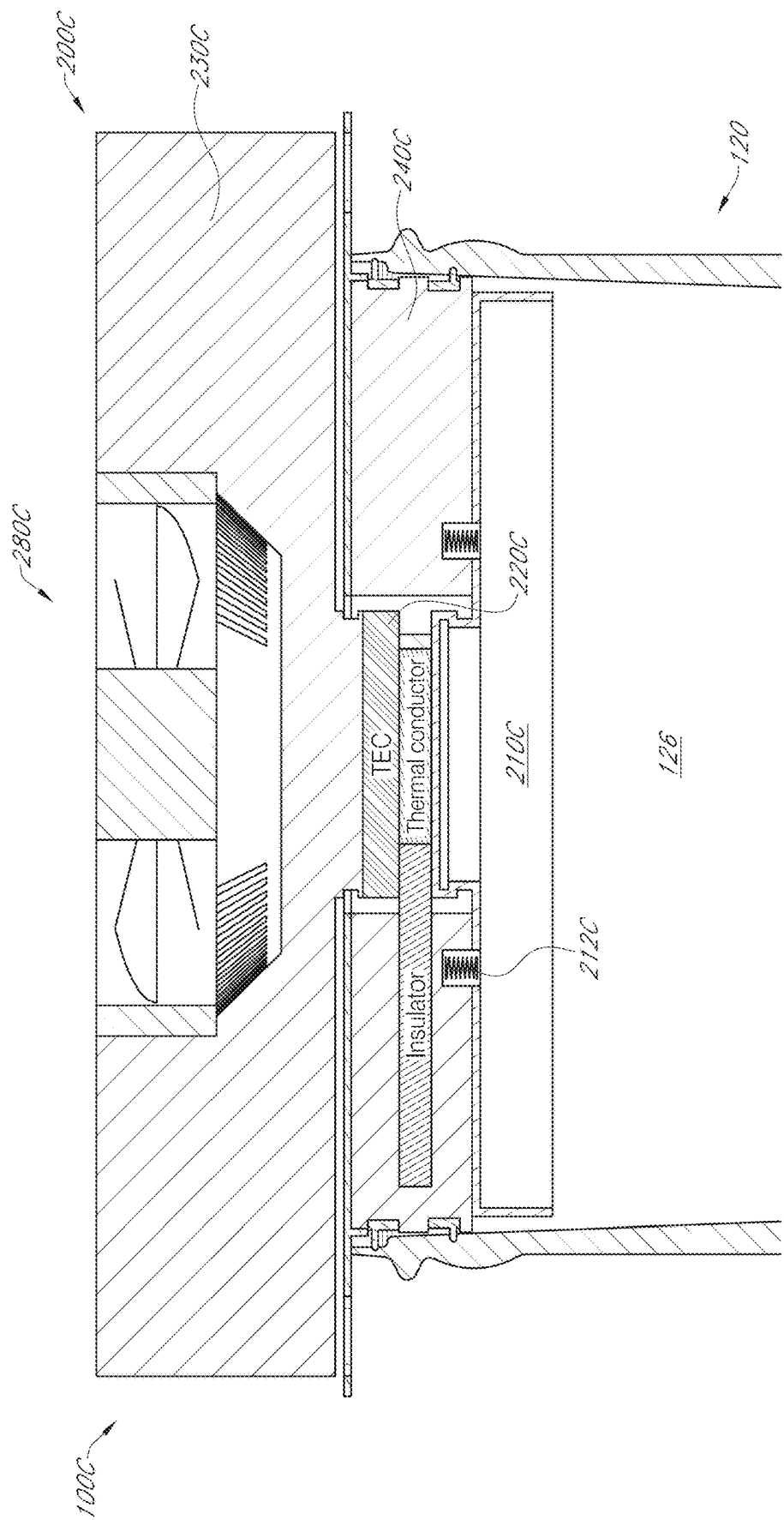


FIG. 3A

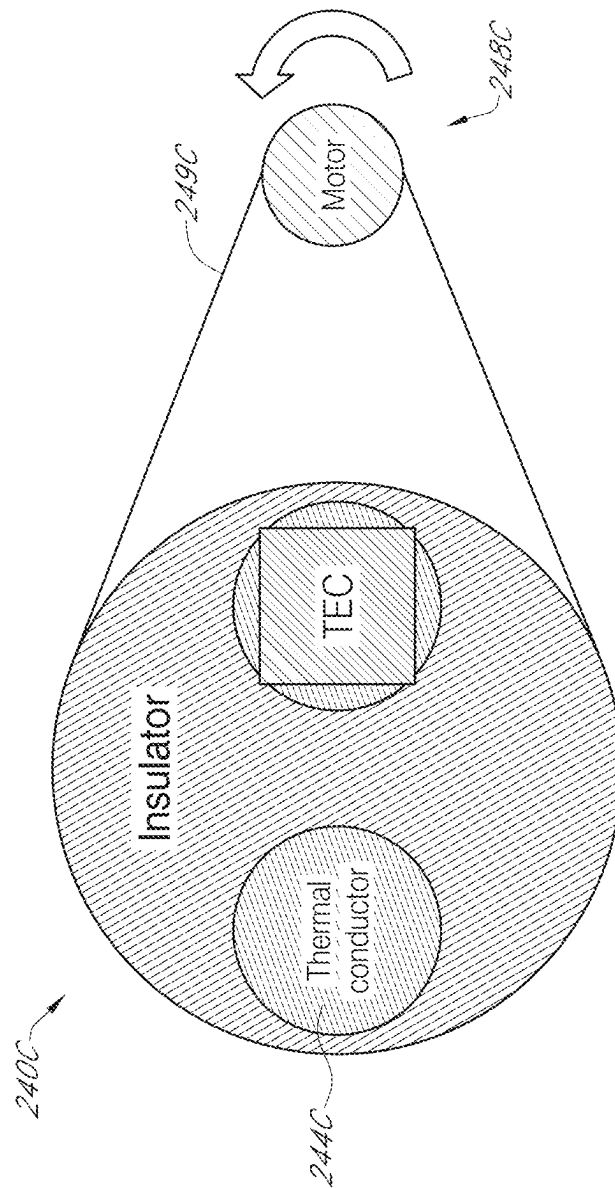


FIG. 3B

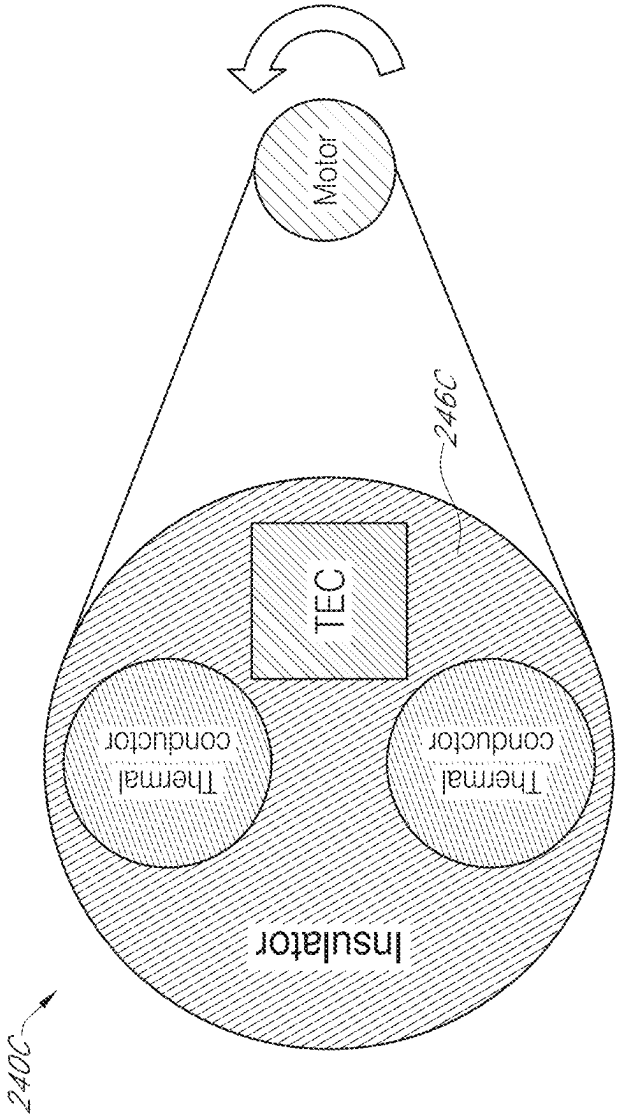


FIG. 3C

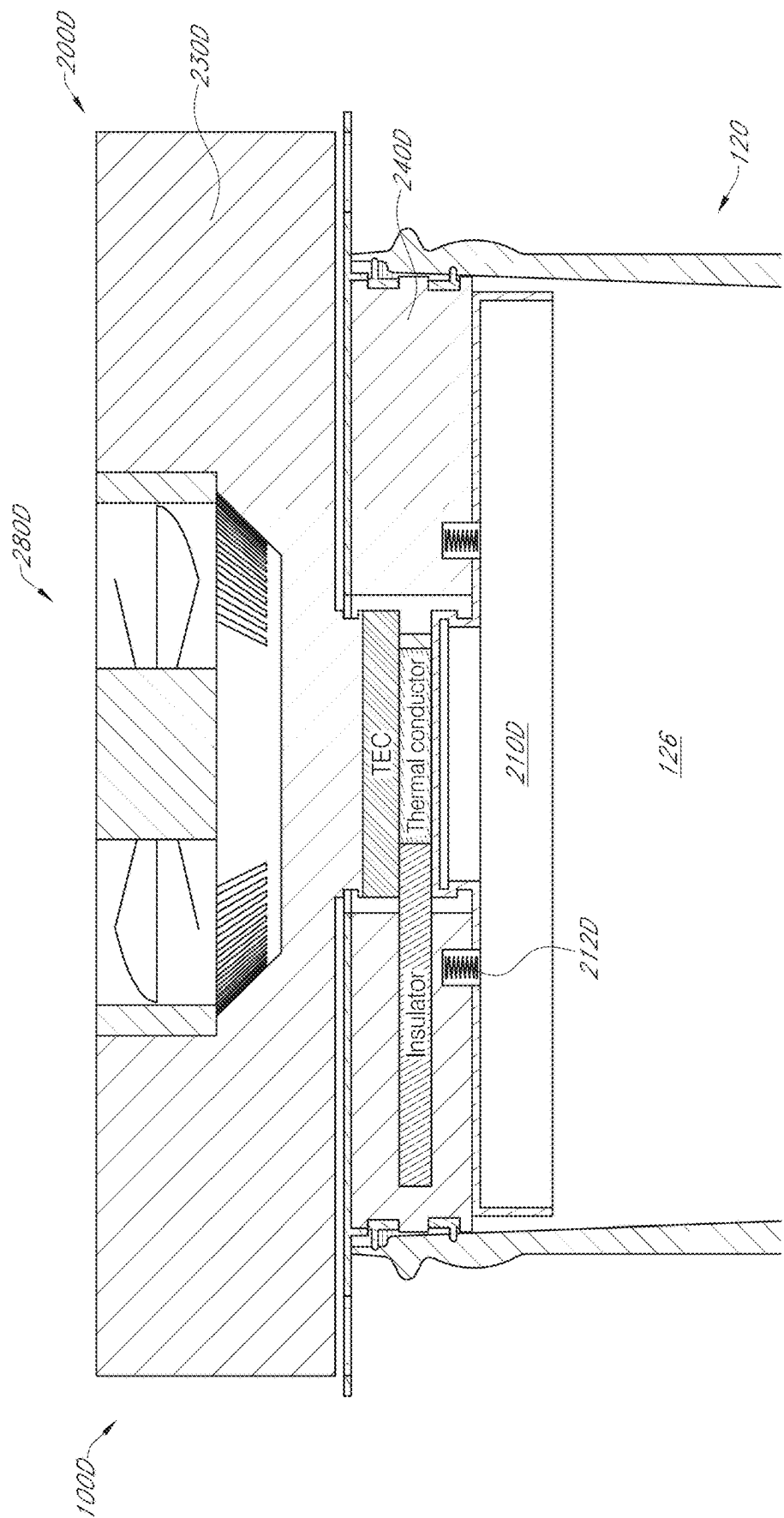


FIG. 4A

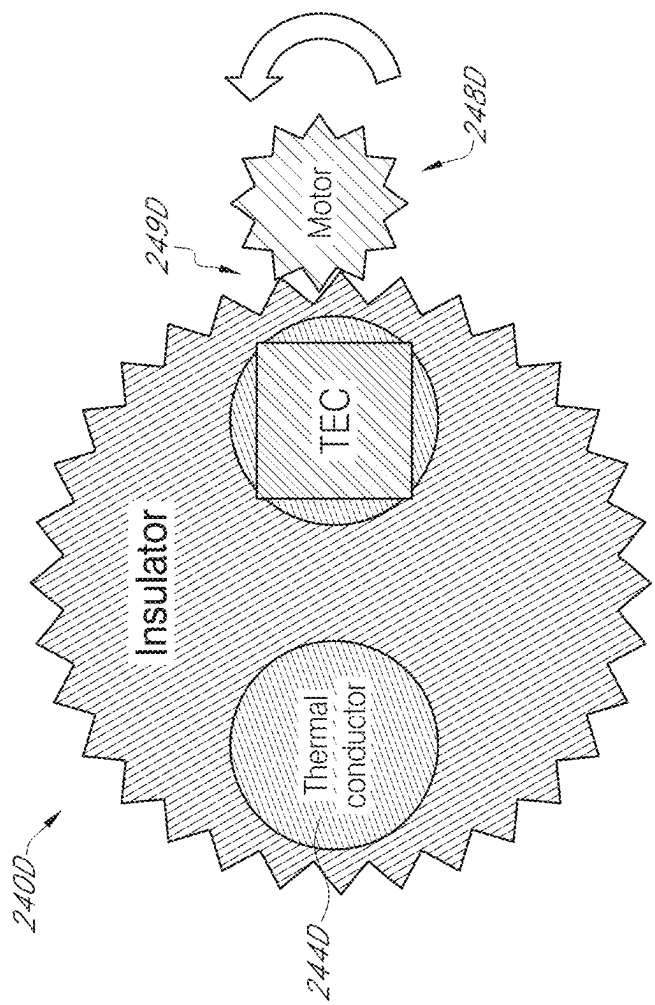


FIG. 4B

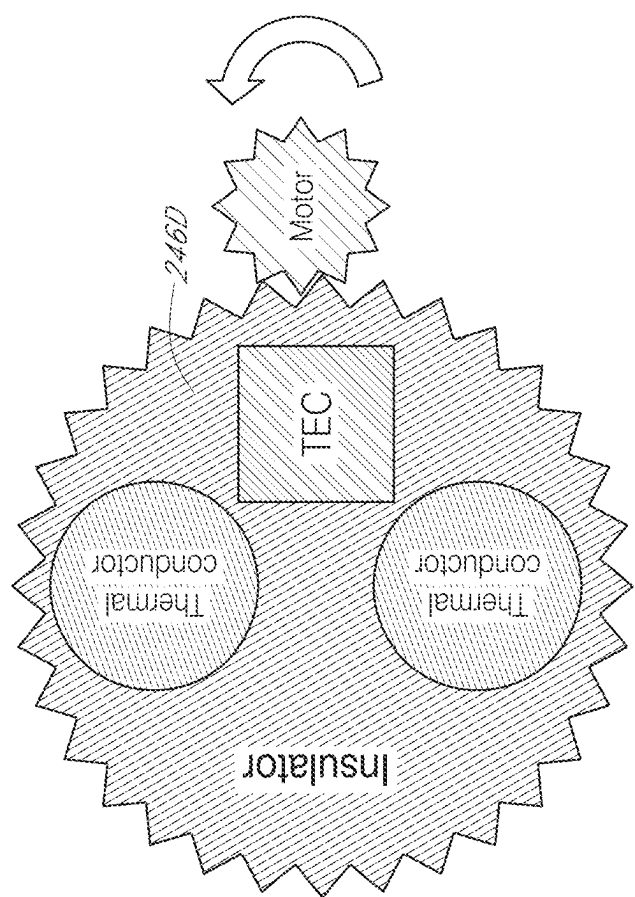


FIG. 4C

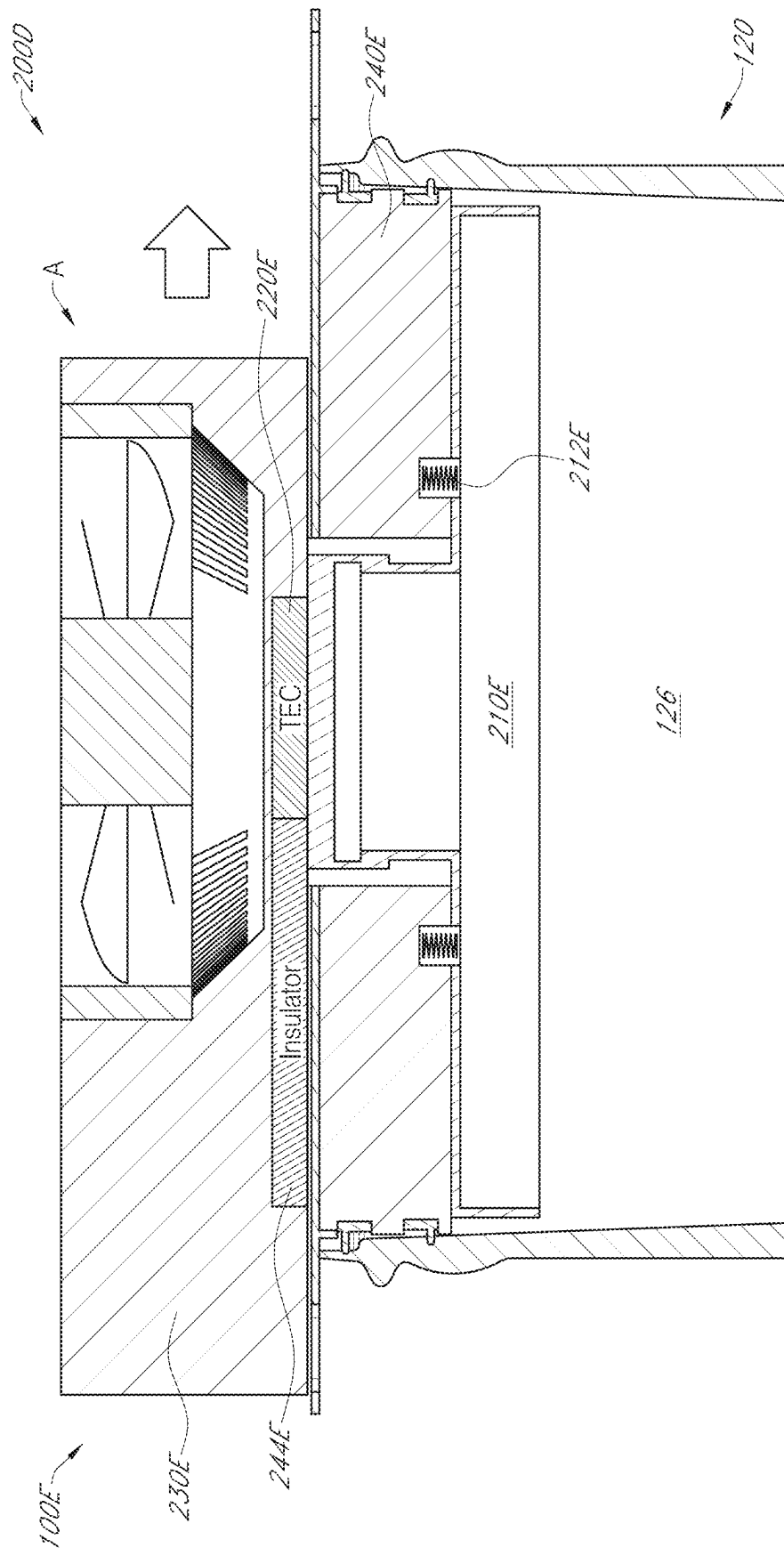


FIG. 5A

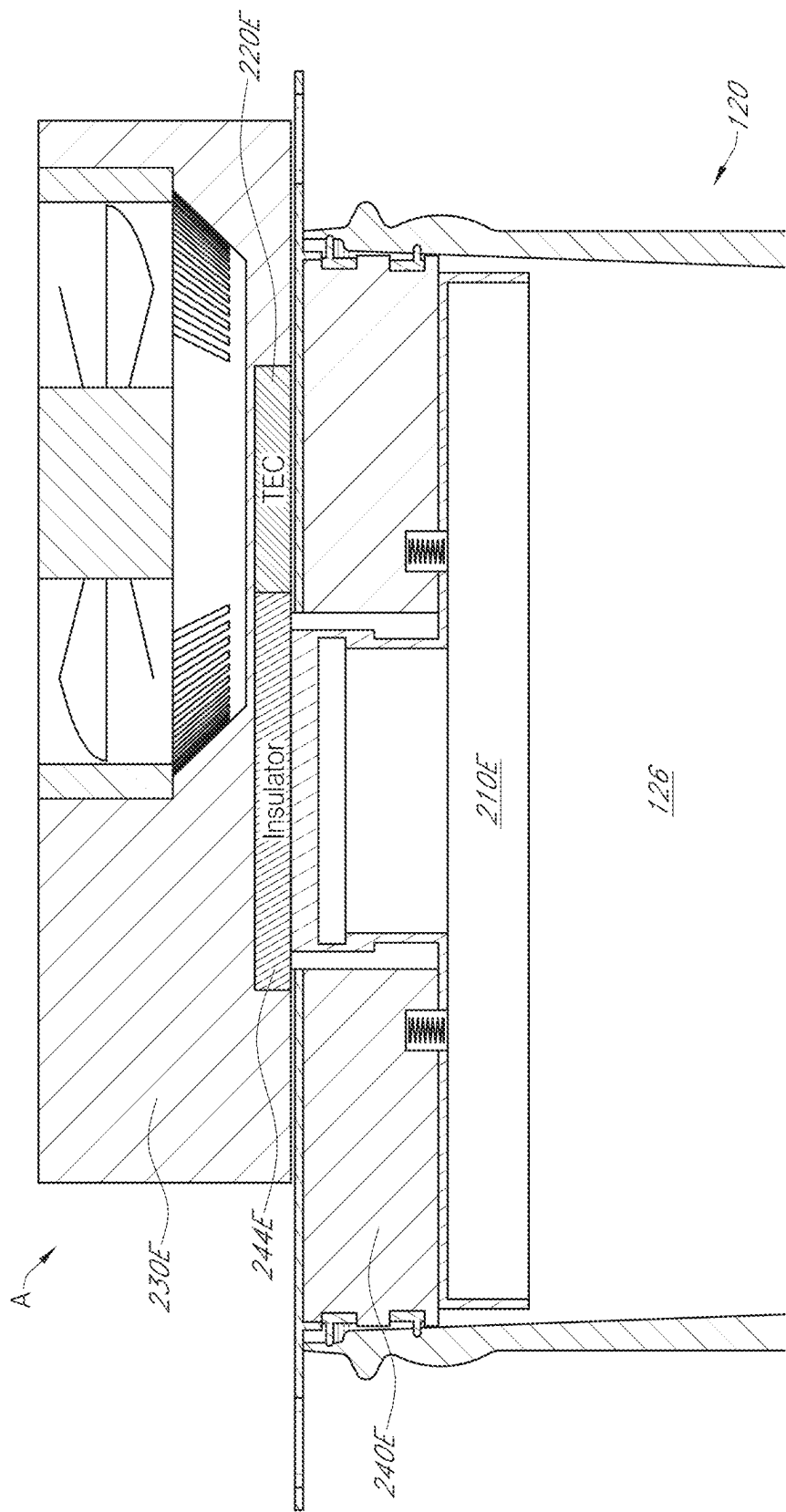


FIG. 5B

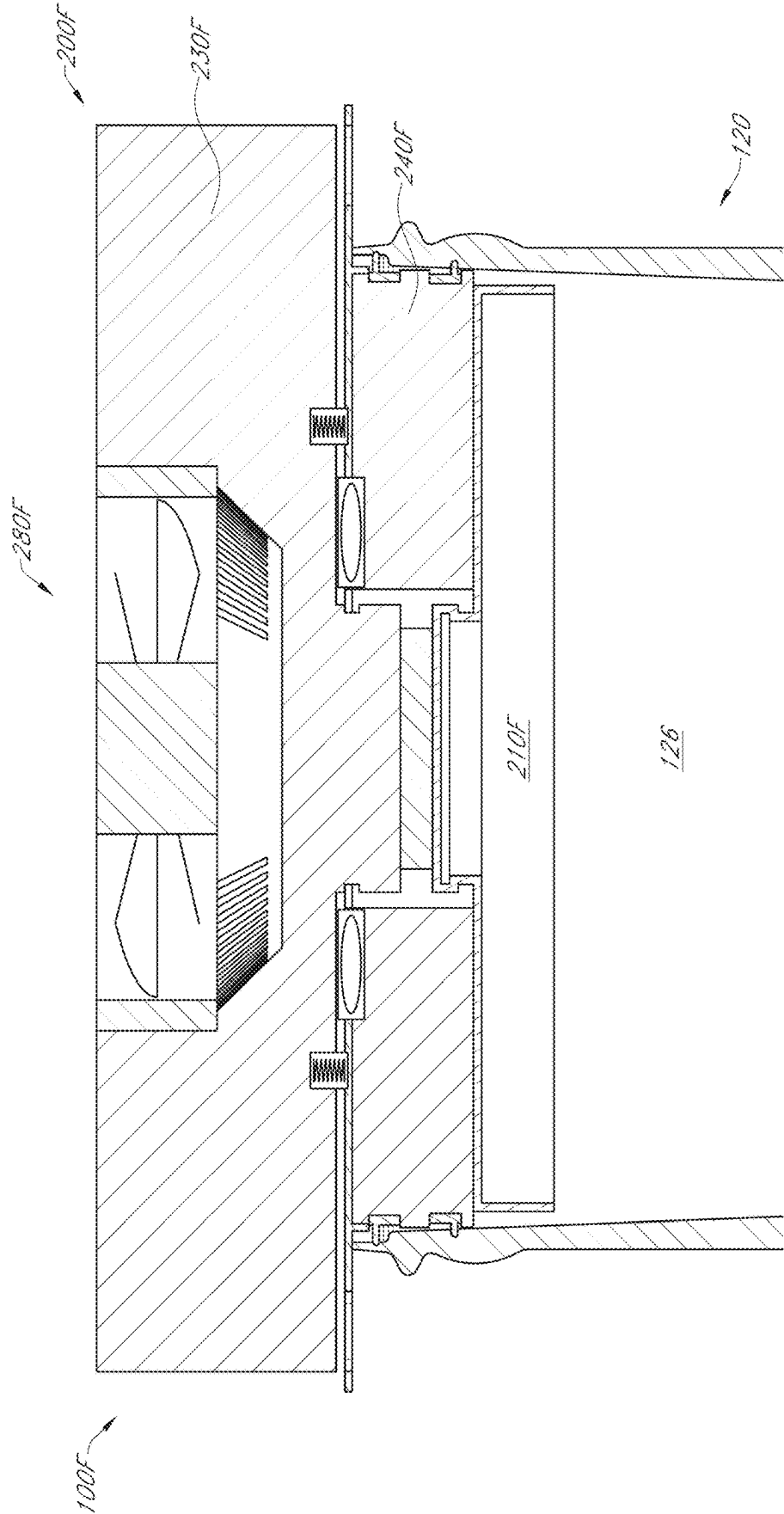


FIG. 6A

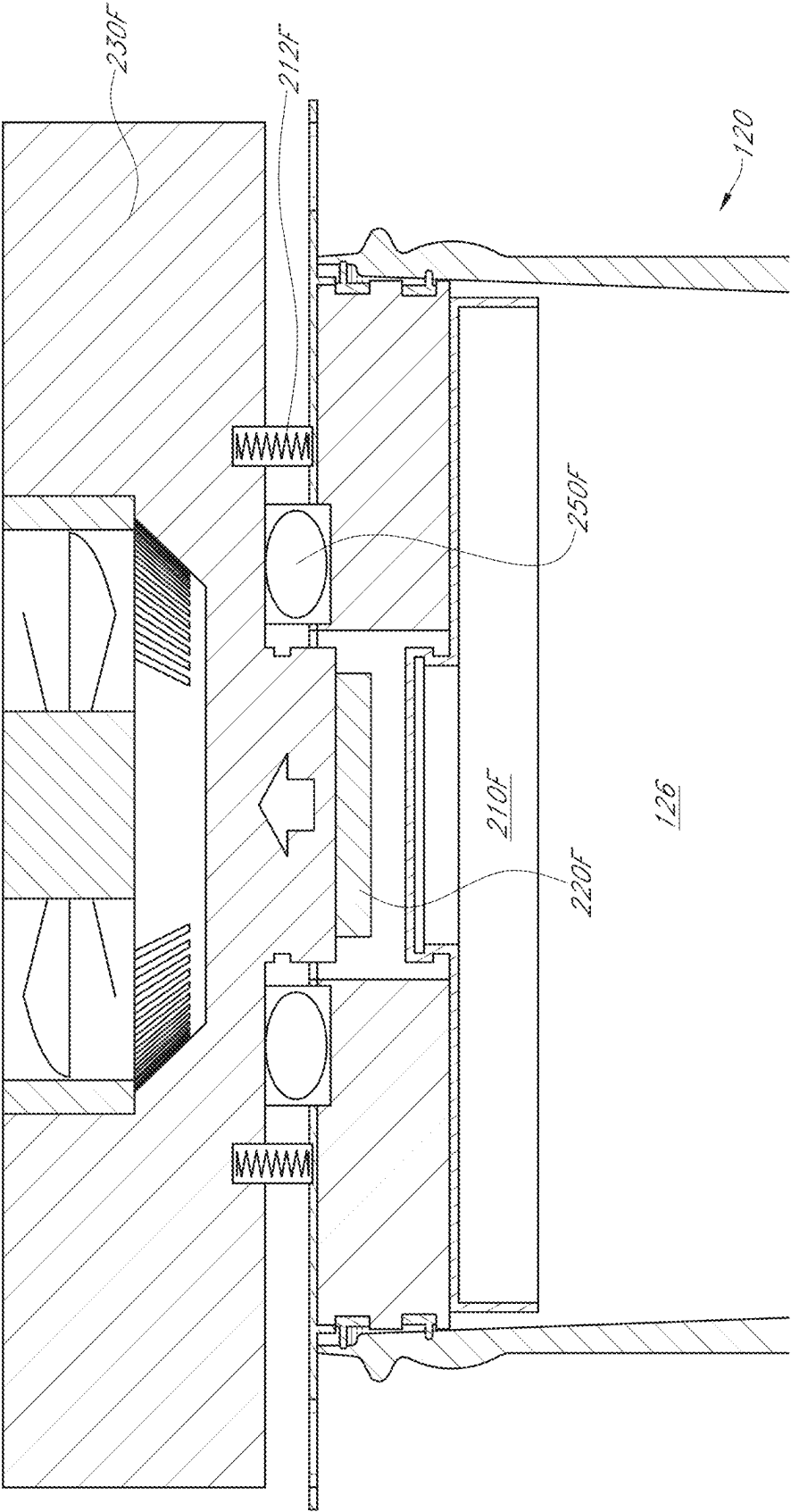


FIG. 6B

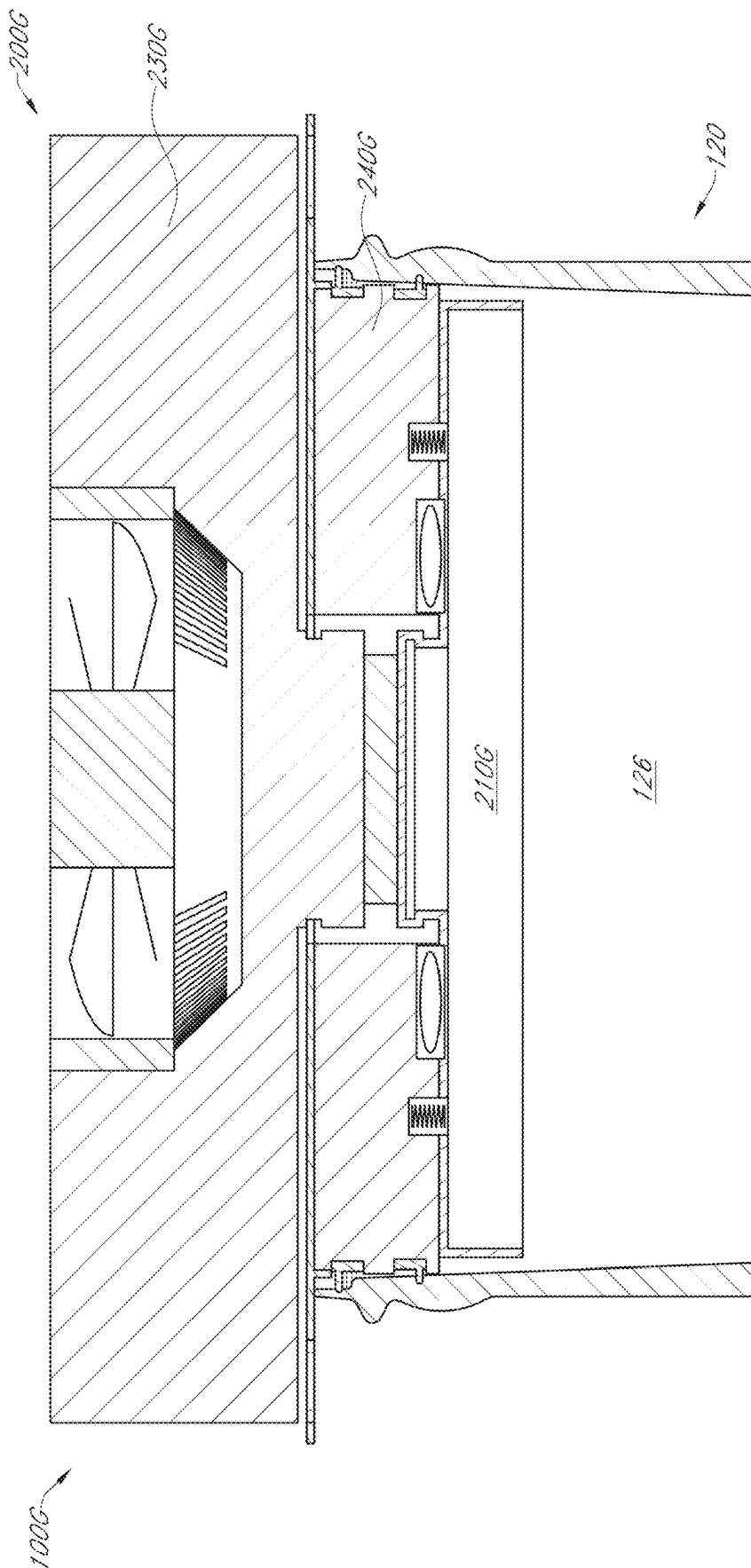


FIG. 7A

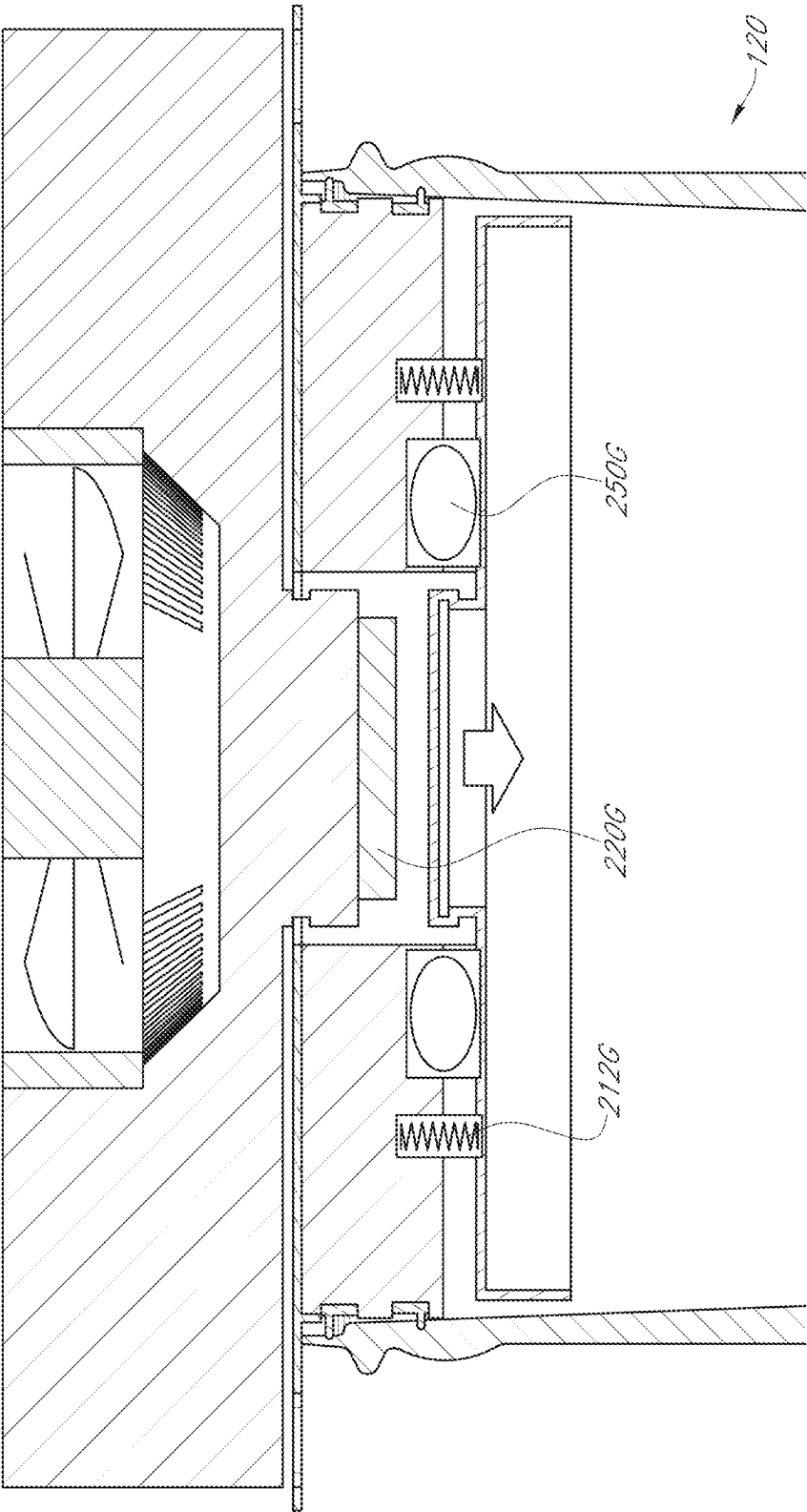


FIG. 7B

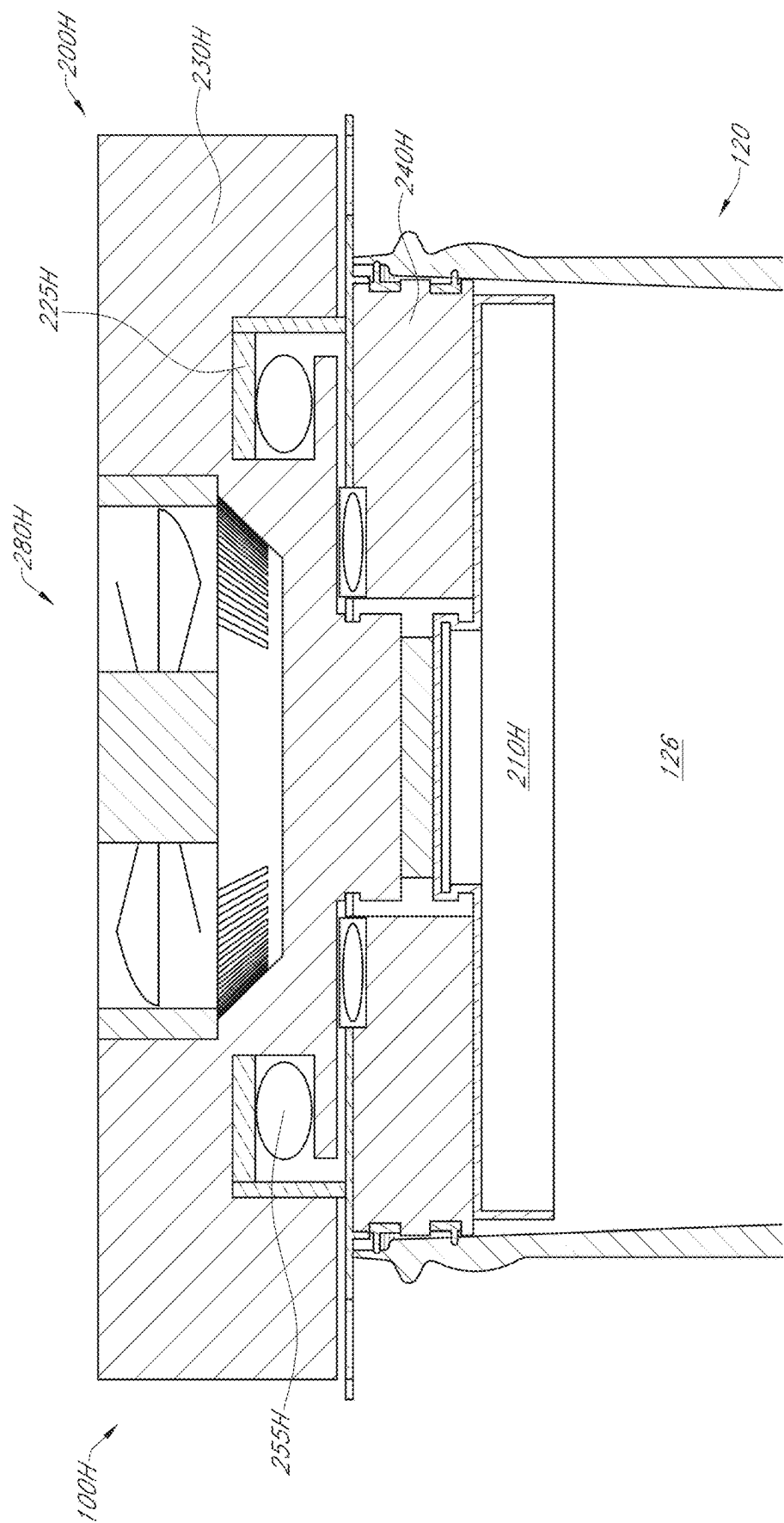


FIG. 8A

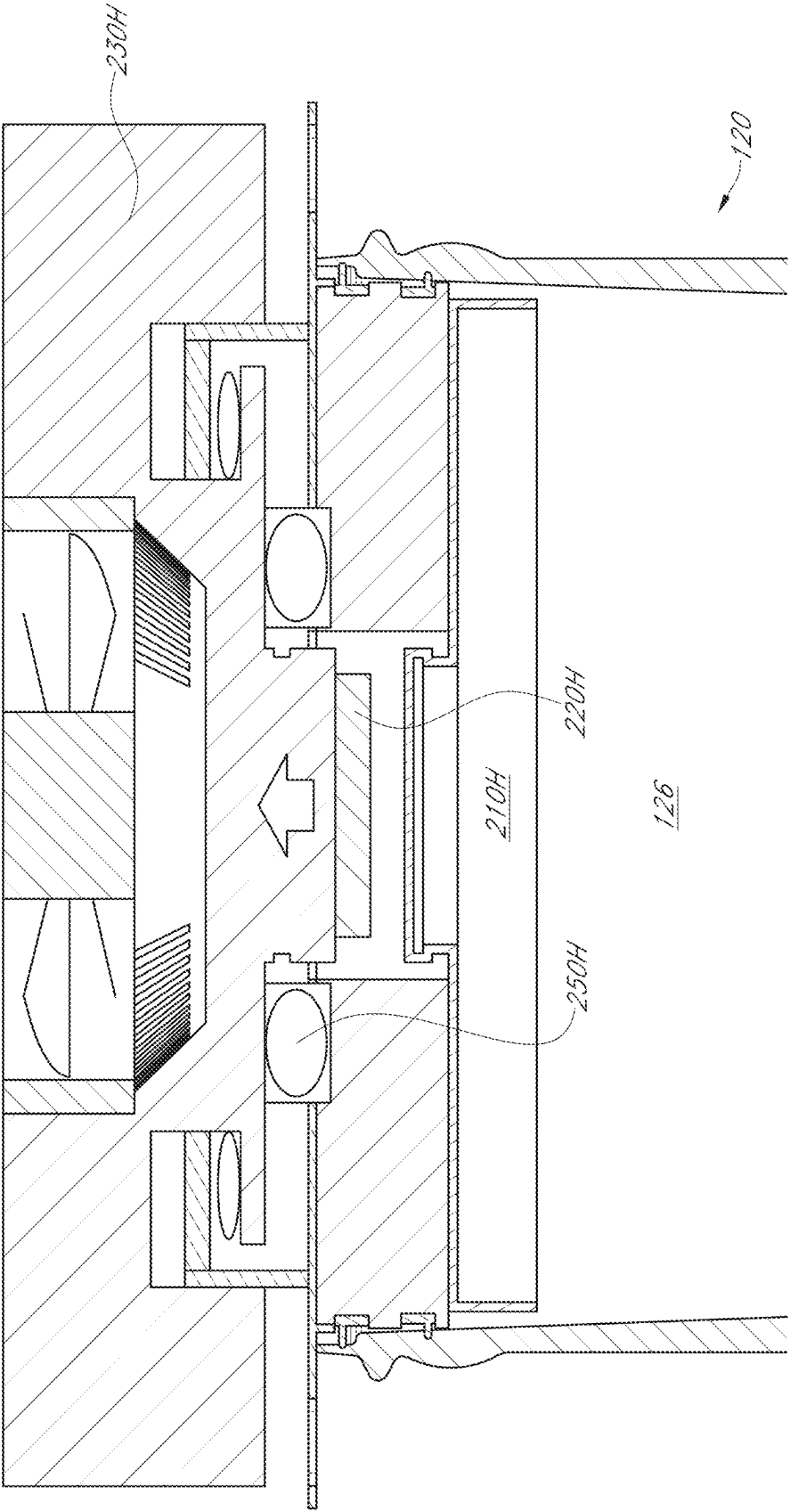


FIG. 8B

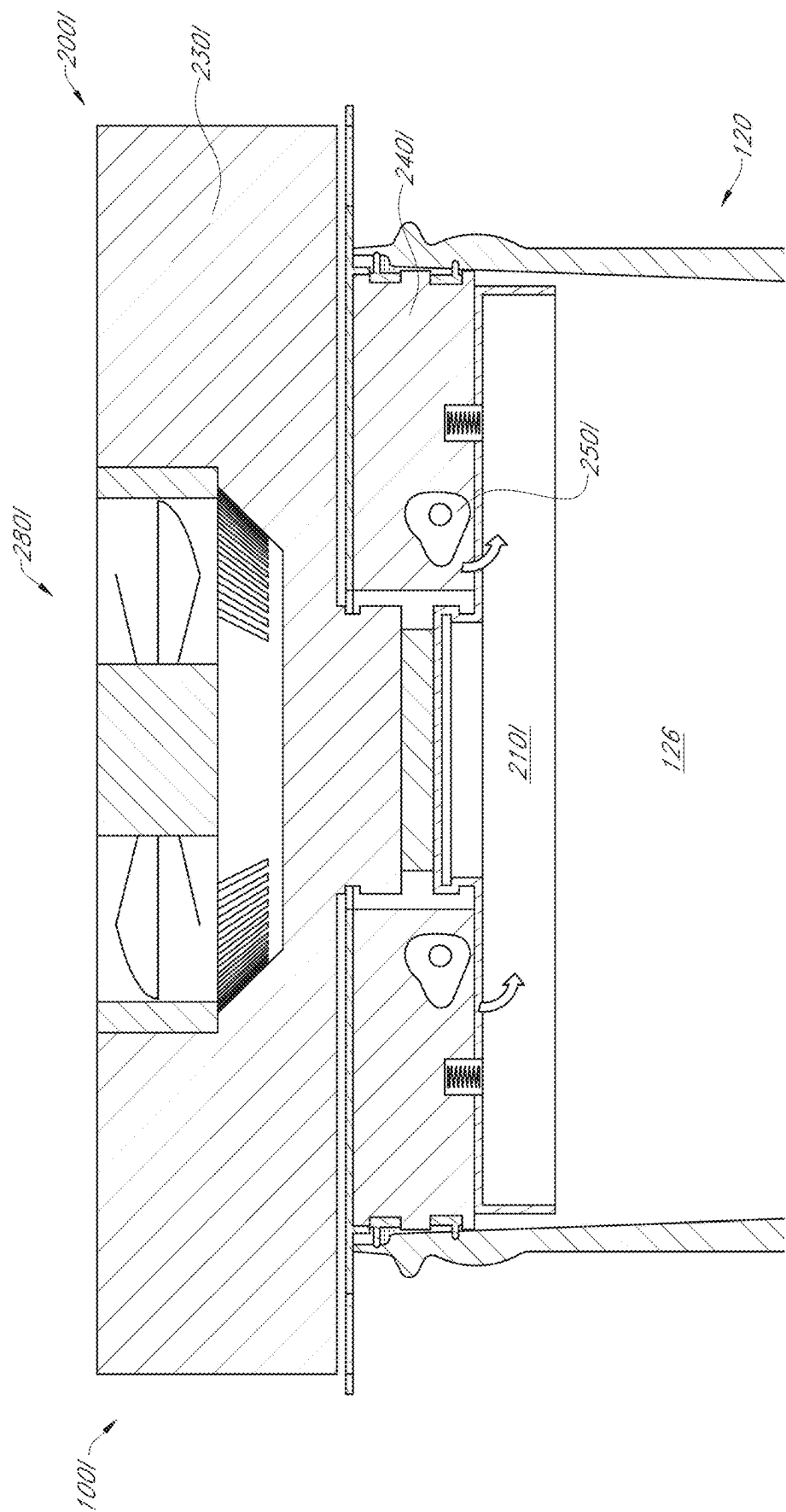


FIG. 9A

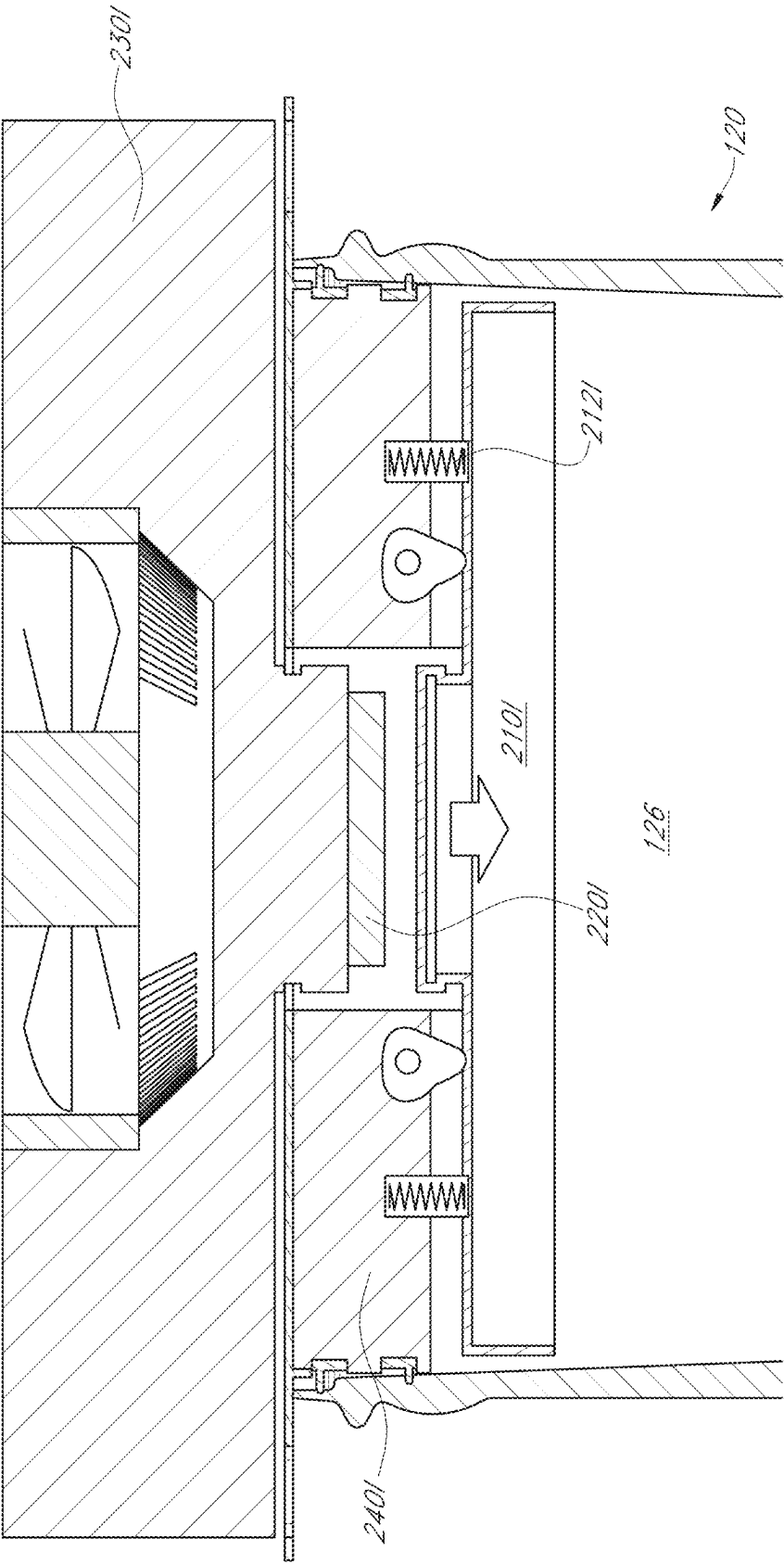


FIG. 9B

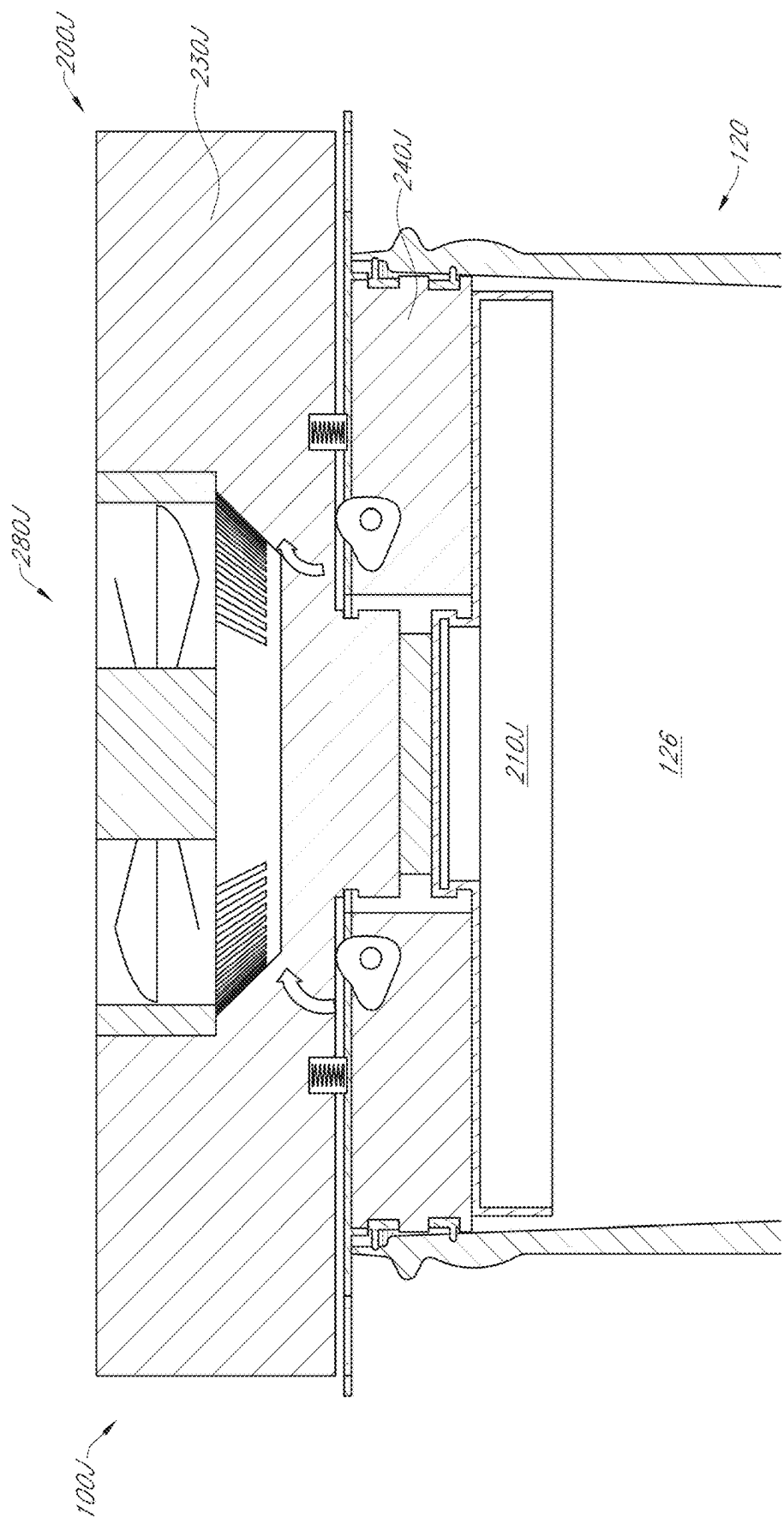


FIG. 10A

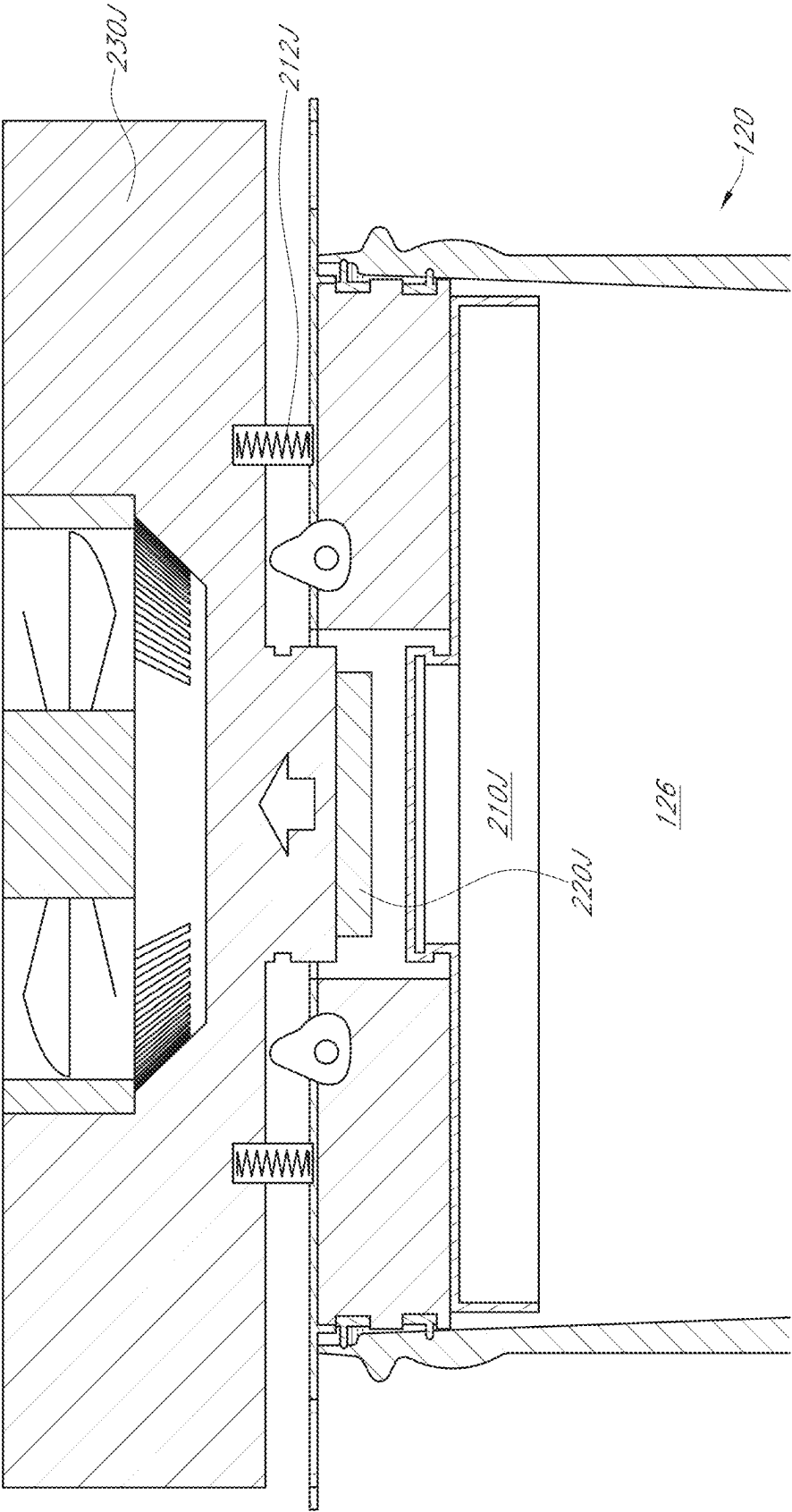


FIG. 10B

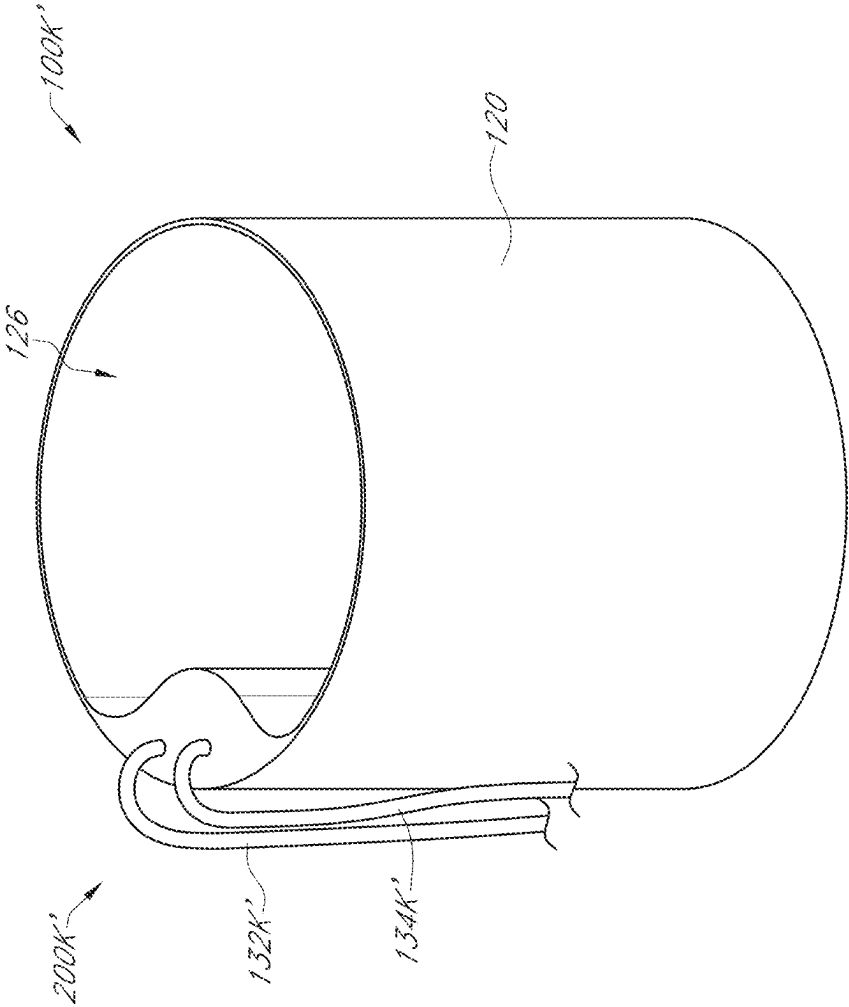


FIG. 11B

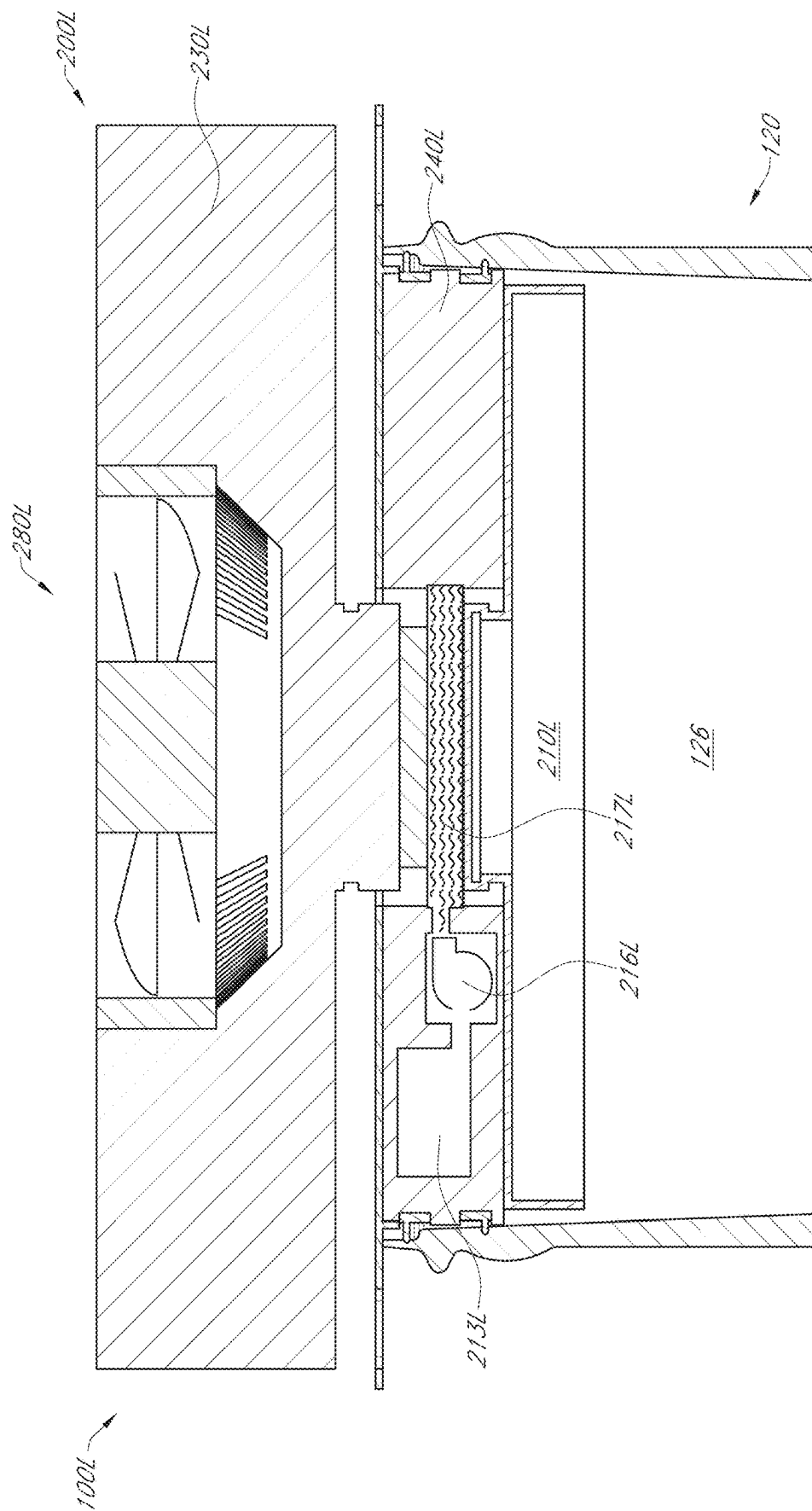


FIG. 12A

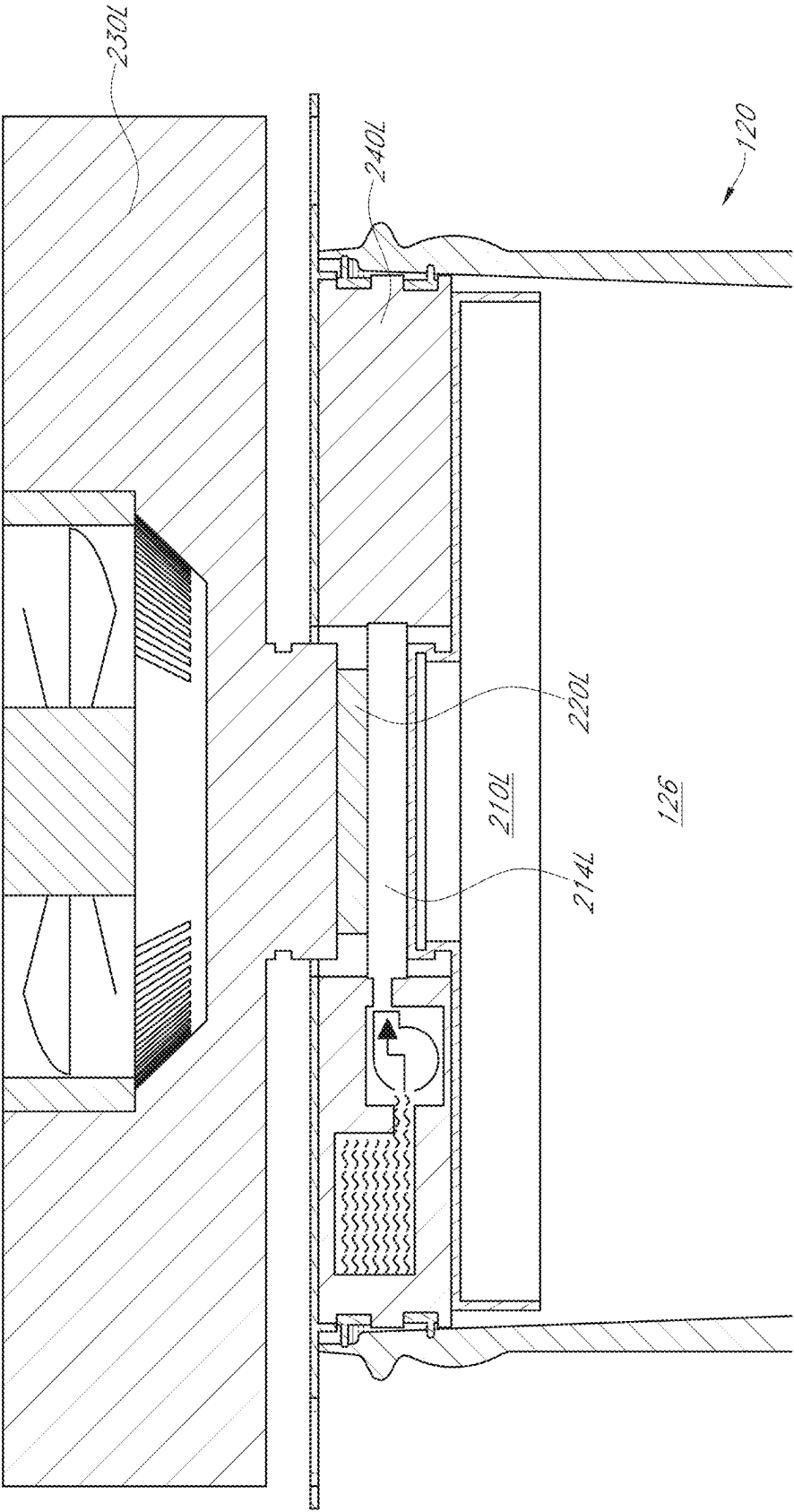


FIG. 12B

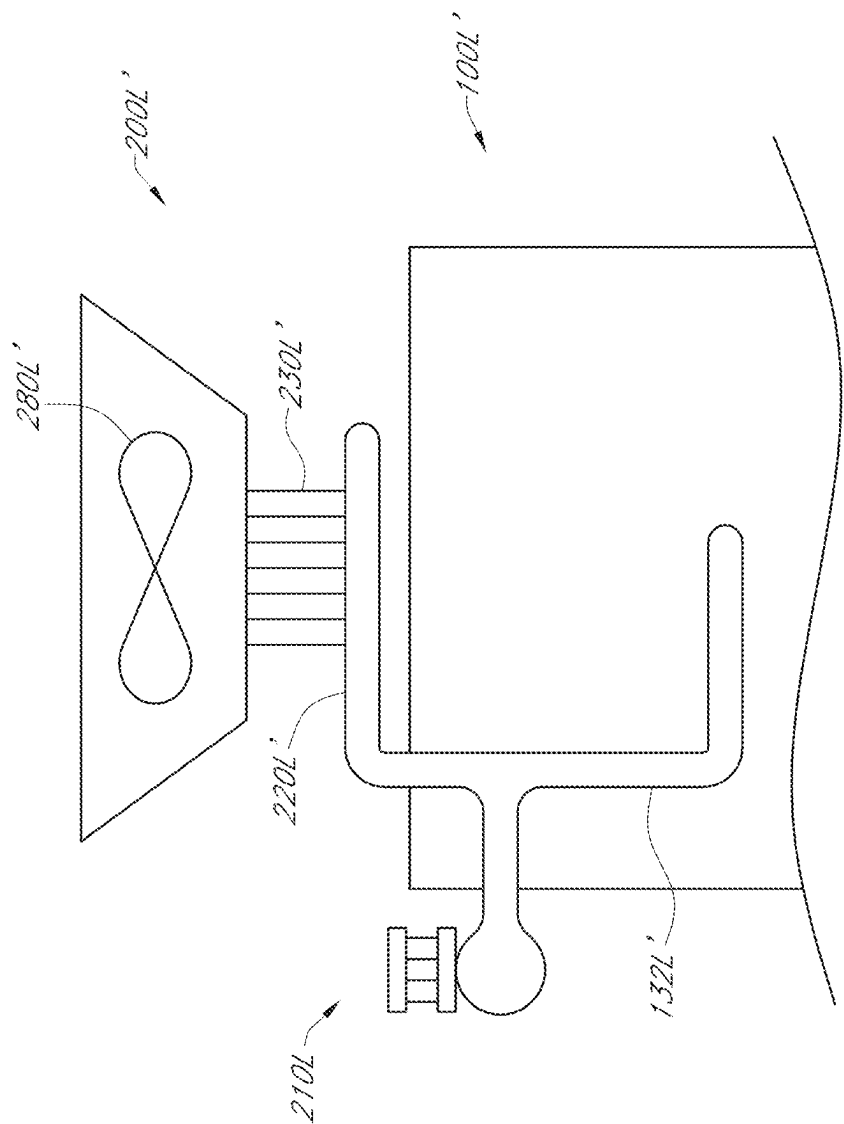


FIG. 12C

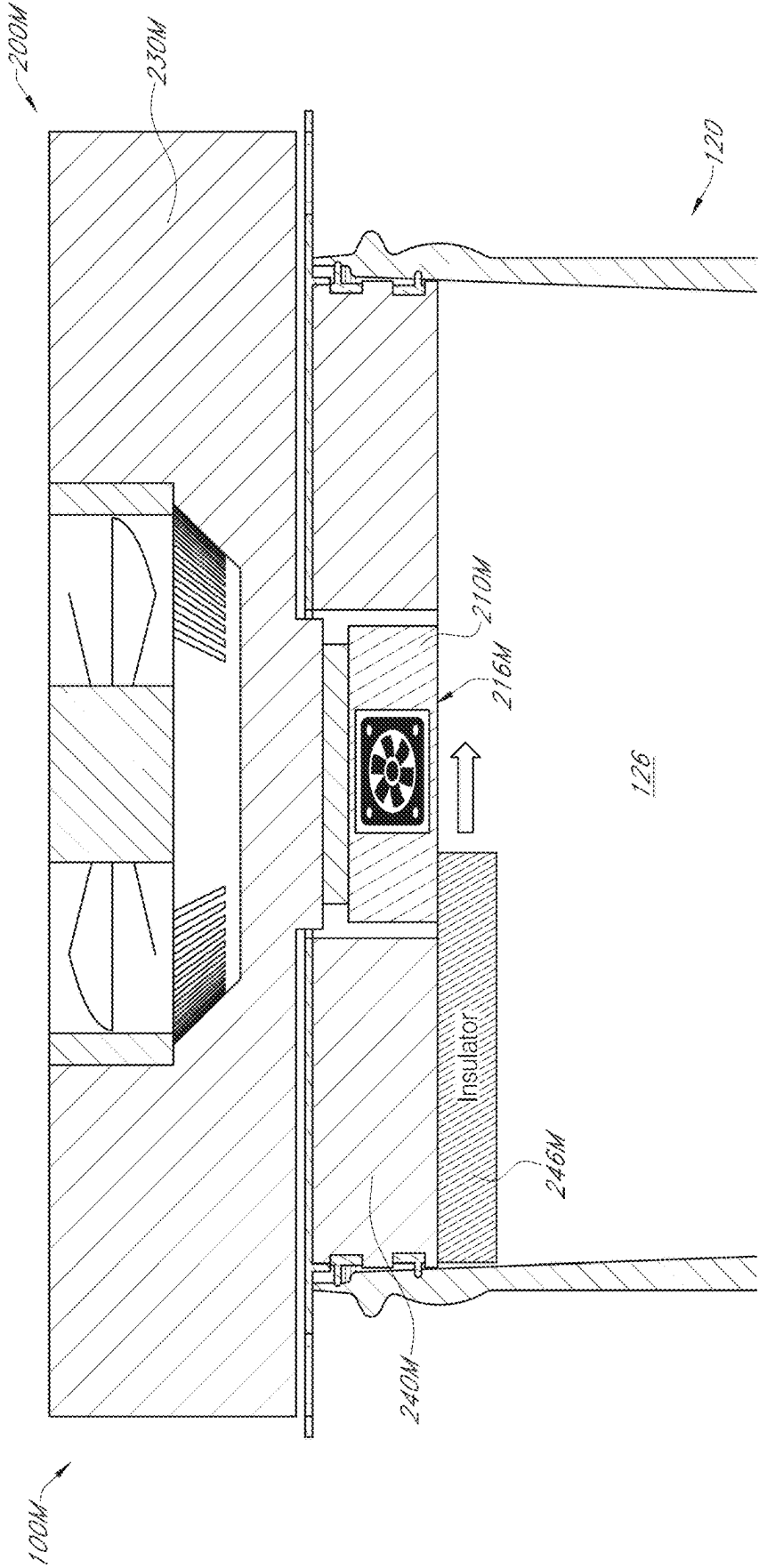


FIG. 13A

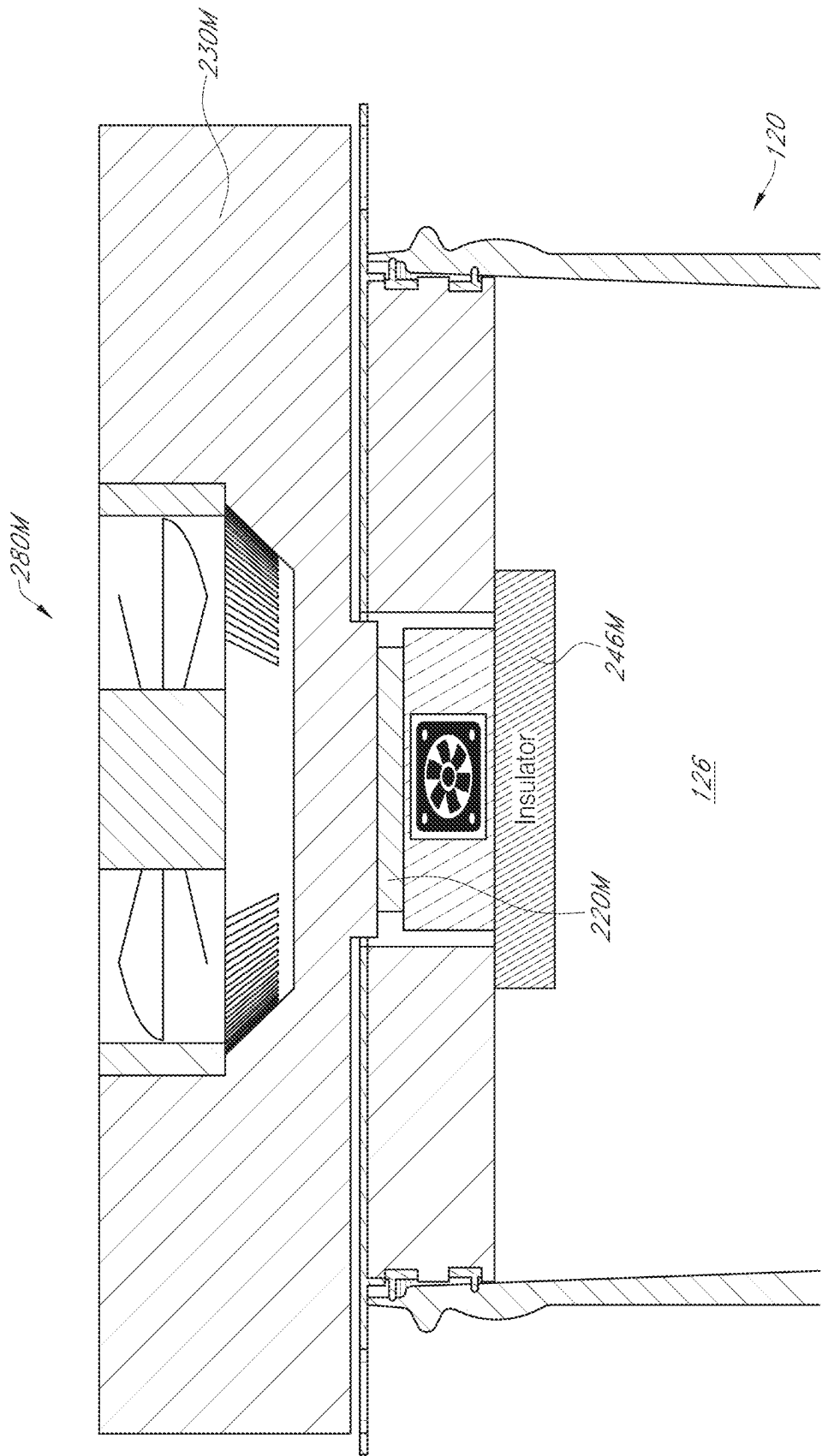


FIG. 13B

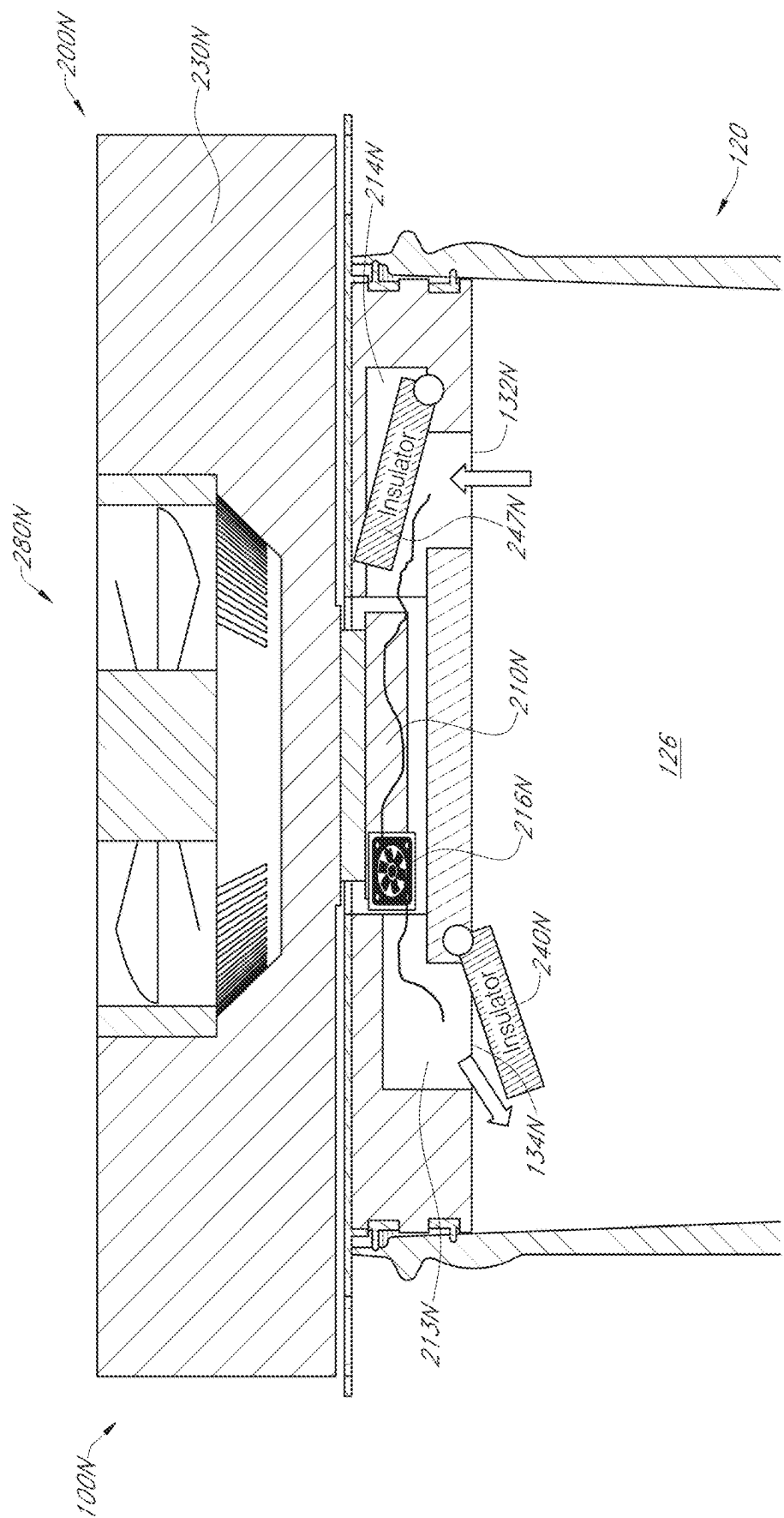


FIG. 14A

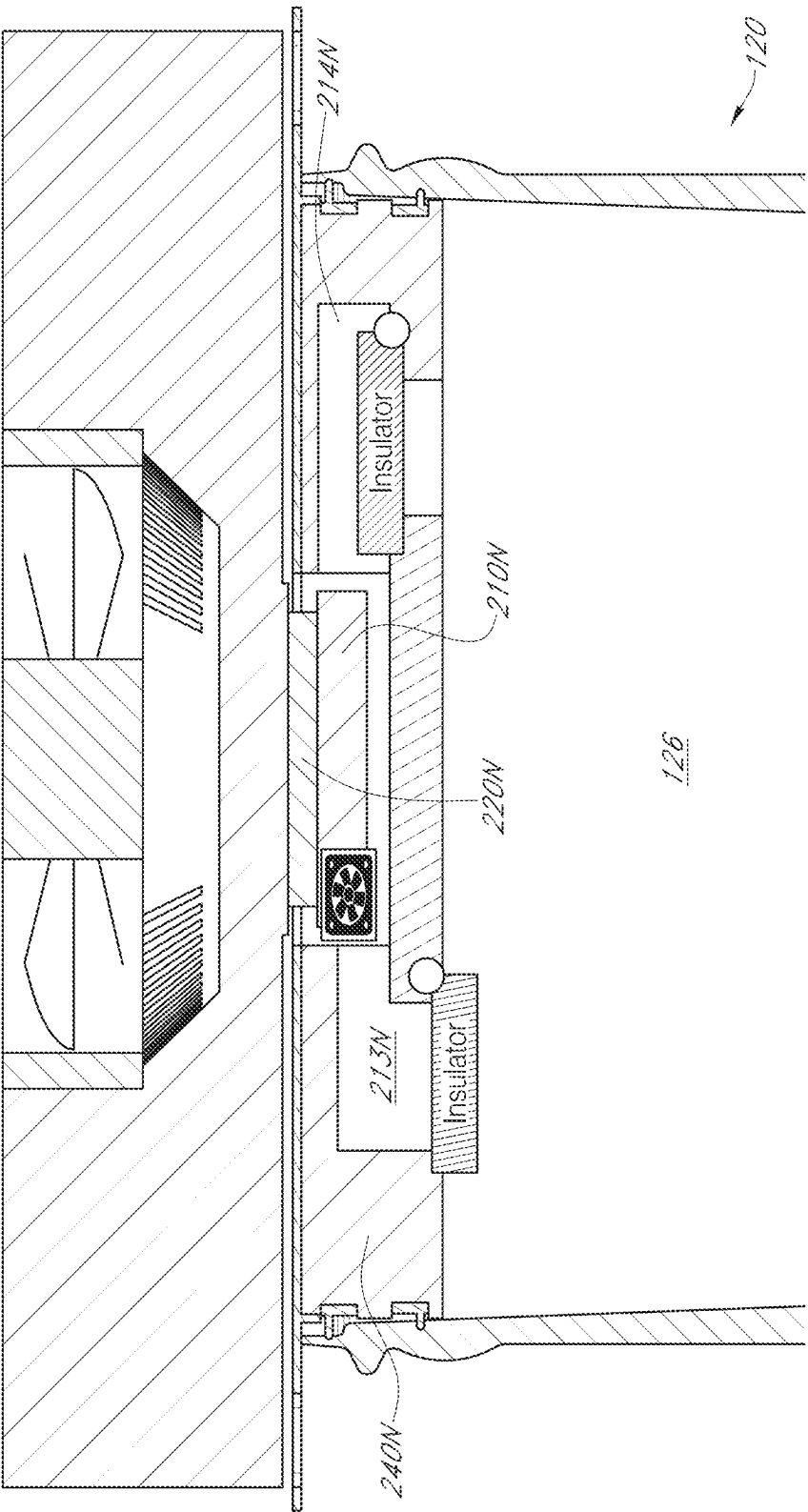


FIG. 14B

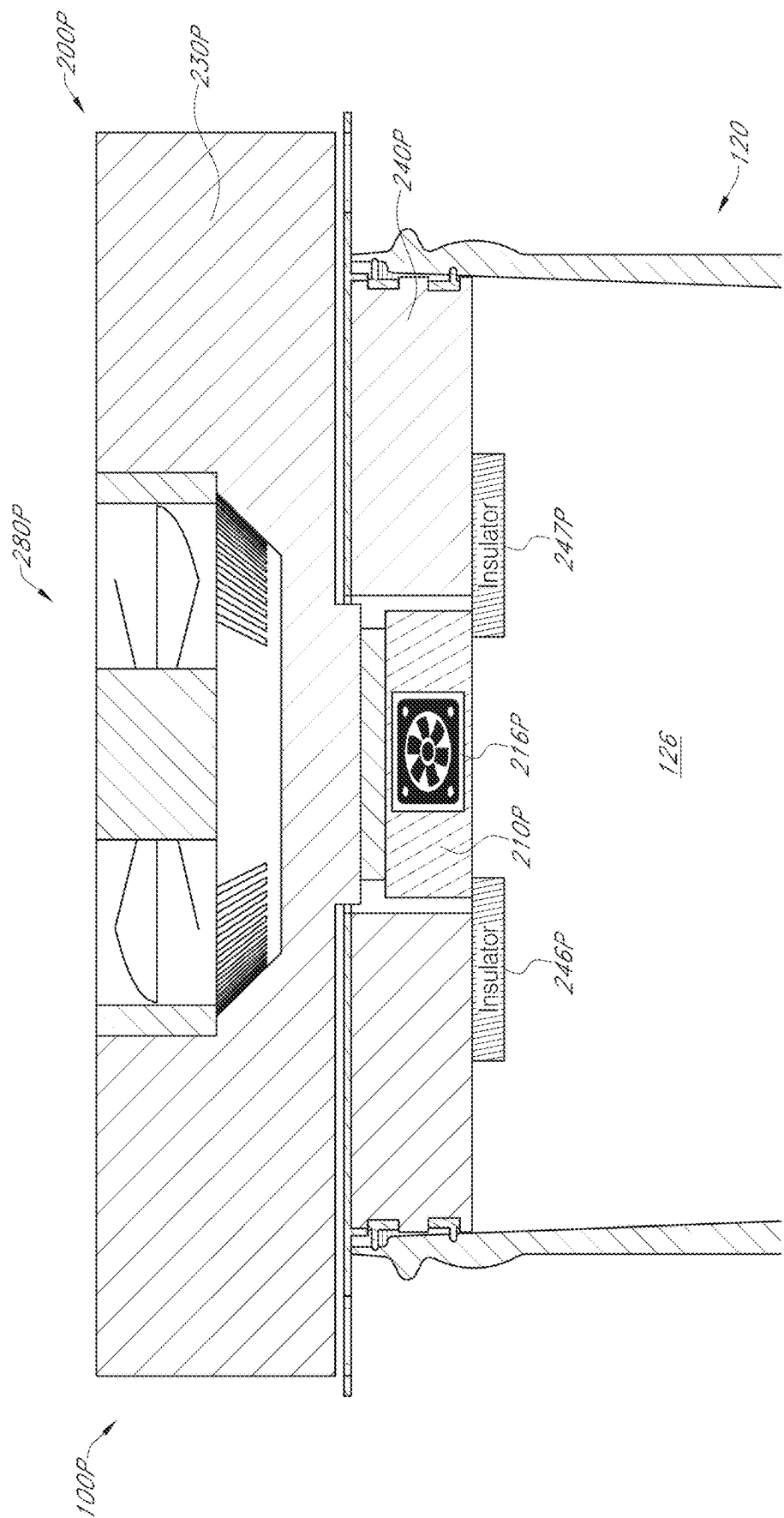


FIG. 15A

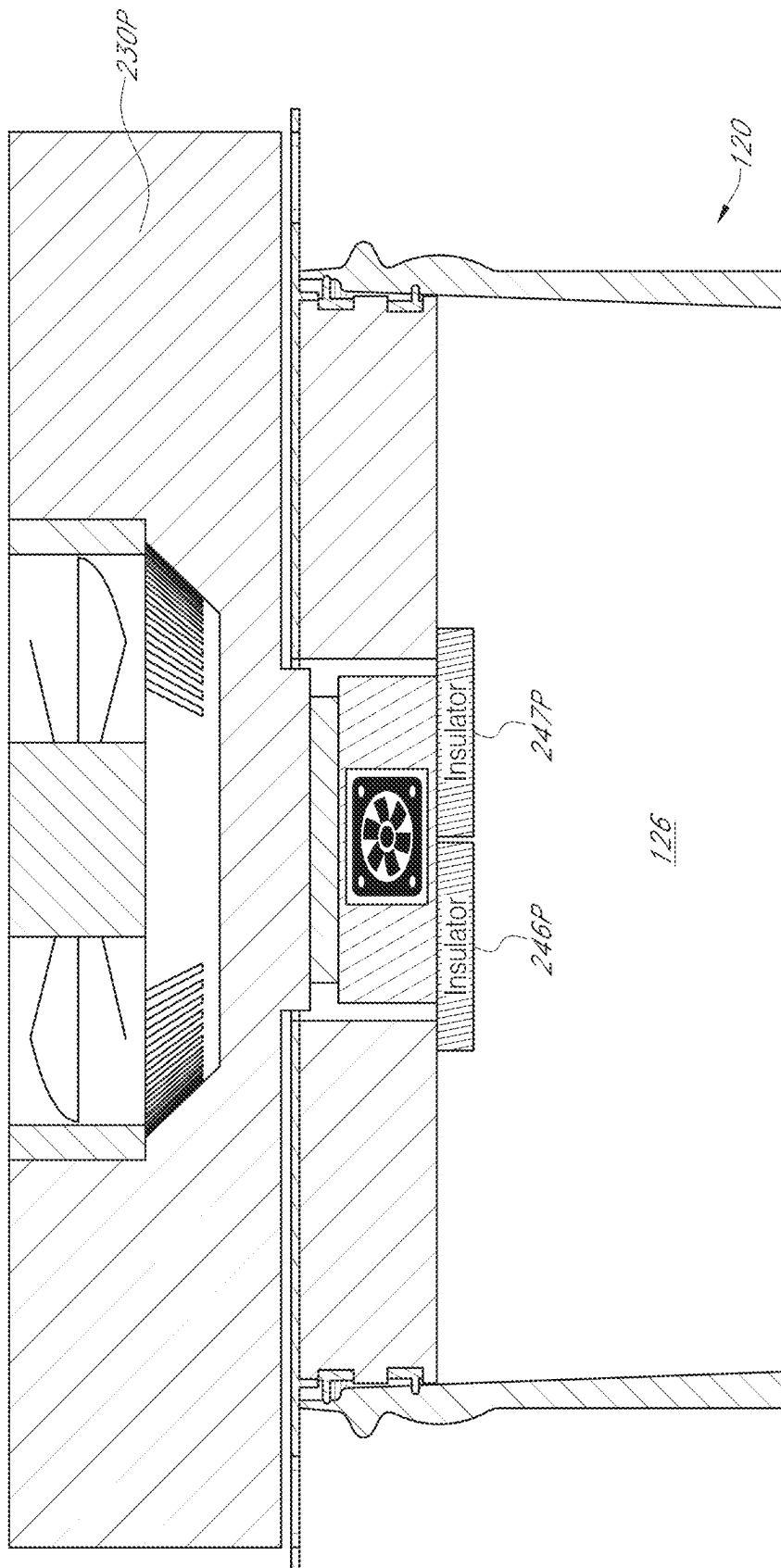


FIG. 15B

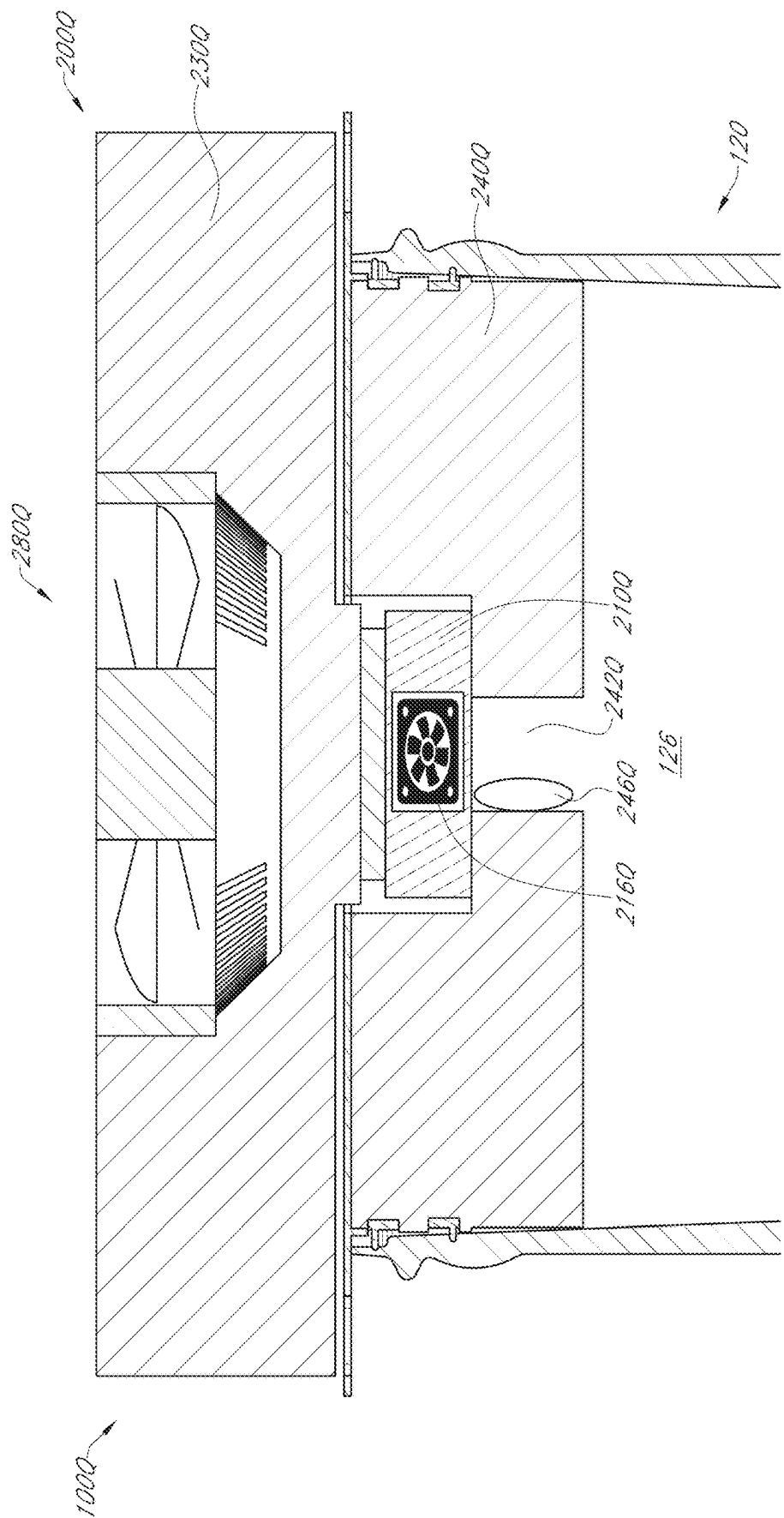


FIG. 16A

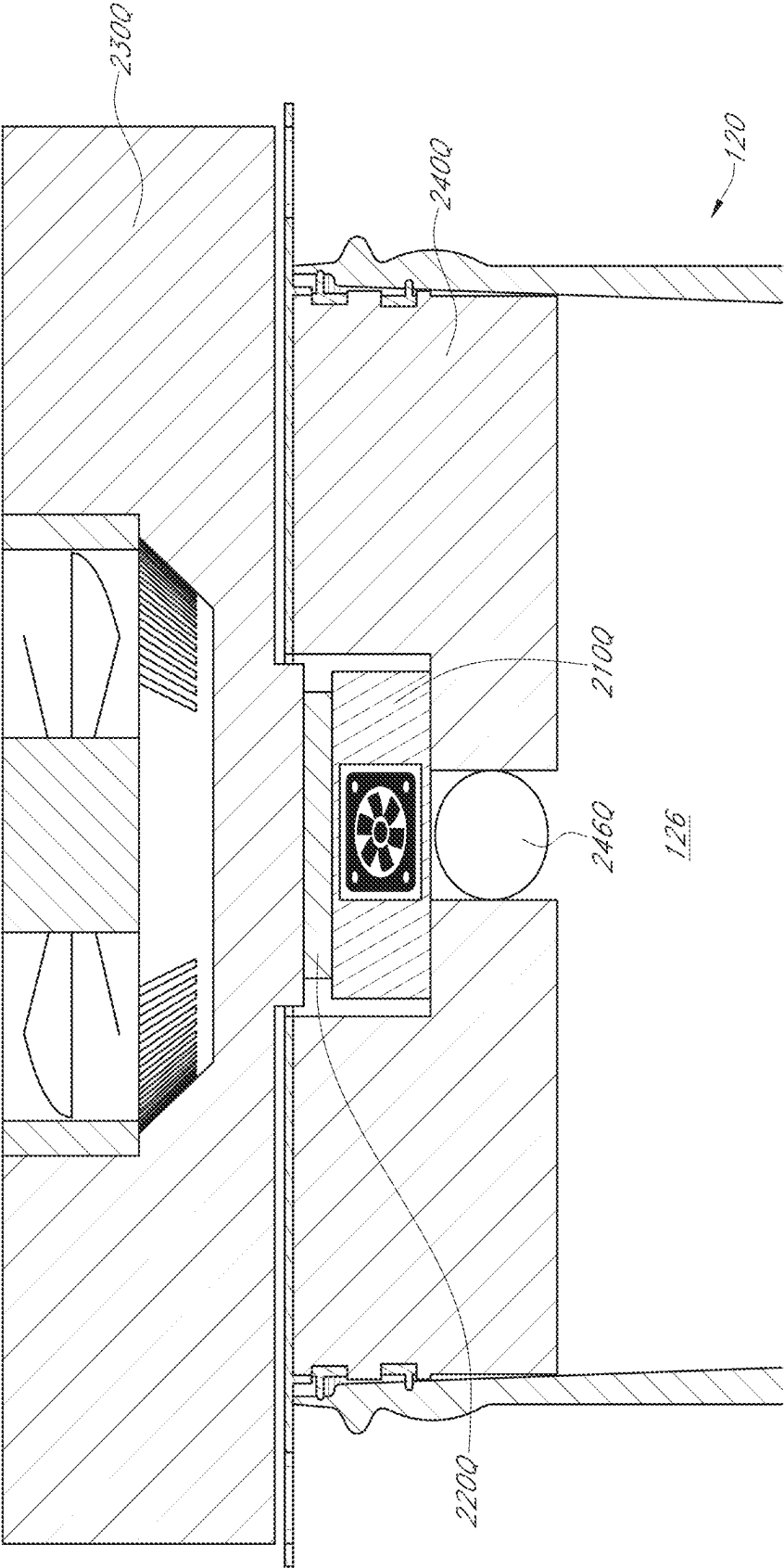


FIG. 16B

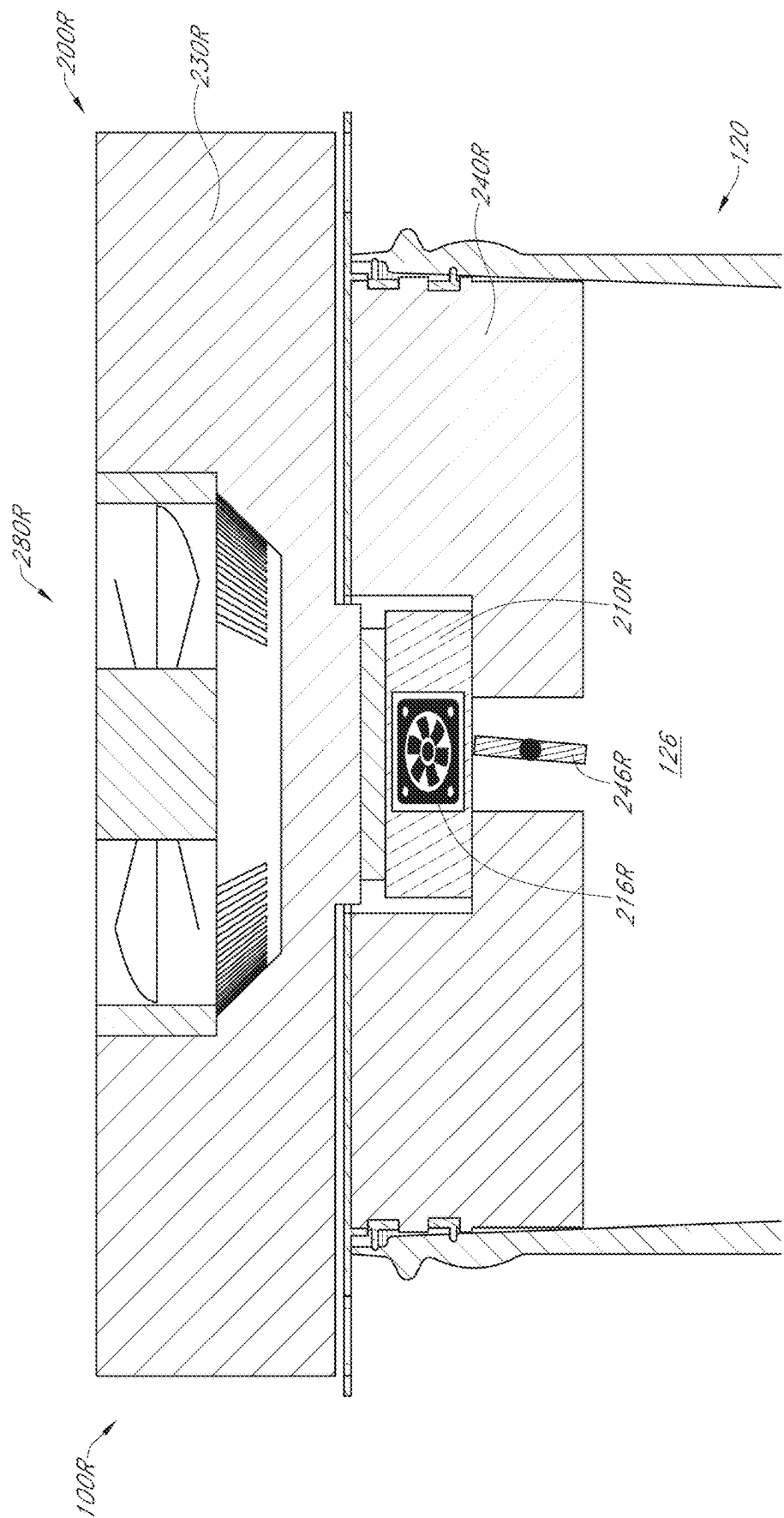


FIG. 17A

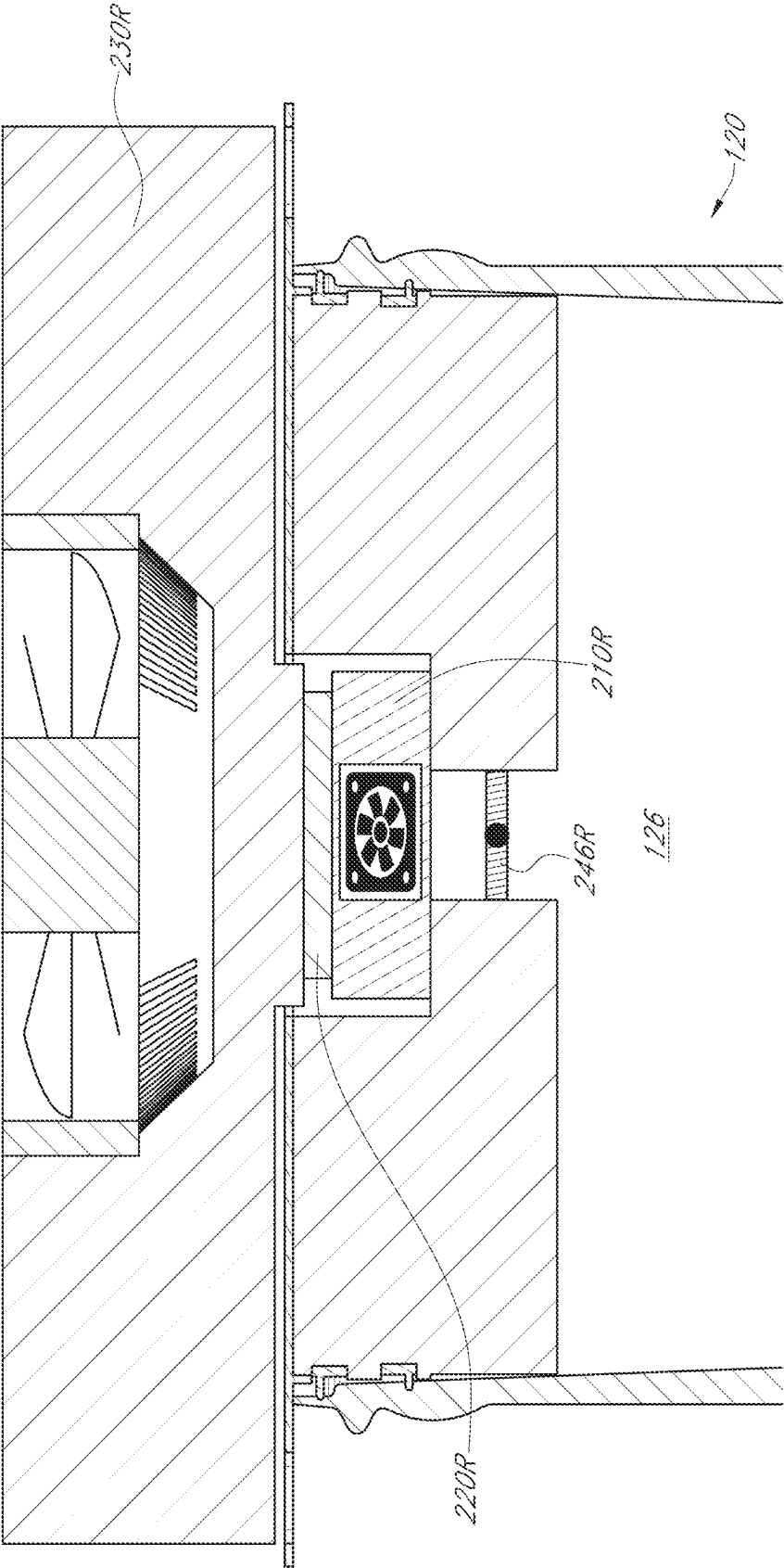


FIG. 17B

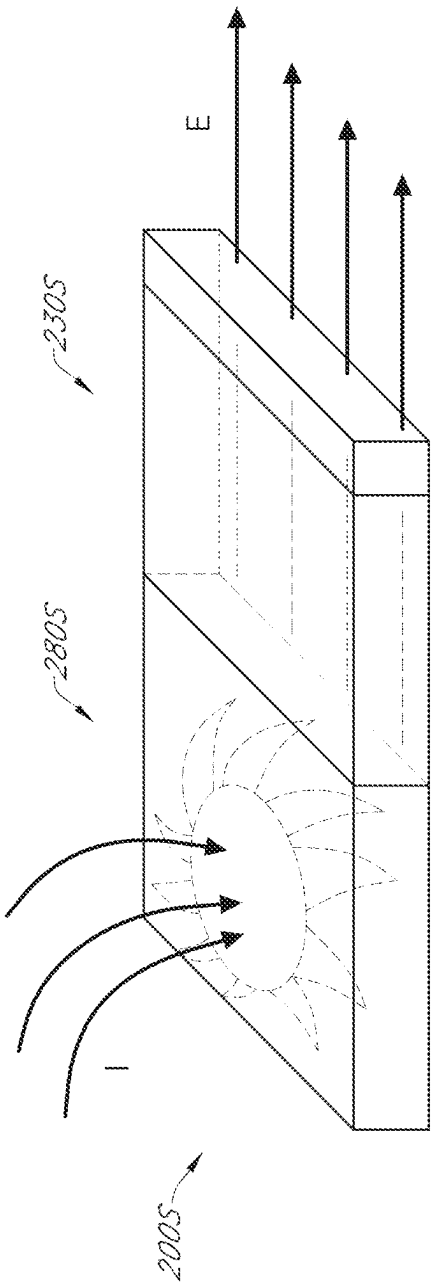


FIG. 18A

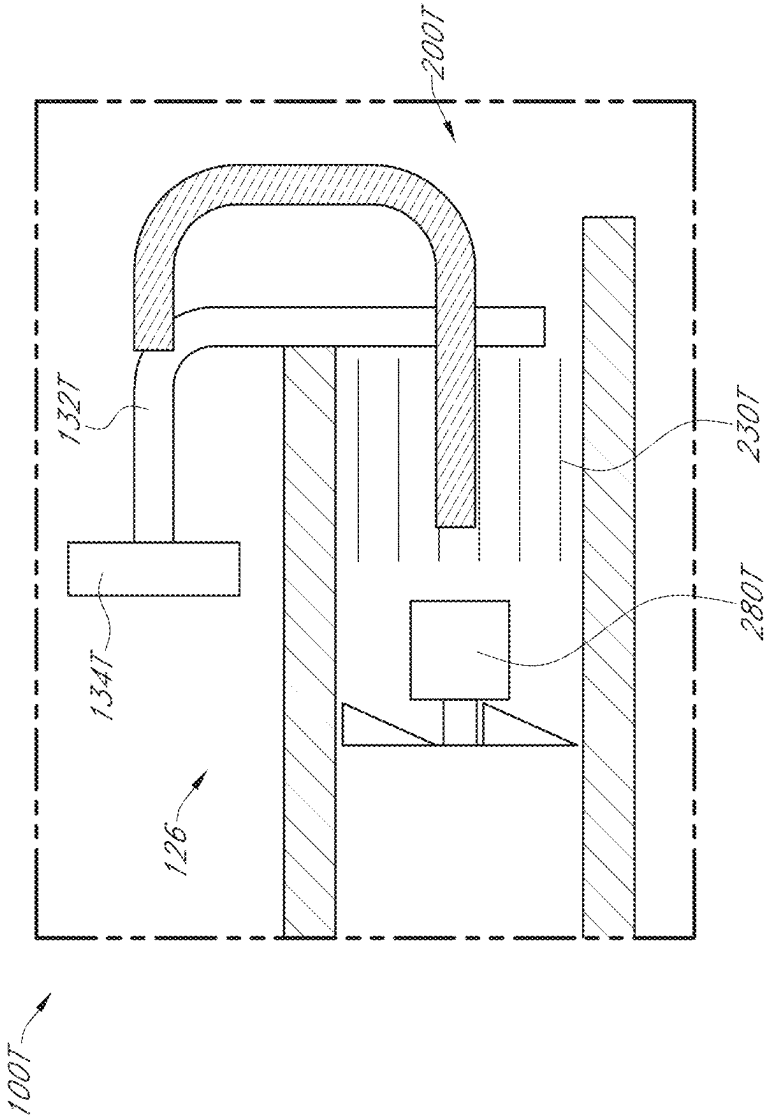


FIG. 18B

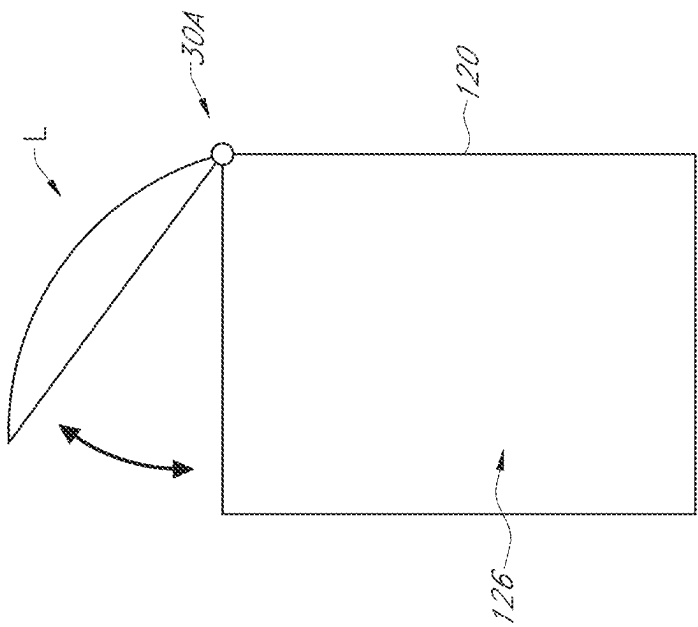


FIG. 18C

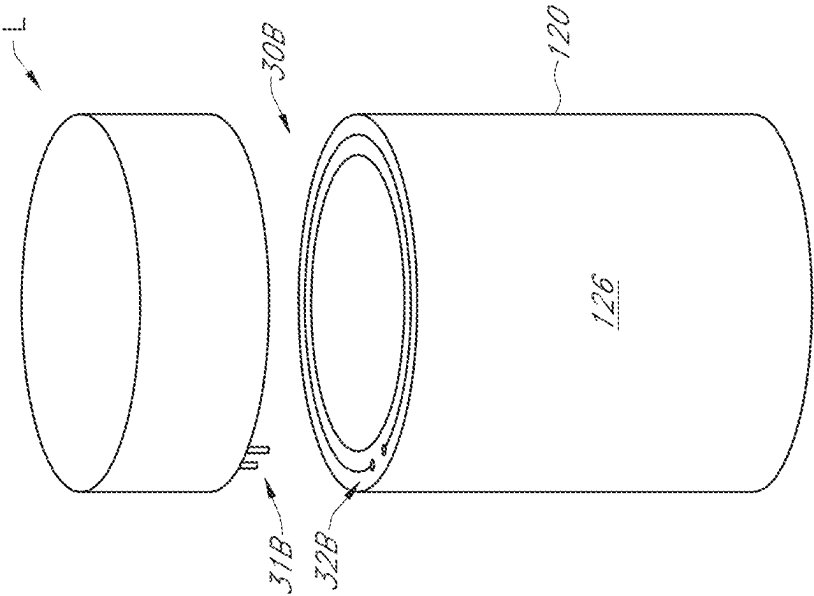


FIG. 18D

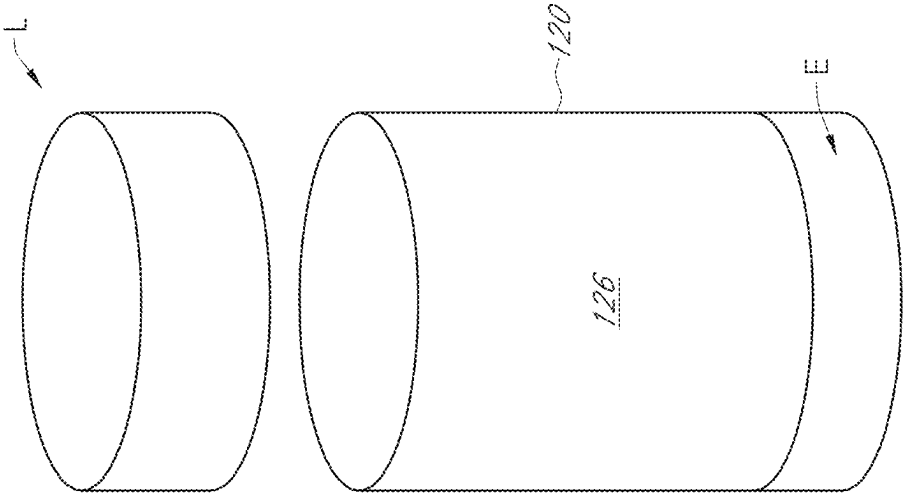


FIG. 18E

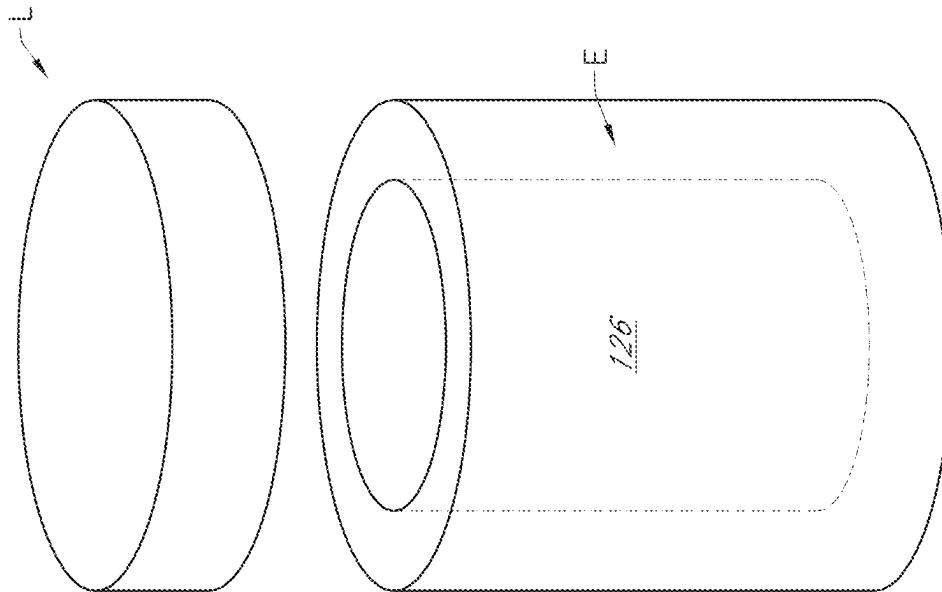


FIG. 18F

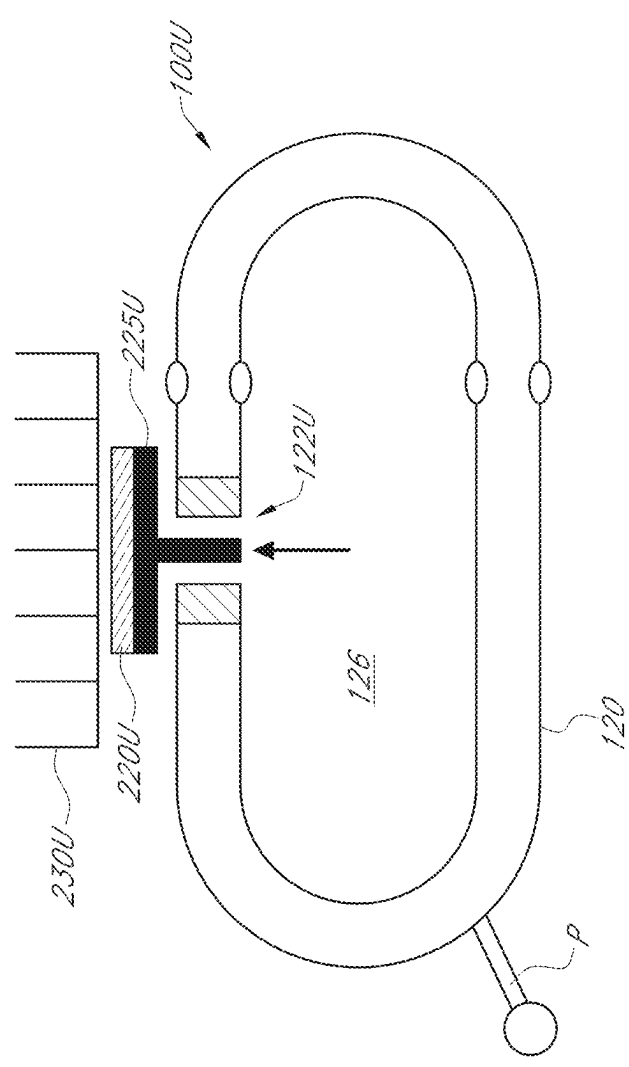


FIG. 19

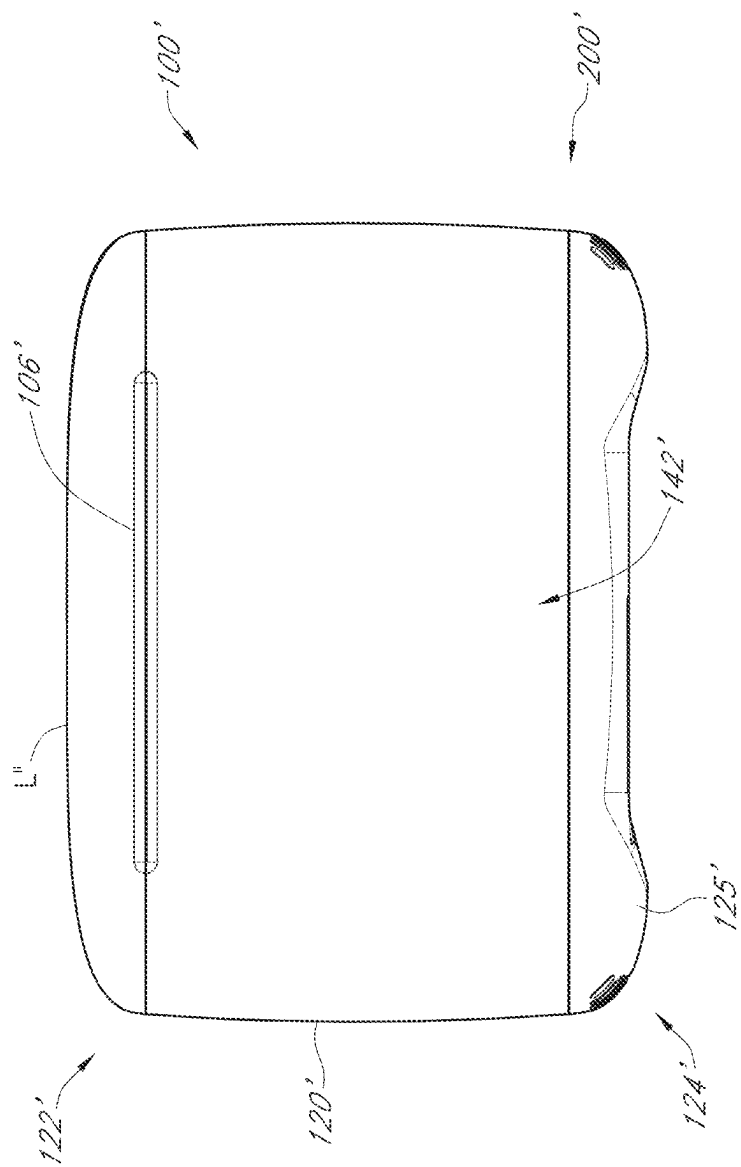


FIG. 20

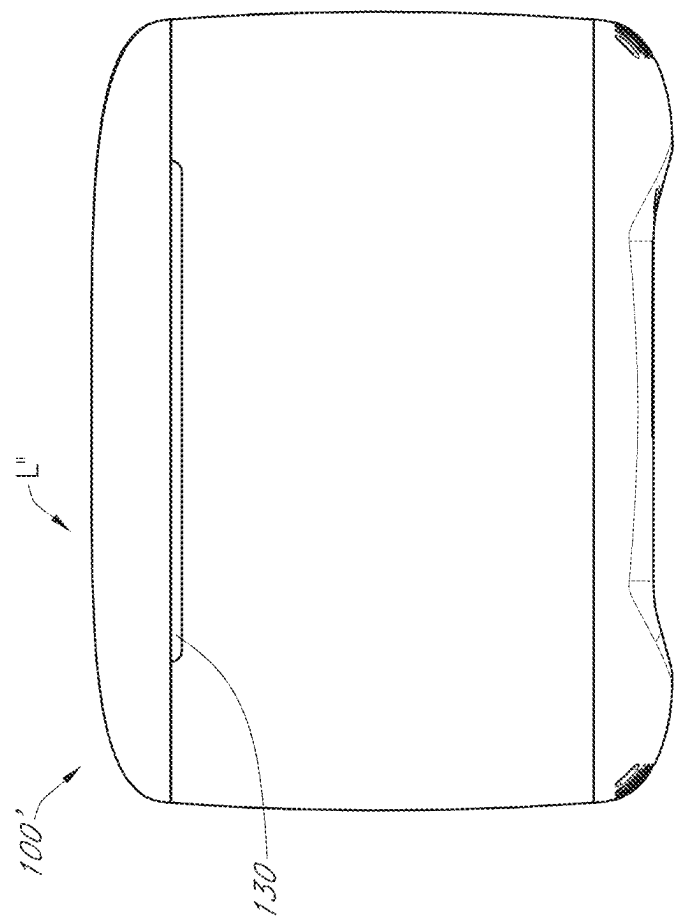


FIG. 21

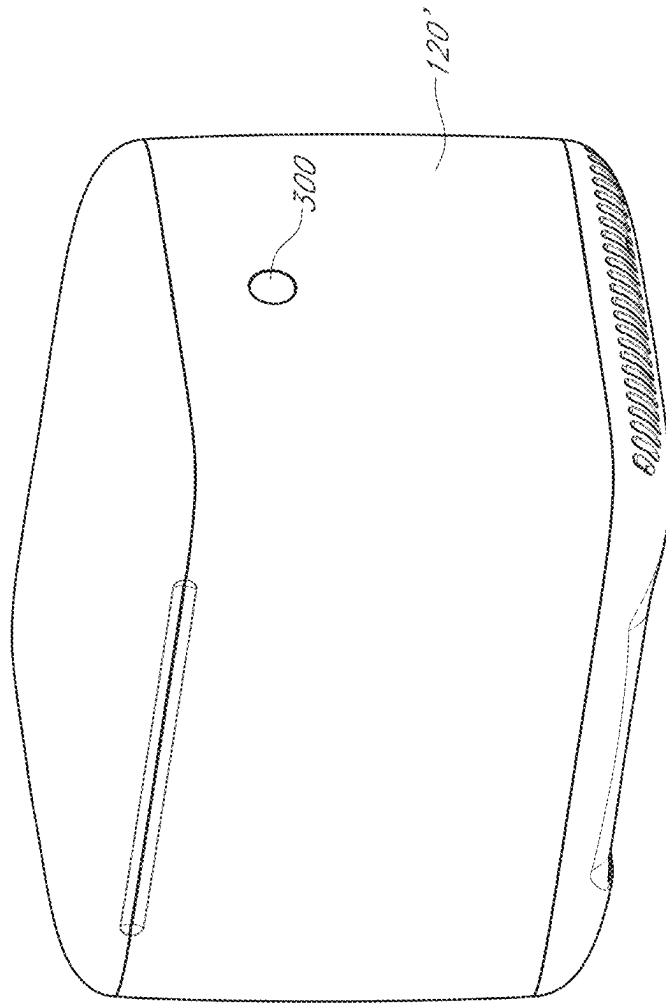


FIG. 22

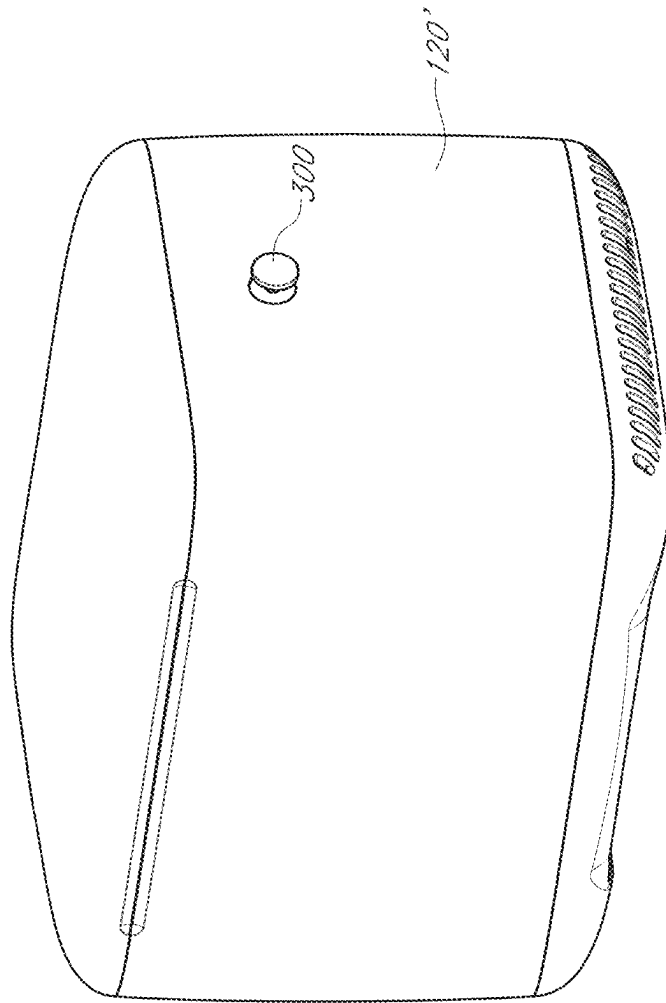


FIG. 23

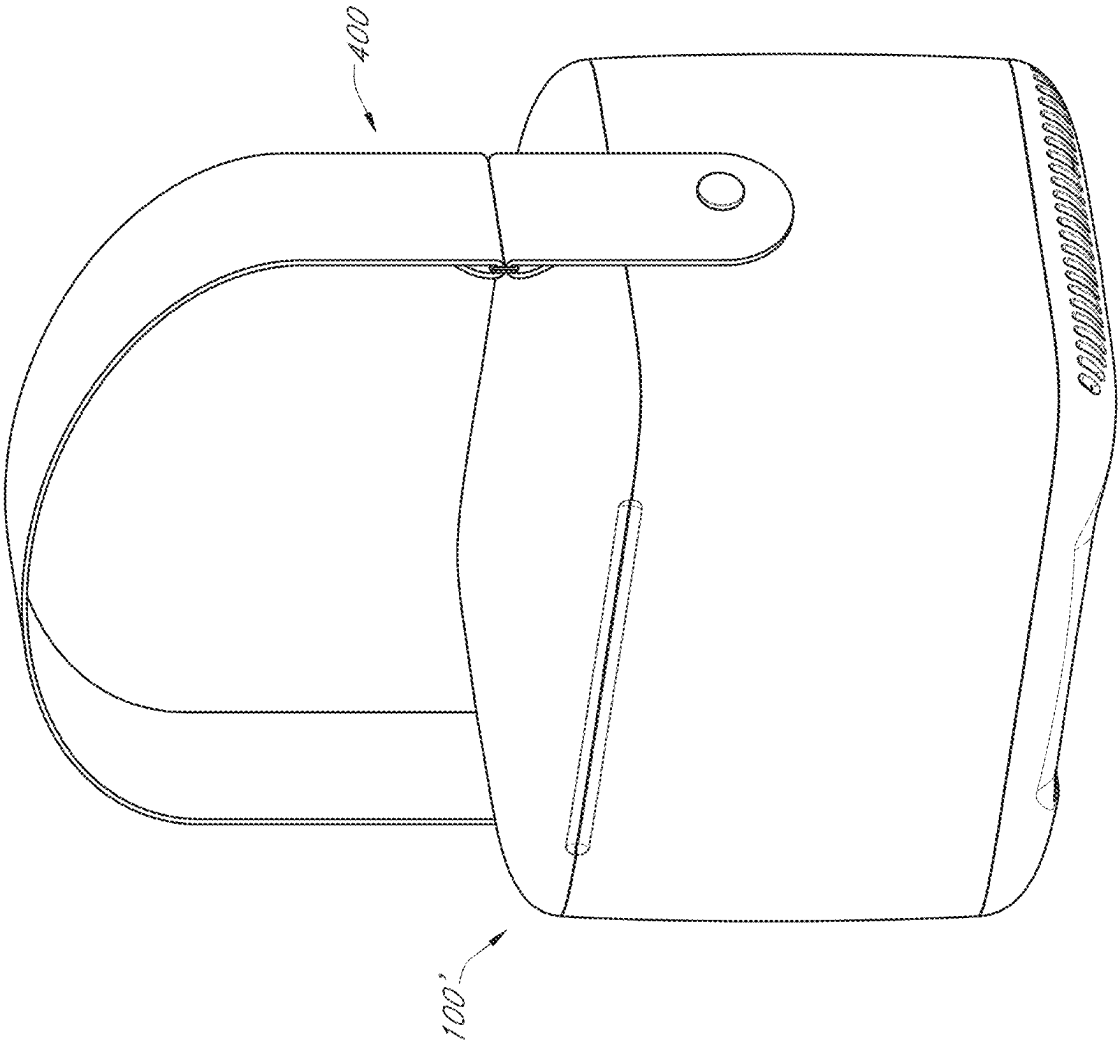


FIG. 24

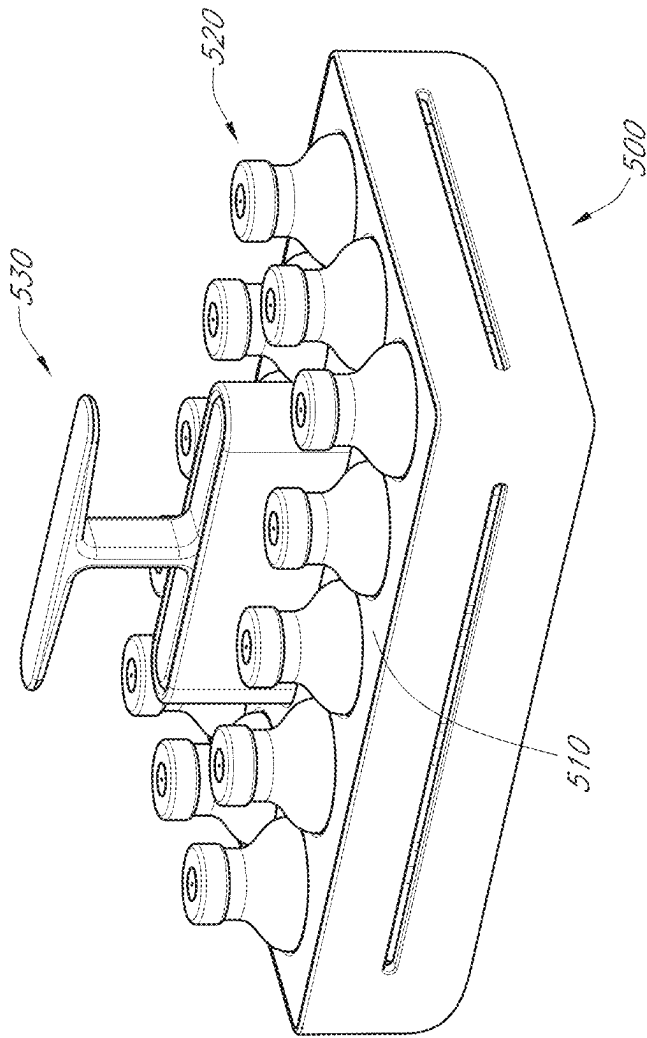


FIG. 25A

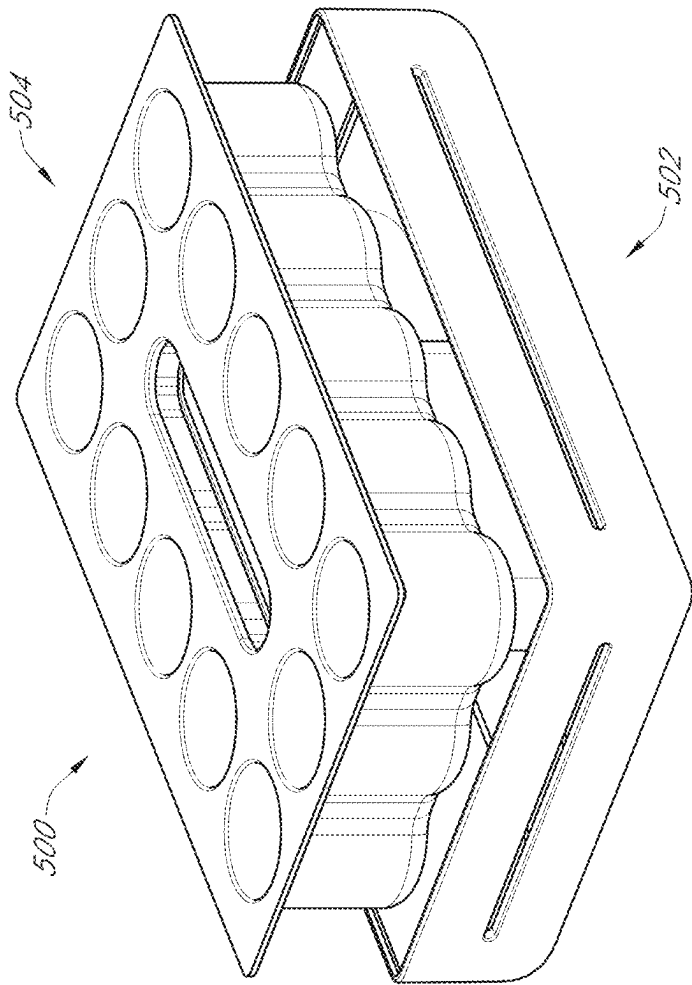
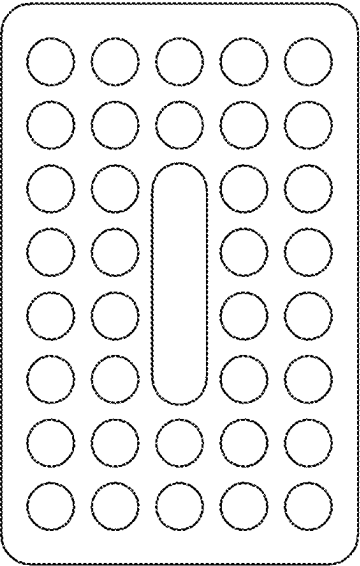


FIG. 25B



504

FIG. 25C

504'

FIG. 25D

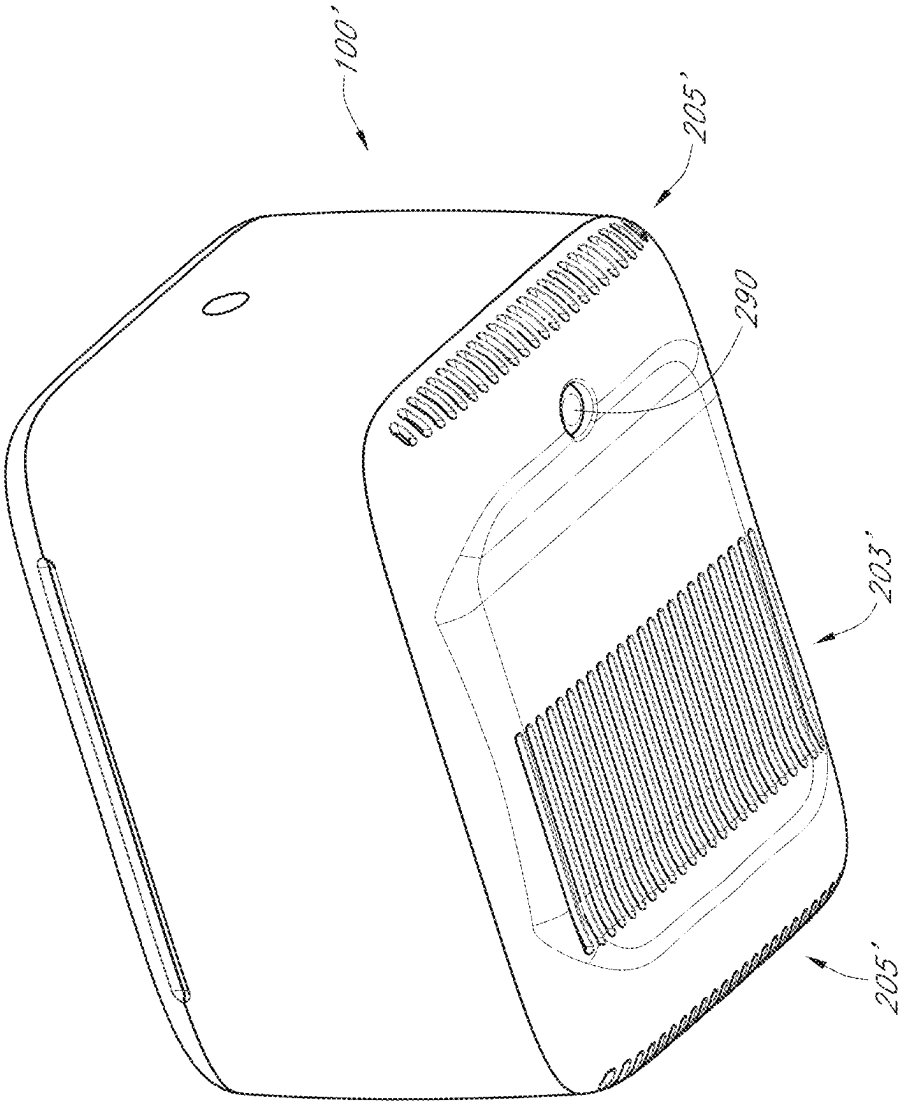


FIG. 26

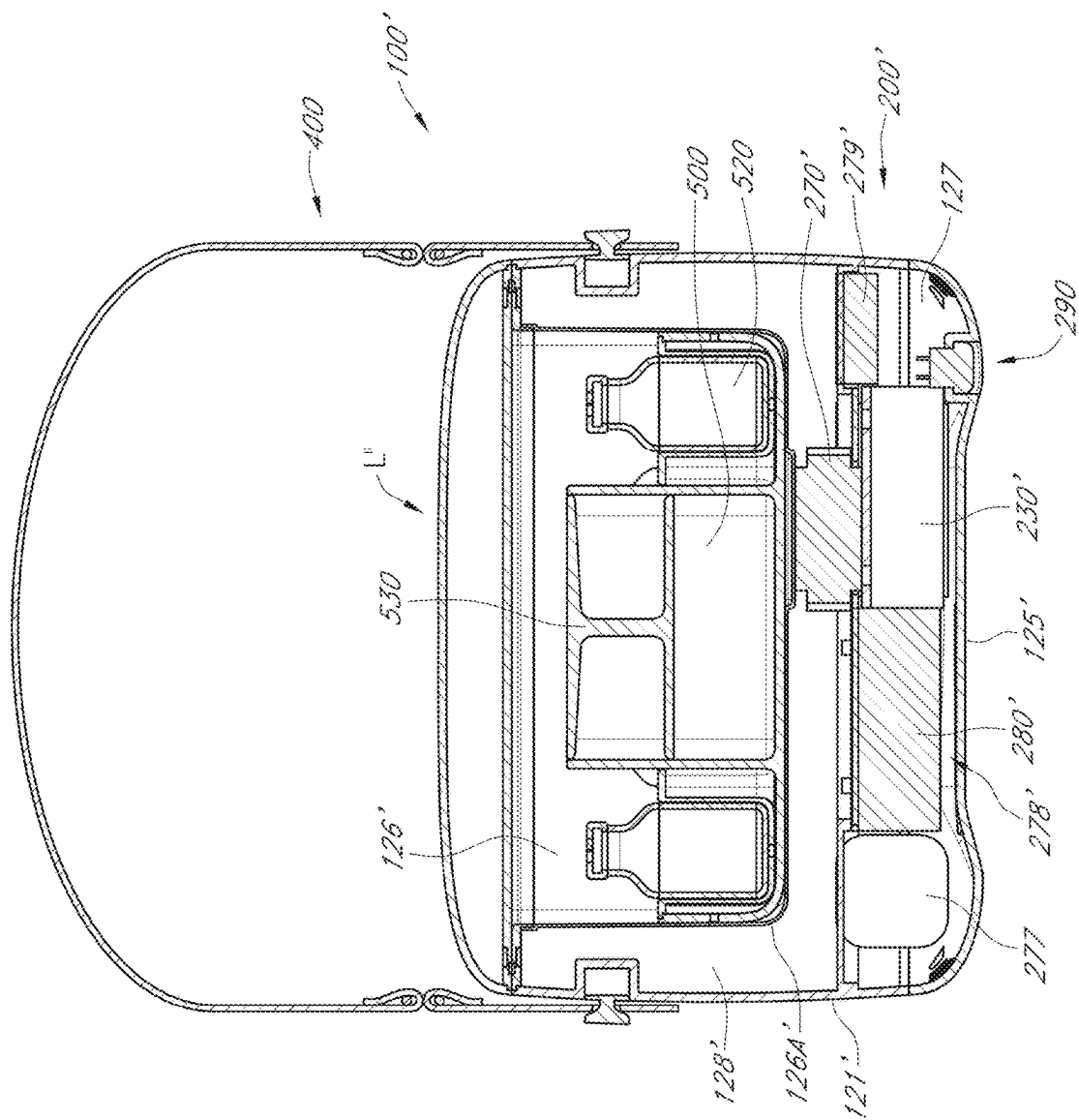


FIG. 27

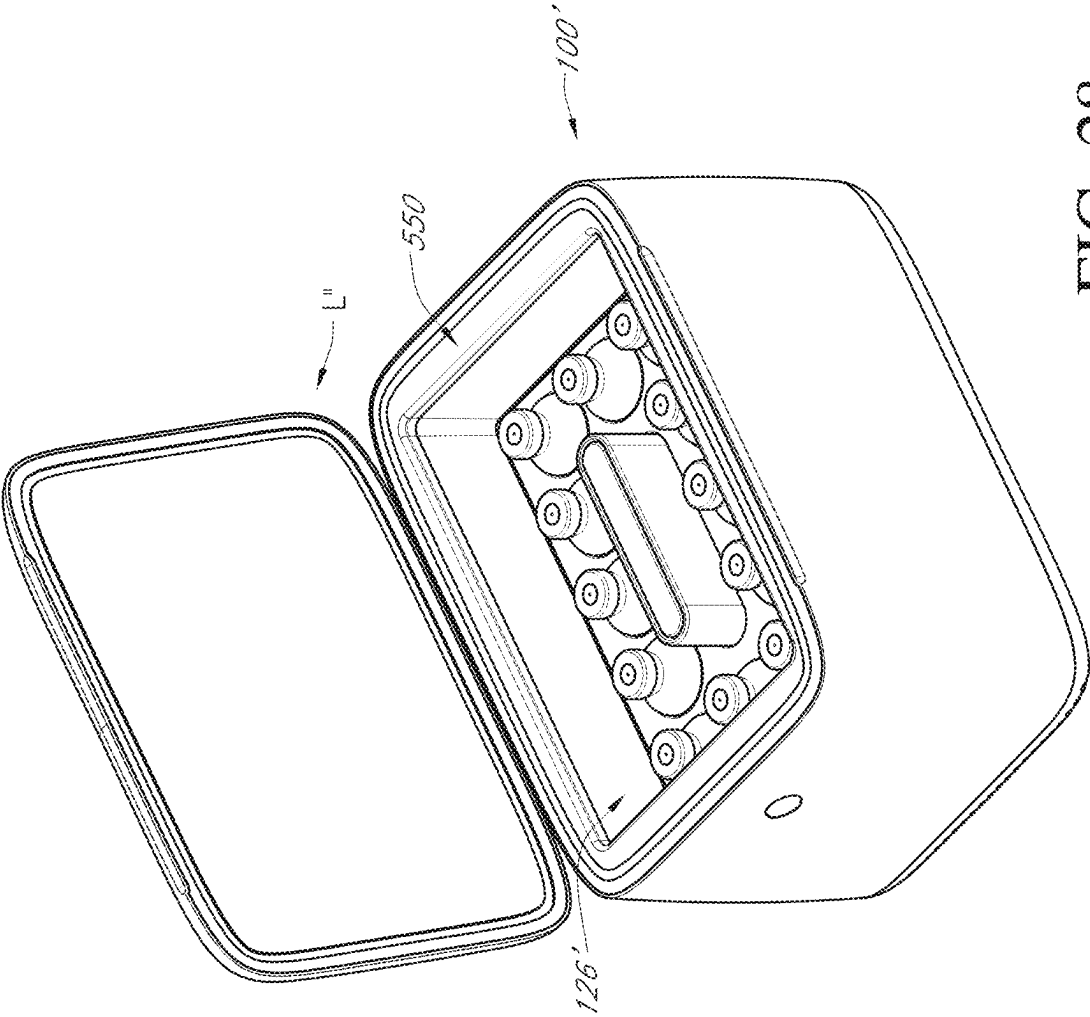


FIG. 28

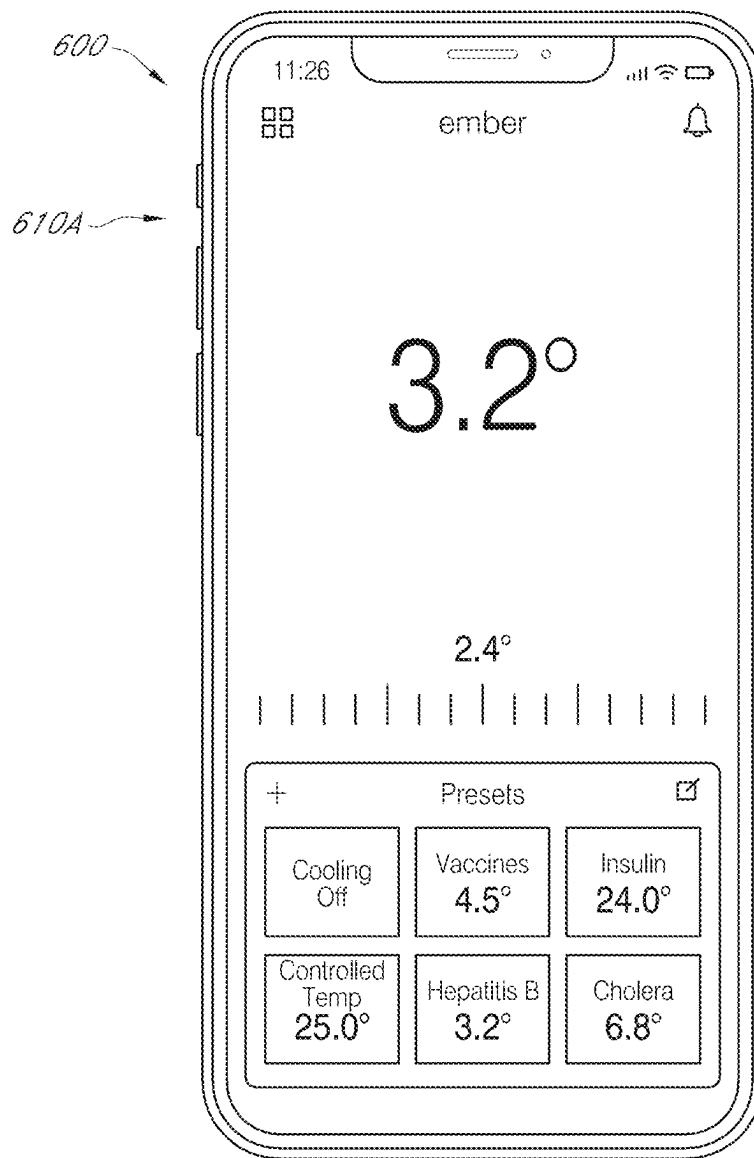


FIG. 29A

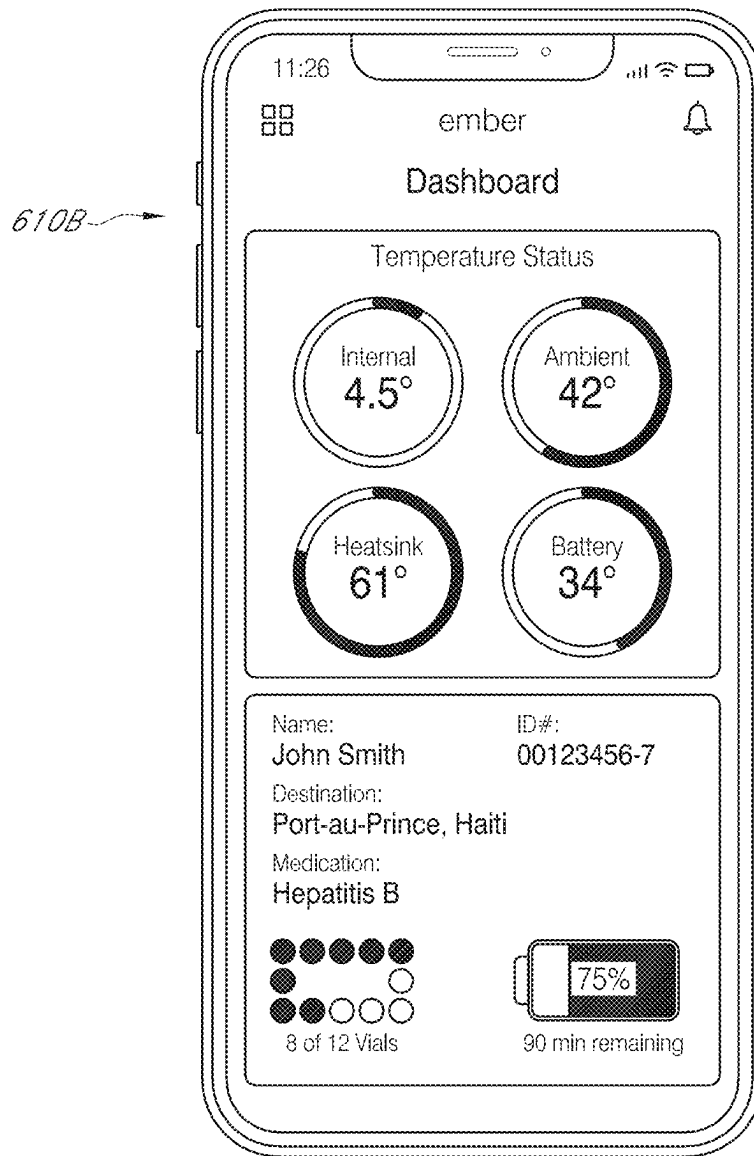


FIG. 29B

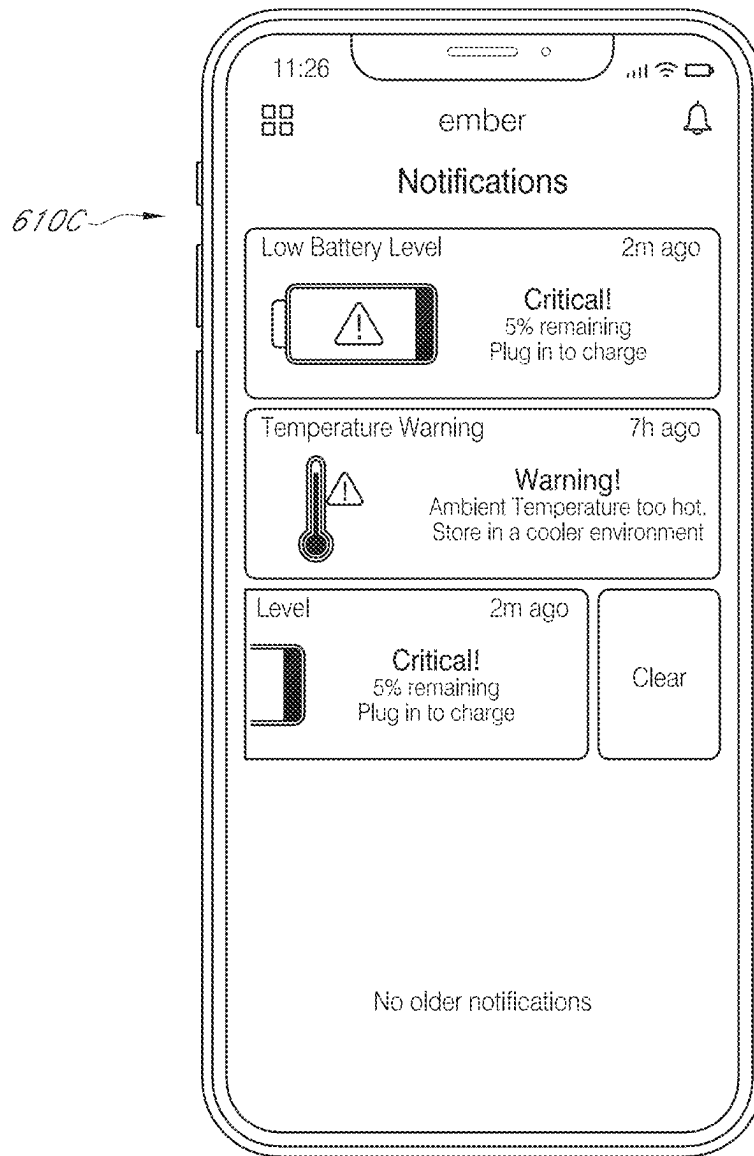


FIG. 29C

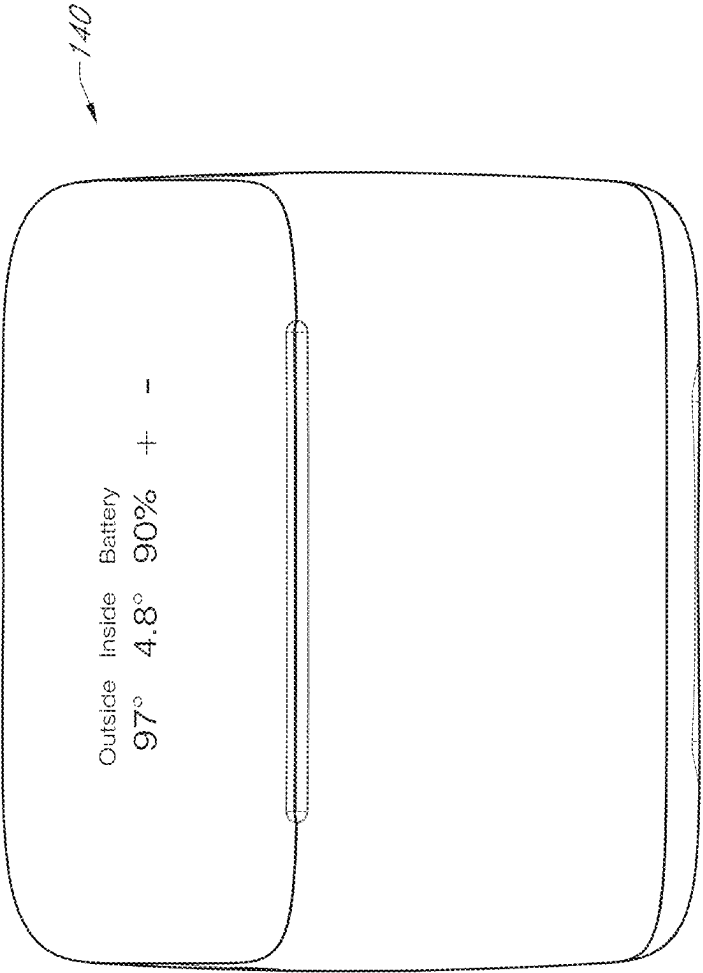


FIG. 30

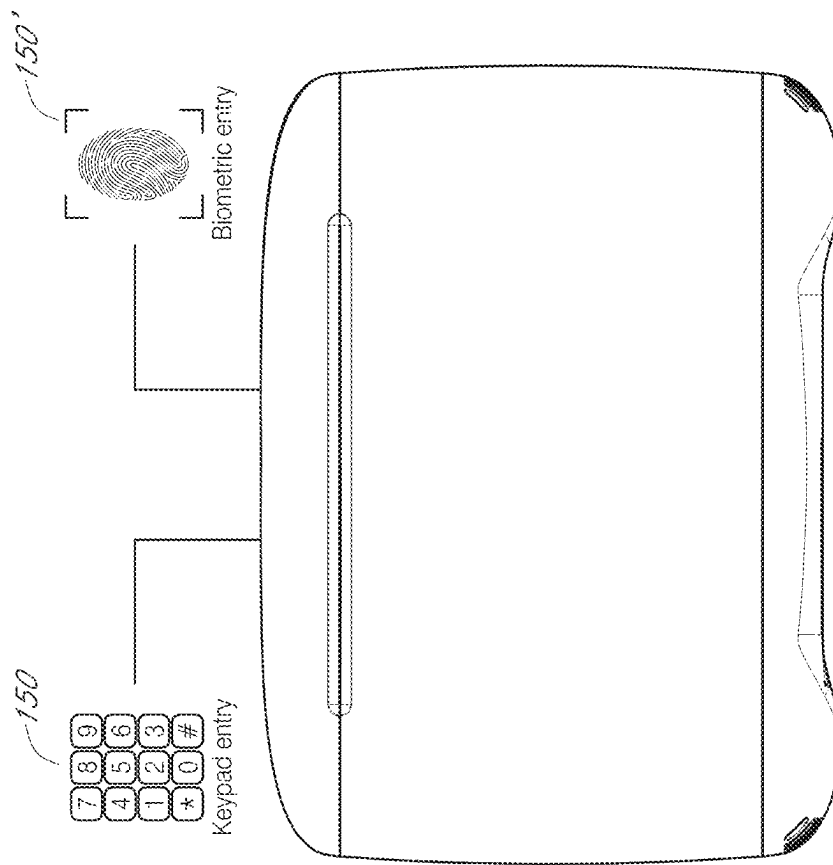


FIG. 31

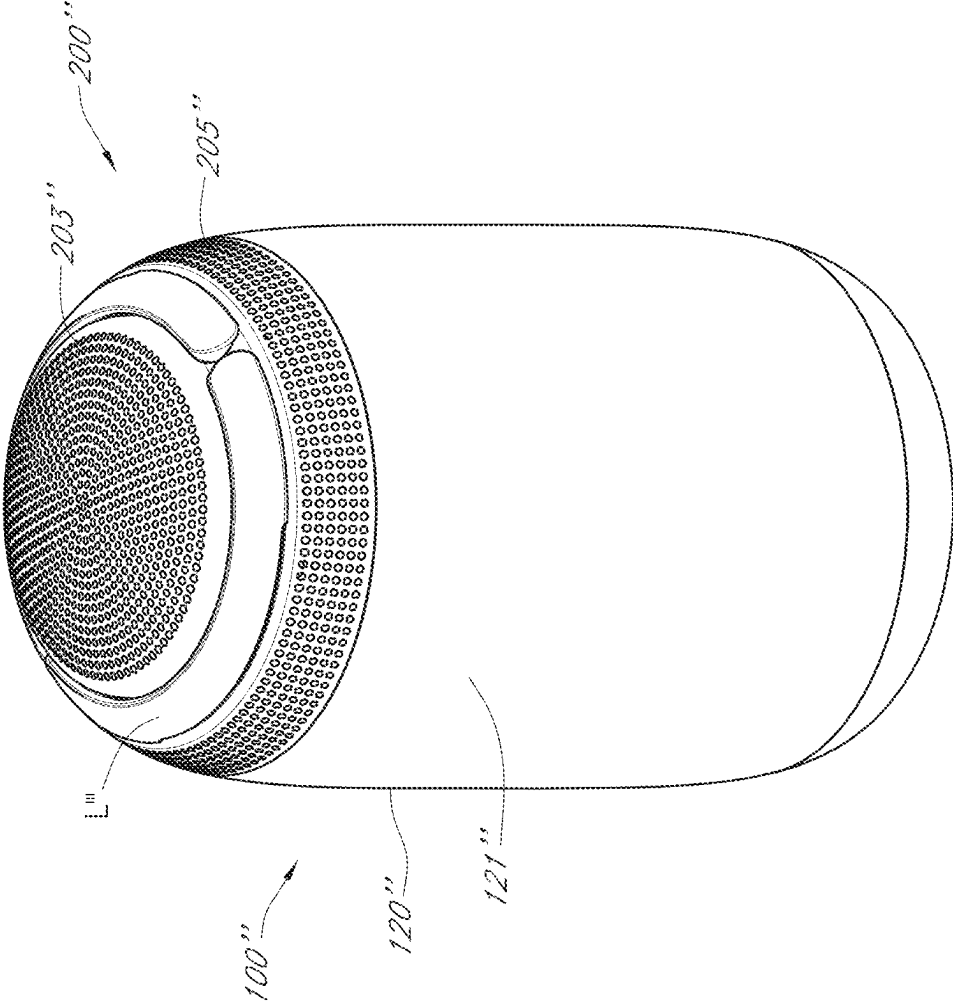


FIG. 32

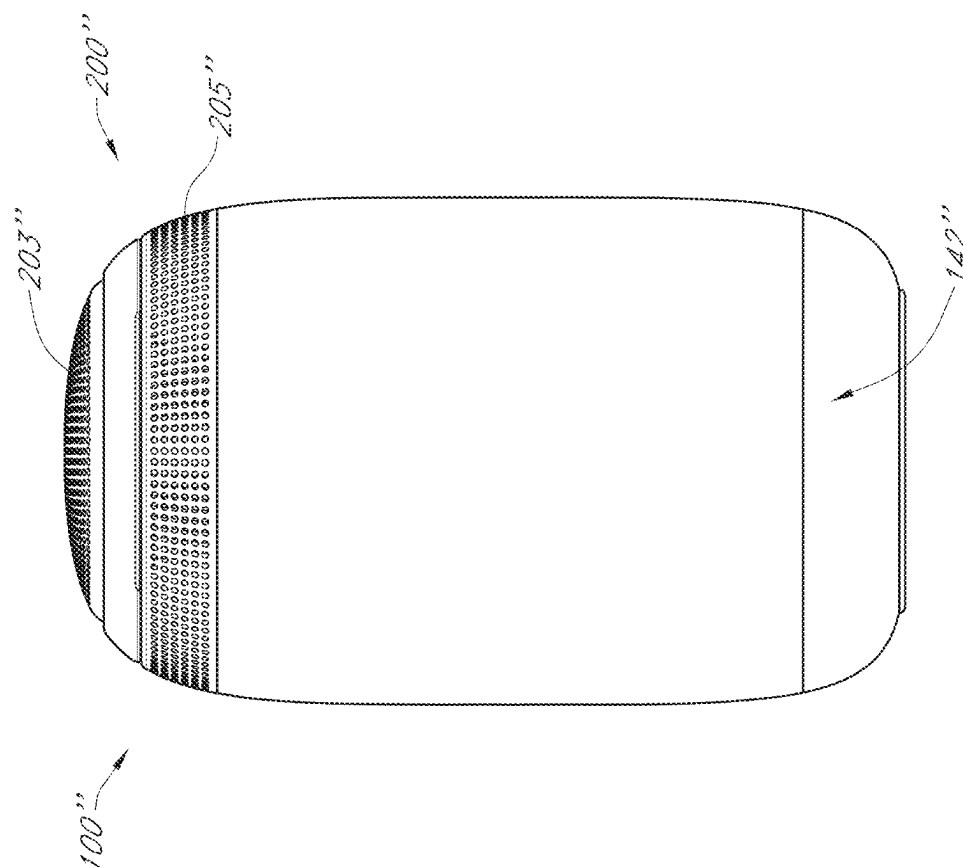


FIG. 33A

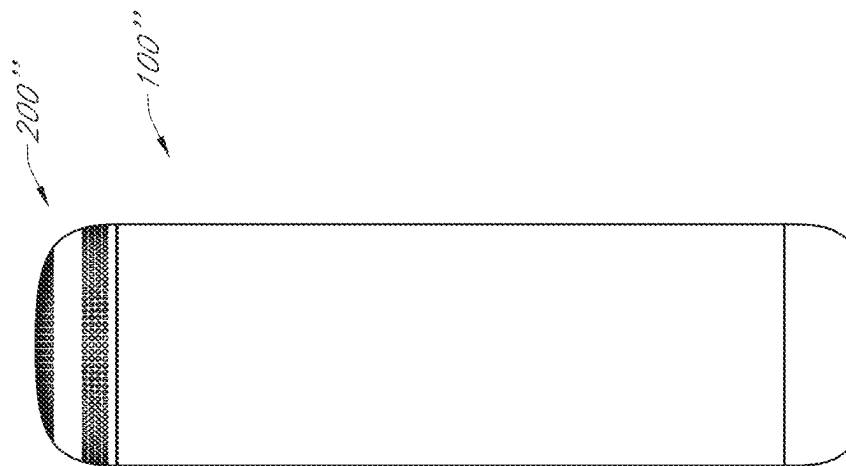


FIG. 33B

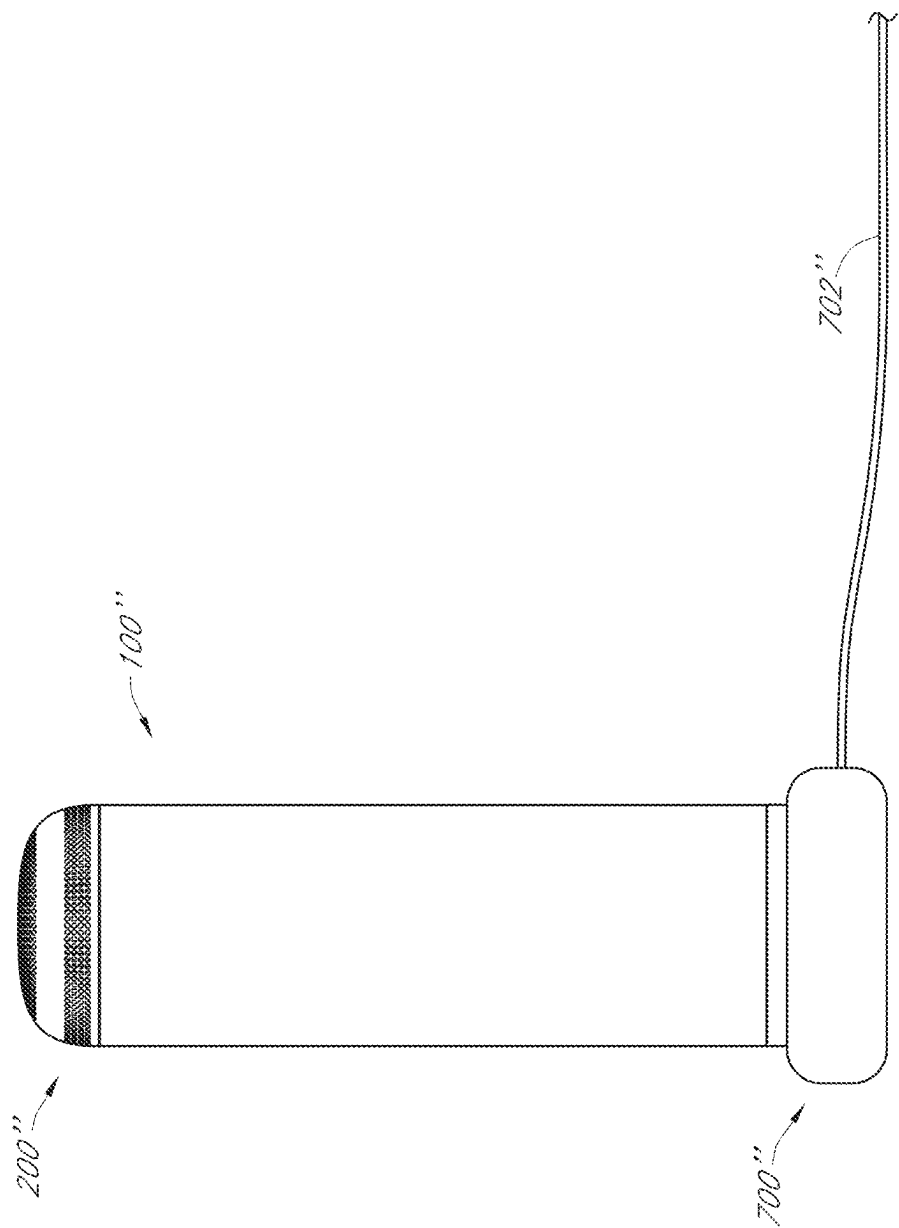


FIG. 34

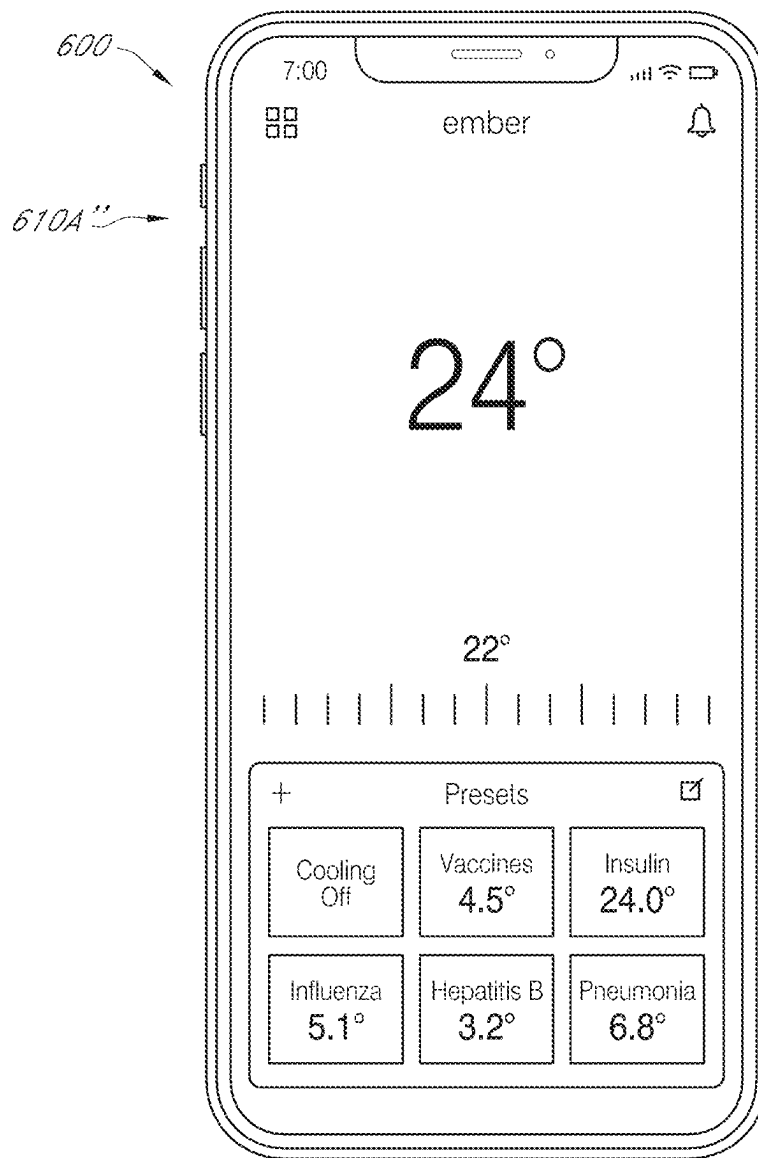


FIG. 35A

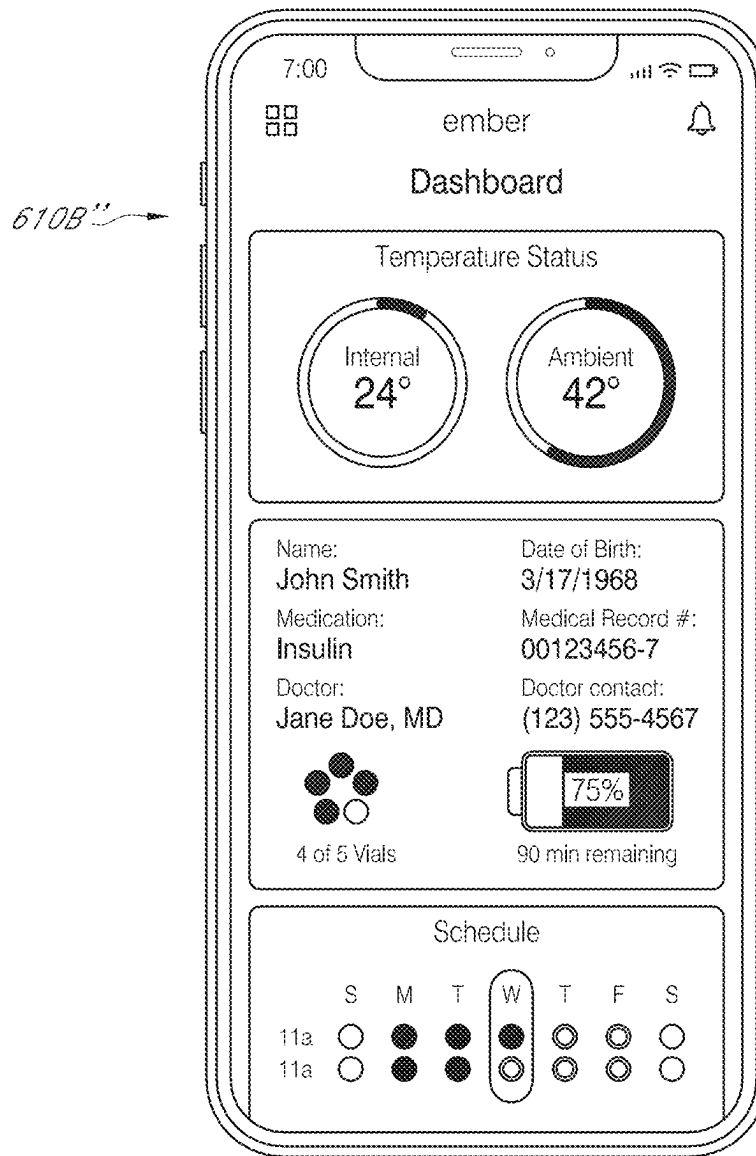


FIG. 35B

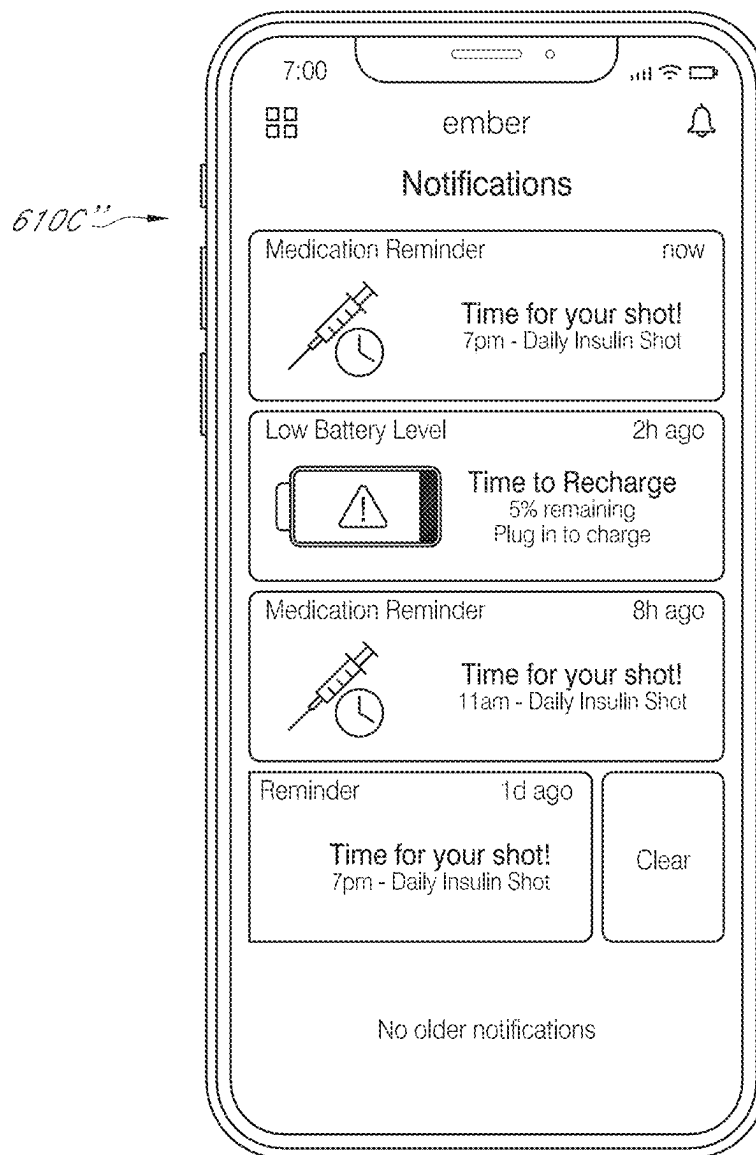


FIG. 35C

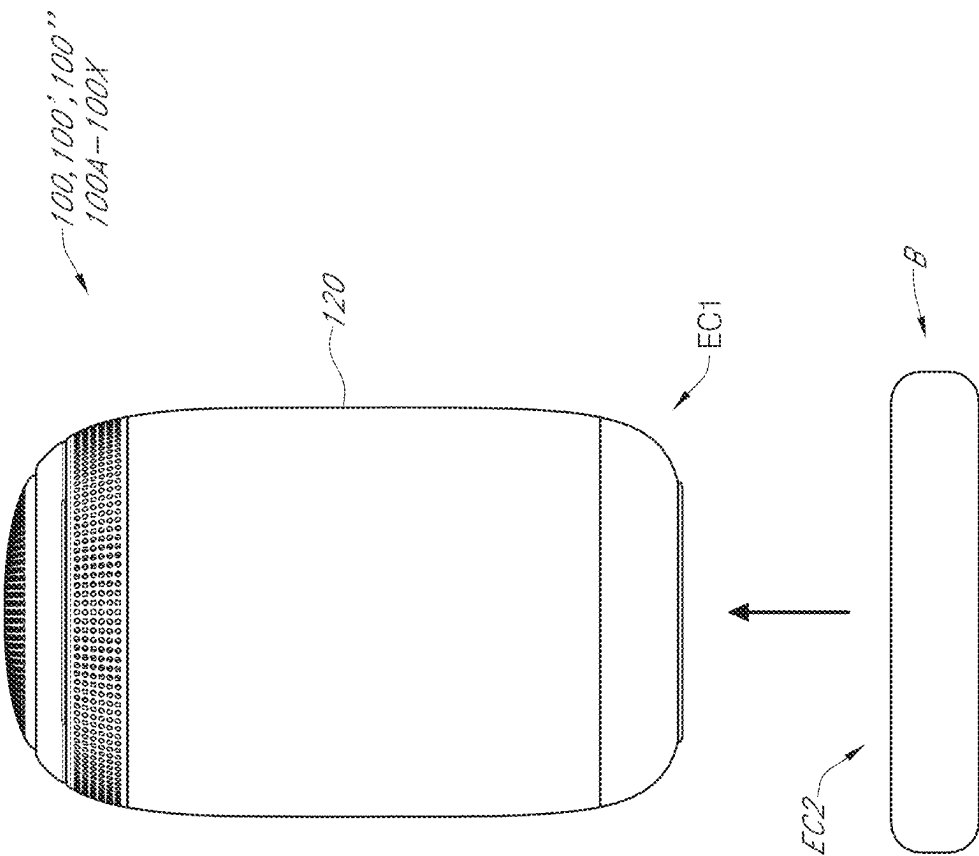


FIG. 36

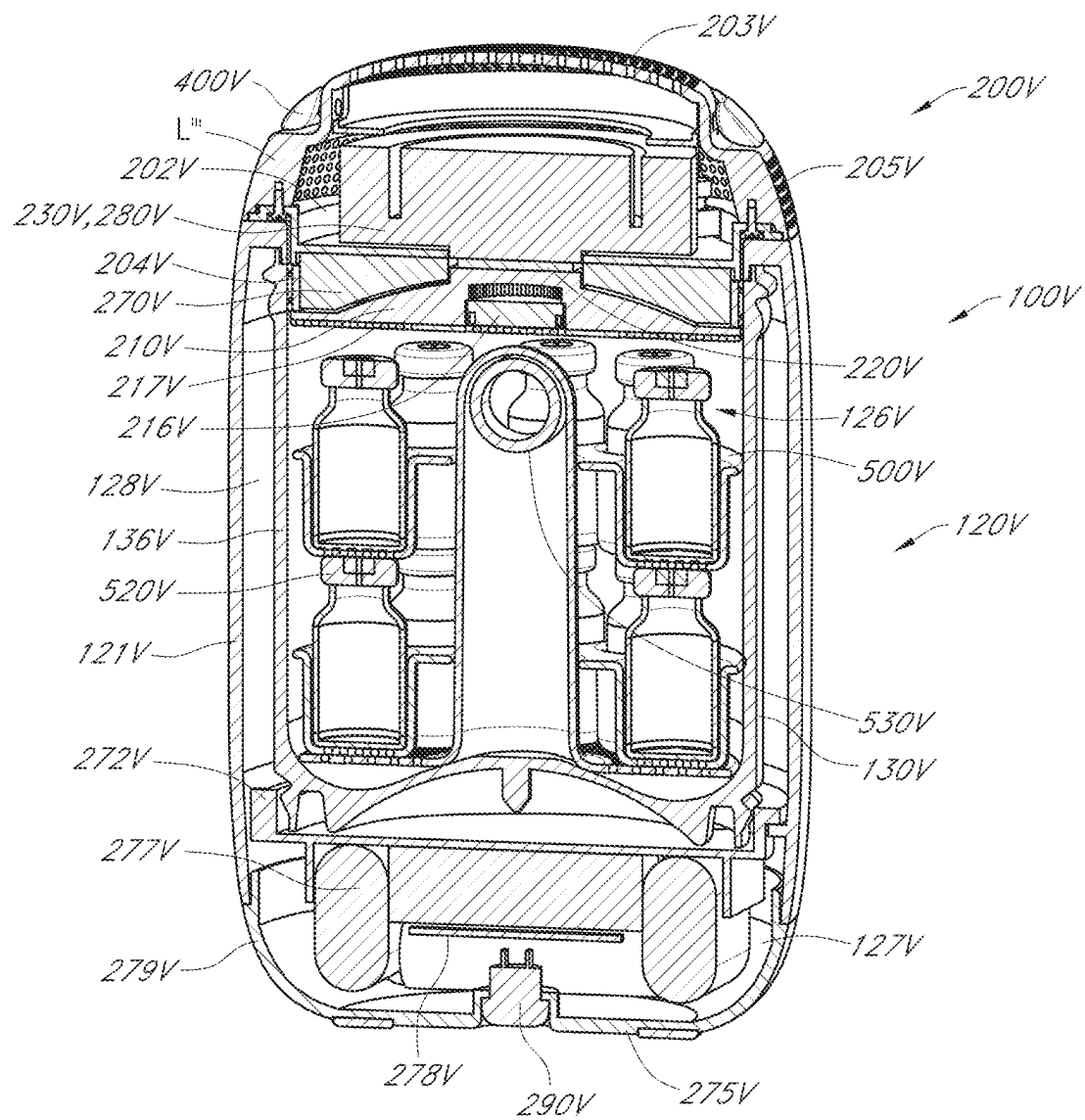


FIG. 37

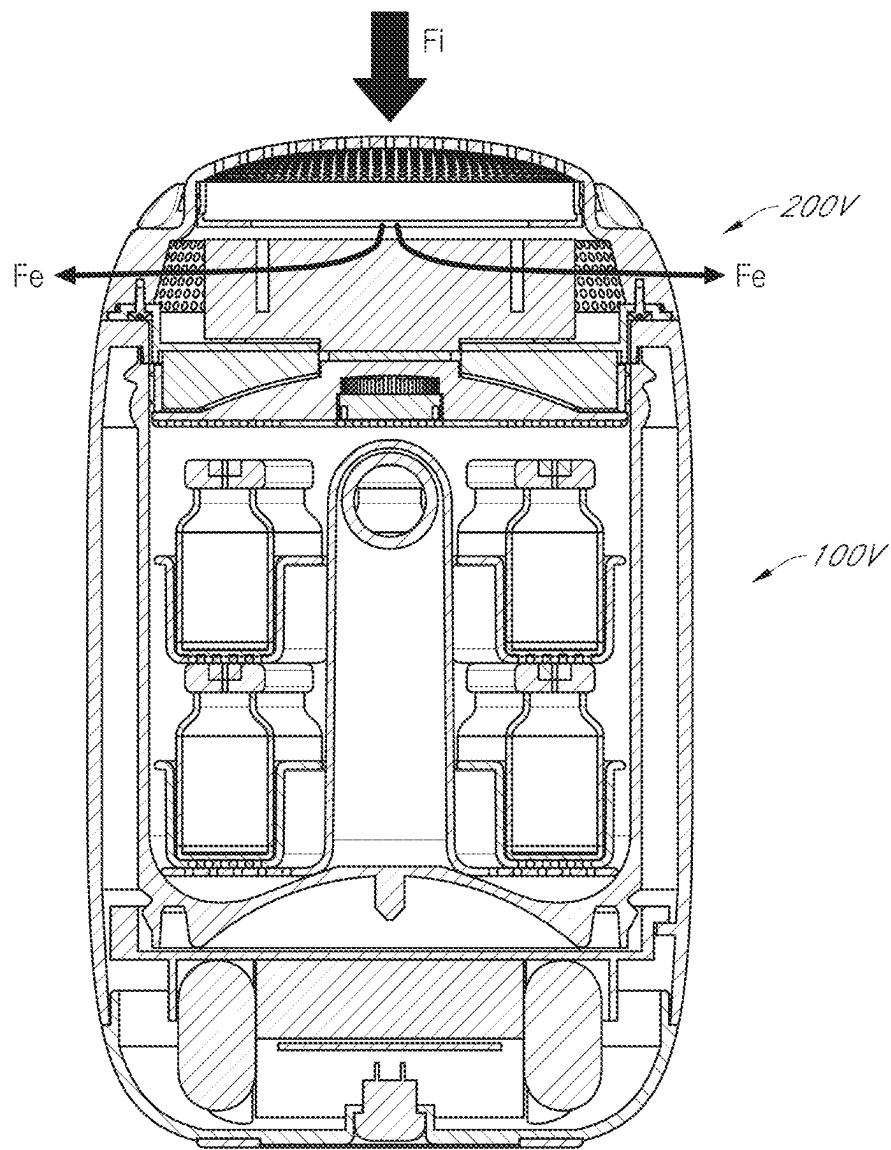


FIG. 38

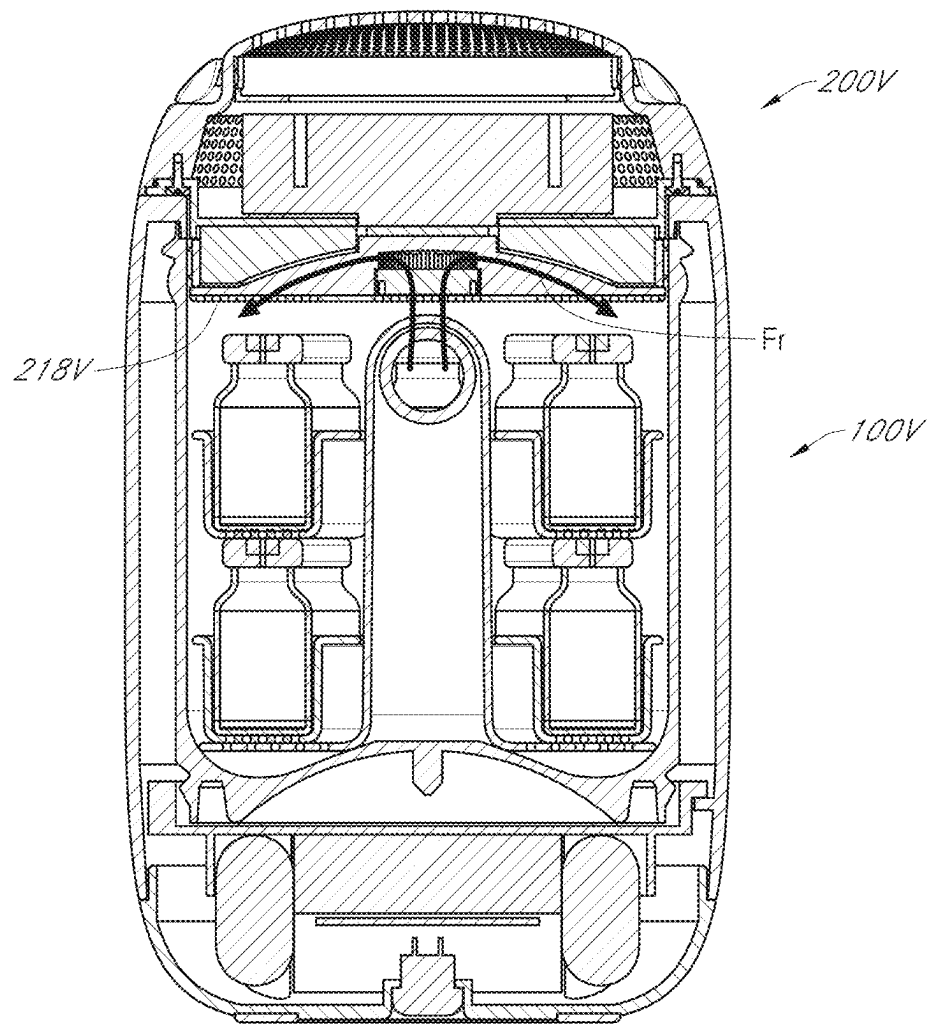


FIG. 39

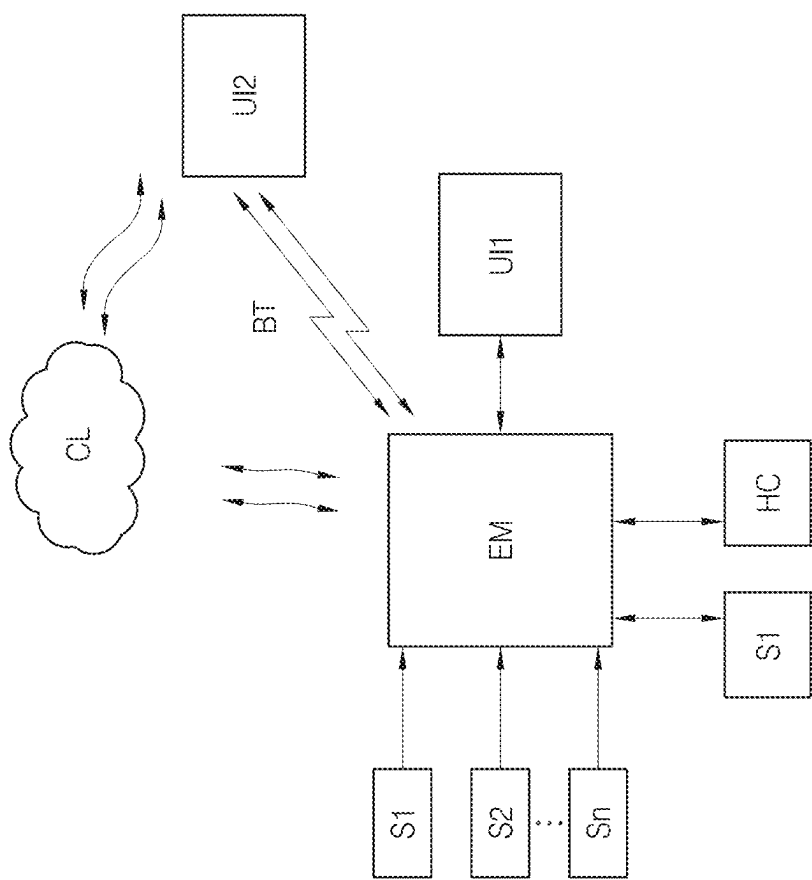


FIG. 40

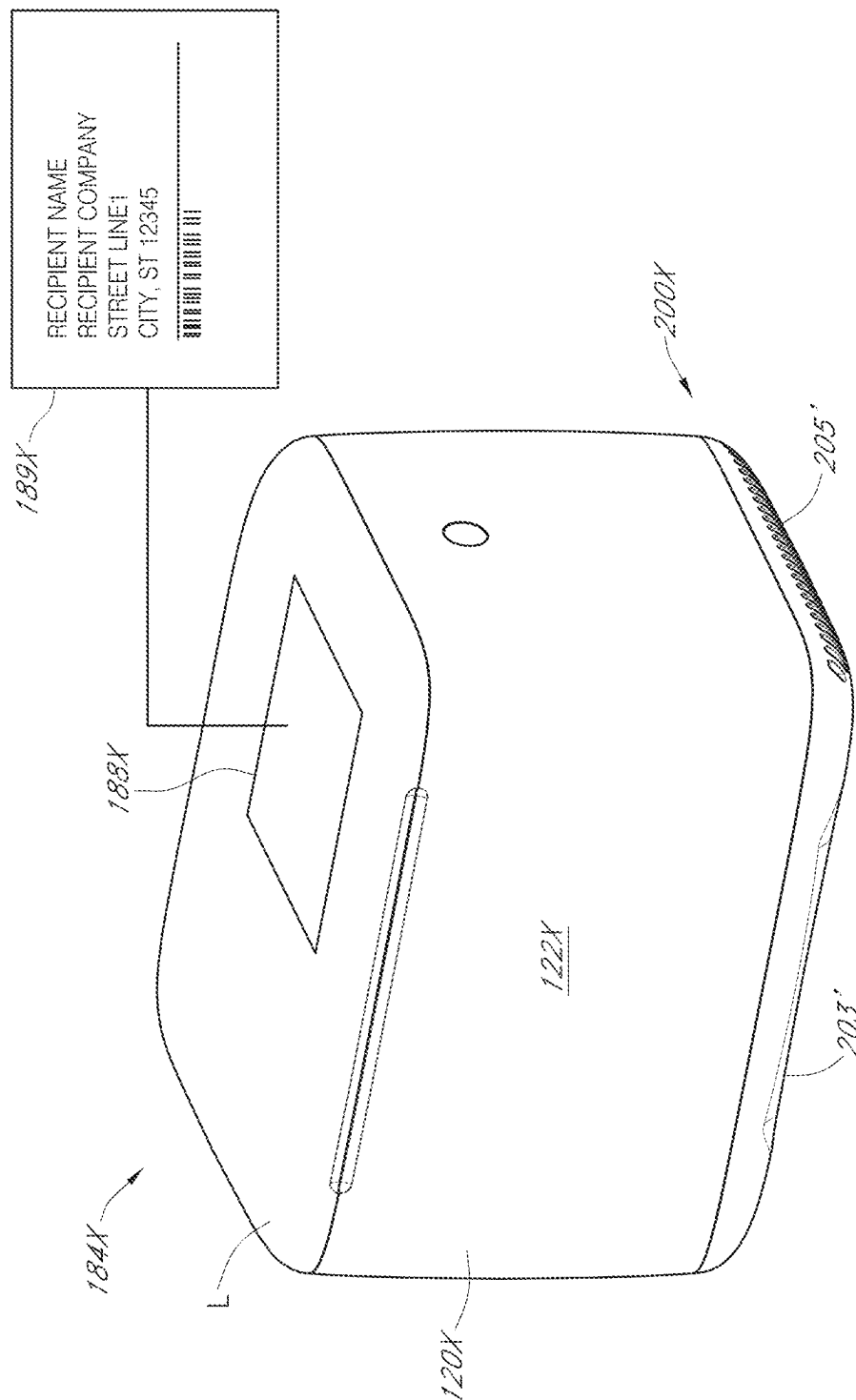


FIG. 41A

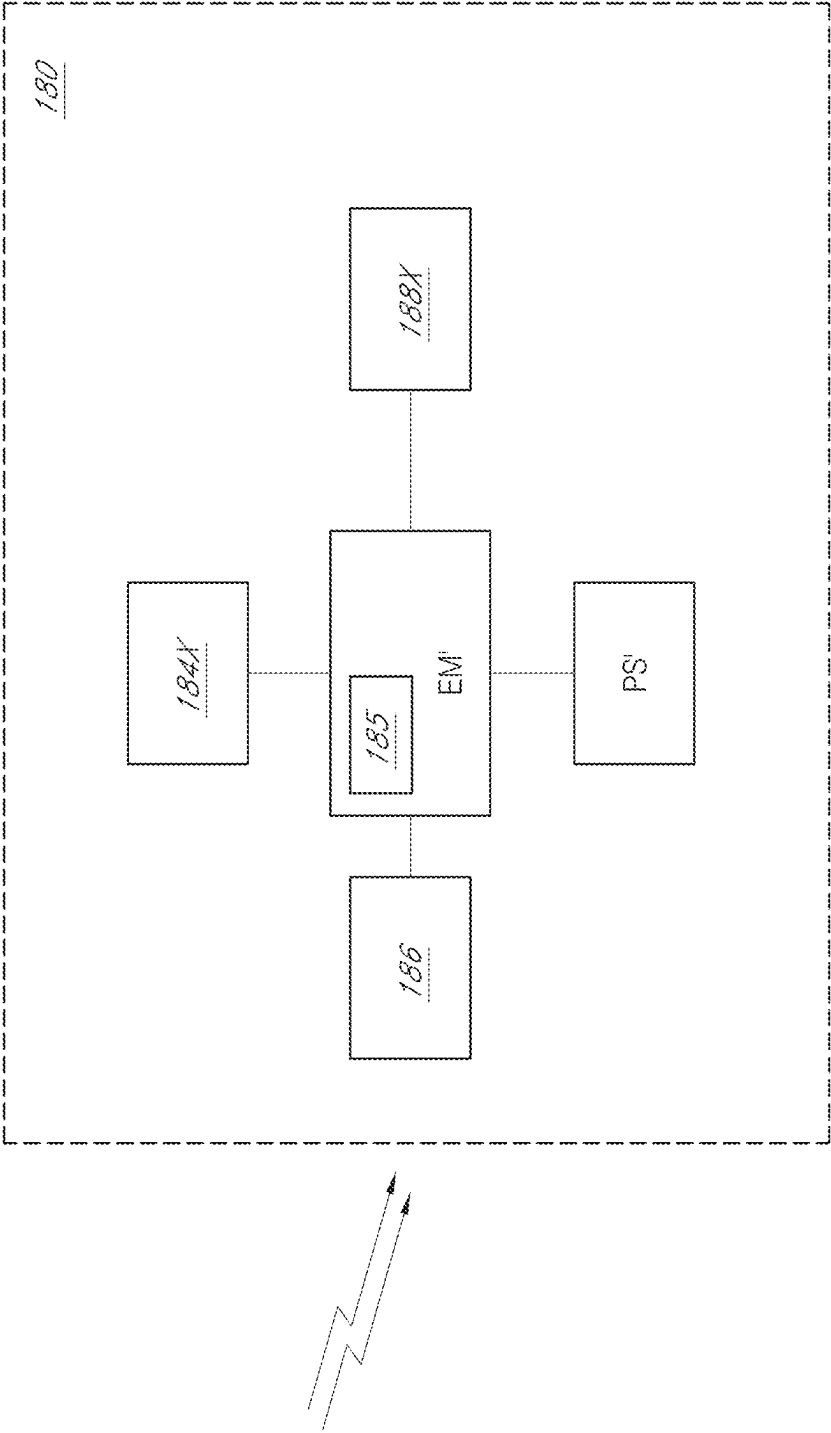


FIG. 41B

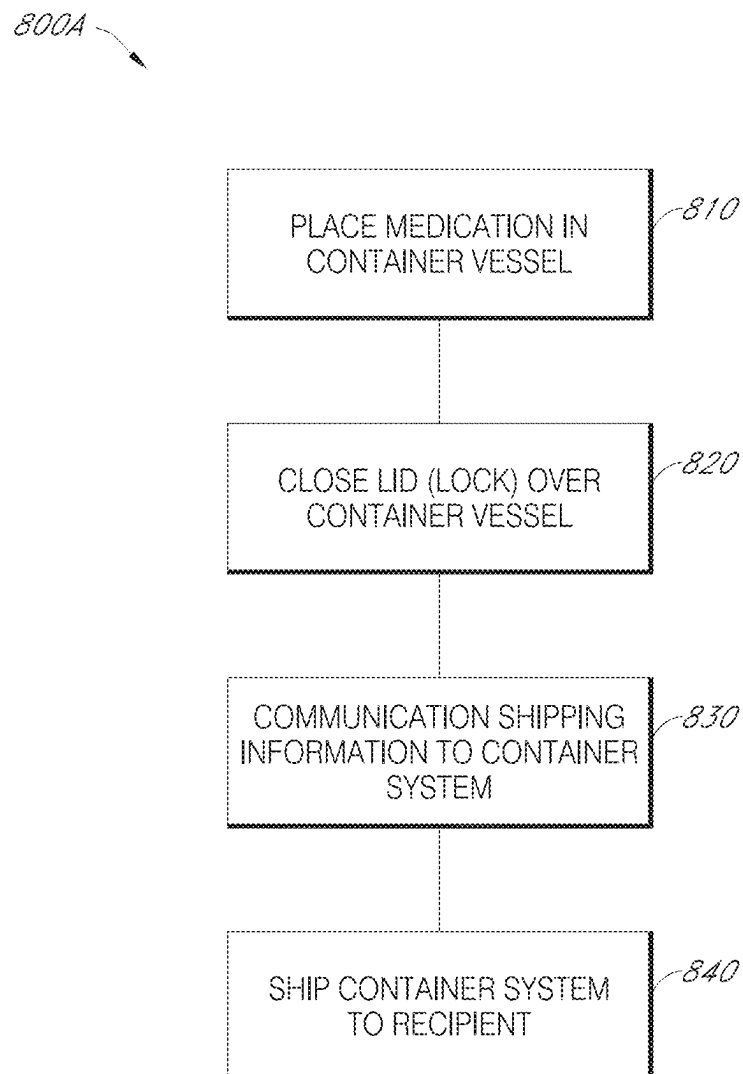


FIG. 42A

800B

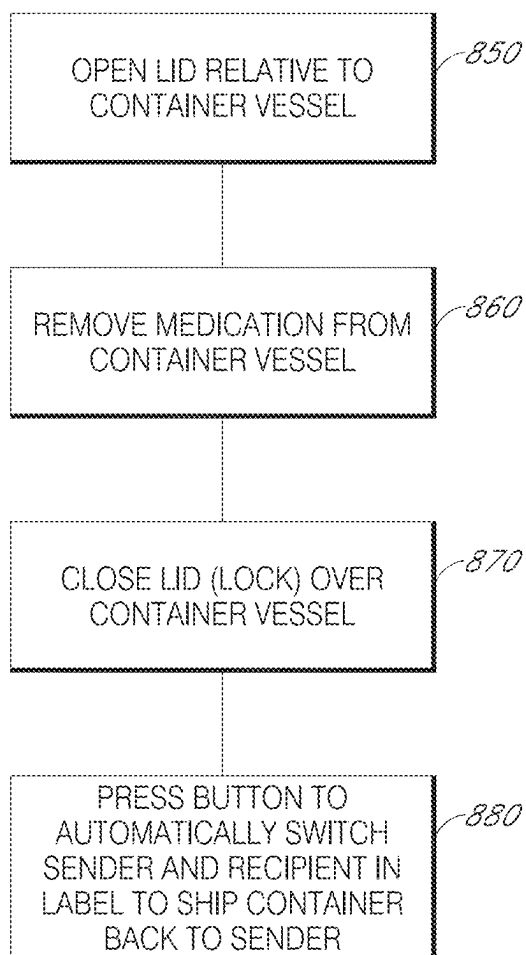


FIG. 42B

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PORTABLE COOLER WITH ACTIVE TEMPERATURE CONTROL

INCORPORATION BY REFERENCE TO ANY PRIORITY APPLICATIONS

Any and all applications for which a foreign or domestic priority claim is identified in the Application Data Sheet as filed with the present application are hereby incorporated by reference under 37 CFR 1.57 and should be considered a part of this specification.

BACKGROUND OF THE INVENTION

Field of the Invention

The invention is directed to a portable cooler (e.g., for medicine such as insulin, vaccines, epinephrine, medicine injectors, cartridges, biological fluids, etc.), and more particularly to a portable cooler with active temperature control.

Description of the Related Art

Certain medicine needs to be maintained at a certain temperature or temperature range to be effective (e.g., to maintain potency). Once potency of medicine (e.g., a vaccine) is lost, it cannot be restored, rendering the medicine ineffective and/or unusable. However, maintaining the cold chain (e.g., a record of the medicine's temperature history as it travels through various distribution channels) can be difficult. Additionally, where medicine is transported to remote locations for delivery (e.g., rural, mountainous, sparsely populated areas without road access), maintaining the medicine in the required temperature range may be difficult, especially when travelling through harsh (e.g., desert) climates. Existing medicine transport coolers are passive and inadequate for proper cold chain control (e.g., when used in extreme weather, such as in desert climates, tropical or subtropical climates, etc.).

SUMMARY

Accordingly, there is a need for improved portable cooler designs (e.g., for transporting medicine, such as vaccines, insulin, epinephrine, vials, cartridges, injector pens, etc.) that can maintain the contents of the cooler at a desired temperature or temperature range. Additionally, there is a need for an improved portable cooler design with improved cold chain control and record keeping of the temperature history of the contents (e.g., medicine, such as vaccines) of the cooler (e.g., during transport to remote locations).

In accordance with one aspect, a portable cooler container with active temperature control system is provided. The active temperature control system is operated to heat or cool a chamber of a vessel to approach a temperature set point suitable for a medication stored in the cooler container.

In accordance with another aspect, a portable cooler is provided that includes a temperature control system operable (e.g., automatically) to maintain the chamber of the cooler at a desired temperature or temperature range for a prolonged period of time. Optionally, the portable cooler is sized to house one or more liquid containers (e.g., medicine vials, cartridges or containers, such as a vaccine vials or insulin vials/cartridges, medicine injectors). Optionally, the portable cooler automatically logs (e.g., stores on a memory of the cooler) and/or communicates data on one or more sensed parameters (e.g., of the temperature of the chamber)

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to a remote electronic device (e.g., remote computer, mobile electronic device such as a smartphone or tablet computer, remote server, etc.). Optionally, the portable cooler can automatically log and/or transmit the data to the remote electronic device (e.g., automatically in real time, periodically at set intervals, etc.).

In accordance with another aspect, a portable cooler container with active temperature control is provided. The container comprises a container body having a chamber configured to receive and hold one or more volumes of perishable liquid, the chamber defined by a base and an inner peripheral wall of the container body. The container also comprises a temperature control system comprising one or more thermoelectric elements configured to actively heat or cool at least a portion of the chamber, and circuitry configured to control an operation of the one or more thermoelectric elements to heat or cool at least a portion of the chamber to a predetermined temperature or temperature range.

Optionally, the container can include one or more batteries configured to provide power to one or both of the circuitry and the one or more thermoelectric elements.

Optionally, the circuitry is further configured to wirelessly communicate with a cloud-based data storage system and/or a remote electronic device.

Optionally, the container includes a first heat sink in communication with the chamber, the first sink being selectively thermally coupled to the one or more thermoelectric elements.

Optionally, the container includes a second heat sink in communication with the one or more thermoelectric elements (TECs), such that the one or more TECs are disposed between the first heat sink and the second heat sink.

Optionally, the second heat sink is in thermal communication with a fan operable to draw heat from the second heat sink.

In one implementation, such as where the ambient temperature is above the predetermined temperature or temperature range, the temperature control system is operable to draw heat from the chamber via the first heat sink, which transfers said heat to the one or more TECs, which transfer said heat to the second heat sink, where the optional fan dissipates heat from the second heat sink.

In another implementation, such as where the ambient temperature is below the predetermined temperature or temperature range, the temperature control system is operable to add heat to the chamber via the first heat sink, which transfers said heat from the one or more TECs.

In accordance with one aspect of the disclosure, a portable cooler container with active temperature control is provided. The portable cooler container comprises a container body having a chamber configured to receive and hold one or more containers (e.g., of medicine). The portable cooler container also comprises a lid removably coupleable to the container body to access the chamber, and a temperature control system. The temperature control system comprises one or more thermoelectric elements configured to actively heat or cool at least a portion of the chamber, one or more batteries and circuitry configured to control an operation of the one or more thermoelectric elements to heat or cool at least a portion of the chamber to a predetermined temperature or temperature range. A display screen is disposed on one or both of the container body and the lid, the display screen configured to selectively display shipping information for the portable cooler container using electronic ink.

In accordance with another aspect of the disclosure, a portable cooler container with active temperature control is provided. The portable cooler container comprises a con-

tainer body having a chamber configured to receive and hold one or more containers (e.g., of medicine), the chamber defined by a base and an inner peripheral wall of the container body. A lid is removably coupleable to the container body to access the chamber. The portable cooler container also comprises a temperature control system. The temperature control system comprises one or more thermoelectric elements and one or more fans, one or both of the thermoelectric elements and fans configured to actively heat or cool at least a portion of the chamber, one or more batteries and circuitry configured to control an operation of the one or more thermoelectric elements to heat or cool at least a portion of the chamber to a predetermined temperature or temperature range.

In accordance with another aspect of the disclosure, a portable cooler container with active temperature control is provided. The portable cooler container comprises a container body having a chamber configured to receive and hold one or more volumes of perishable liquid, the chamber defined by a base and an inner peripheral wall of the container body, and a lid movably coupled to the container body by one or more hinges. The portable cooler container also comprises a temperature control system that comprises one or more thermoelectric elements configured to actively heat or cool at least a portion of the chamber, and one or more power storage elements. The temperature control system also comprises circuitry configured to control an operation of the one or more thermoelectric elements to heat or cool at least a portion of the chamber to a predetermined temperature or temperature range, the circuitry further configured to wirelessly communicate with a cloud-based data storage system or a remote electronic device. An electronic display screen is disposed on one or both of the container body and the lid, the display screen configured to selectively display shipping information for the portable cooler container.

BRIEF DESCRIPTION OF THE DRAWINGS

FIGS. 1A-1D are schematic views of one embodiment of a cooler container.

FIGS. 2A-2B are schematic partial views of another embodiment of a cooler container.

FIG. 2C is a schematic view of another embodiment of a cooler container.

FIGS. 3A-3C are schematic partial views of another embodiment of a cooler container.

FIGS. 4A-4C are schematic partial views of another embodiment of a cooler container.

FIGS. 5A-5B are schematic partial views of another embodiment of a cooler container.

FIGS. 6A-6B are schematic partial views of another embodiment of a cooler container.

FIGS. 7A-7B are schematic partial views of another embodiment of a cooler container.

FIGS. 8A-8B are schematic partial views of another embodiment of a cooler container.

FIGS. 9A-9B are schematic partial views of another embodiment of a cooler container.

FIGS. 10A-10B are schematic partial views of another embodiment of a cooler container.

FIG. 11A is a schematic view of another embodiment of a cooler container.

FIG. 11B is a schematic view of another embodiment of a cooler container.

FIGS. 12A-12B are schematic partial views of another embodiment of a cooler container.

FIG. 12C is a schematic view of another embodiment of a cooler container.

FIGS. 13A-13B are schematic partial views of another embodiment of a cooler container.

FIGS. 14A-14B are schematic partial views of another embodiment of a cooler container.

FIGS. 15A-15B are schematic partial views of another embodiment of a cooler container.

FIGS. 16A-16B are schematic partial views of another embodiment of a cooler container.

FIGS. 17A-17B are schematic partial views of another embodiment of a cooler container.

FIG. 18A is a schematic view of a portion of another embodiment of a cooler container.

FIG. 18B is a schematic view of a portion of another embodiment of a cooler container.

FIG. 18C is a schematic view of one embodiment of a coupling mechanism between the lid and vessel of the cooler container.

FIG. 18D is a schematic view of another embodiment of a coupling mechanism between the lid and the vessel of the cooler container.

FIG. 18E is a schematic view of one embodiment of a vessel for the cooler container.

FIG. 18F is a schematic view of another embodiment of a vessel for the cooler container.

FIG. 19 is a schematic view of another embodiment of a cooler container.

FIG. 20 is a schematic front view of another embodiment of a cooler container.

FIG. 21 is a schematic rear view of the cooler container of FIG. 20.

FIG. 22 is a schematic perspective view of the cooler container of FIG. 20.

FIG. 23 is a schematic perspective view of the cooler container of FIG. 20.

FIG. 24 is a schematic perspective view of the cooler container of FIG. 20.

FIG. 25A is a schematic view of a tray removed from the container.

FIG. 25B is a schematic view of an interchangeable tray system for use with the container.

FIG. 25C is a schematic top view of one embodiment of a tray for use in the container of FIG. 20.

FIG. 25D is a schematic top view of another embodiment of a tray for use in the container of FIG. 20.

FIG. 26 is a schematic bottom view of the cooler container of FIG. 20.

FIG. 27 is a schematic cross-sectional view of the cooler container of FIG. 20 with the tray disposed in the container.

FIG. 28 is a schematic view of the container in an open position with one or more lighting elements.

FIGS. 29A-29C are schematic views of a graphical user interface for use with the container.

FIG. 30 is a schematic view of a visual display of the container.

FIG. 31 is a schematic view of security features of the container.

FIG. 32 is a schematic perspective view of another embodiment of a cooler container.

FIGS. 33A-33B are schematic side views of various containers of different sizes.

FIG. 34 is a schematic view a container disposed on a power base.

FIGS. 35A-35C are schematic views of a graphical user interface for use with the container.

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FIG. 36 is a schematic view of another embodiment of a cooler container.

FIG. 37 is a schematic cross-sectional view of the cooler container of FIG. 32.

FIG. 38 is a schematic cross-sectional view of the cooler container of FIG. 37 with one fan in operation.

FIG. 39 is a schematic cross-sectional view of the cooler container of FIG. 37 with another fan in operation.

FIG. 40 is a schematic block diagram showing communication between the cooler container and a remote electronic device.

FIG. 41A shows a schematic perspective view of a cooler container.

FIG. 41B is a schematic block diagram showing electronics in the cooler container associated with operation of the display screen of the cooler container.

FIGS. 42A-42B show block diagrams of a method for operating the cooler container of FIG. 41A.

DETAILED DESCRIPTION

FIGS. 1A-1D show a schematic cross-sectional view of a container system 100 that includes a cooling system 200. Optionally, the container system 100 has a container vessel 120 that is optionally cylindrical and symmetrical about a longitudinal axis Z, and one of ordinary skill in the art will recognize that the features shown in cross-section in FIGS. 1A-1D are defined by rotating them about the axis Z to define the features of the container 100 and cooling system 200.

The container vessel 120 is optionally a cooler with active temperature control provided by the cooling system 200 to cool the contents of the container vessel 120 and/or maintain the contents of the vessel 120 in a cooled or chilled state. Optionally, the vessel 120 can hold therein one or more (e.g., a plurality of) separate containers (e.g., vials, cartridges, packages, injectors, etc.). Optionally, the one or more (e.g., plurality of) separate containers that can be inserted into the container vessel 120 are medicine containers (e.g., vaccine vials, insulin cartridges, injectors, etc.).

The container vessel 120 has an outer wall 121 that extends between a proximal end 122 that has an opening 123 and a distal end 124 having a base 125. The opening 123 is selectively closed by a lid L removably attached to the proximal end 122. The vessel 120 has an inner wall 126A and a base wall 126B that defines an open chamber 126 that can receive and hold contents to be cooled therein (e.g., one or more volumes of liquid, such as one or more vials, cartridges, packages, injectors, etc.). Optionally, the vessel 120 can be made of metal (e.g., stainless steel). In another implementation, the vessel 120 can be made of plastic. In one implementation, the vessel 120 has a cavity 128 (e.g., annular cavity or chamber) between the inner wall 126A and the outer wall 121. Optionally, the cavity 128 can be under vacuum. In another implementation, the cavity 128 can be filled with air but not be under vacuum. In still another implementation, the cavity 128 can be filled with a thermally insulative material (e.g., foam). In another implementation, the vessel 120 can exclude a cavity so that the vessel 120 is solid between the inner wall 126A and the outer wall 121.

With continued reference to FIGS. 1A-1D, the cooling system 200 is optionally implemented in the lid L that releasably closes the opening 123 of the vessel 120 (e.g., lid L can be attached to vessel 120 to closer the opening 123, and detached or decoupled from the vessel 120 to access the chamber 126 through the opening 123).

The cooling system 200 optionally includes a cold side heat sink 210 that faces the chamber 126, one or more

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thermoelectric elements (TECs) 220 (such as one or more Peltier elements) that selectively contacts the cold side heat sink 210, a hot side heat sink 230 in contact with the thermoelectric element 220 and disposed on an opposite side of the TEC 220 from the cold side heat sink 210, an insulator member 240 disposed between the cold side heat sink 210 and the hot side heat sink 230, one or more distal magnets 250 proximate a surface of the insulator 240, one or more proximal magnets 260 and one or more electromagnets 270 disposed axially between the distal magnets 250 and the proximal magnets 260. The proximal magnets 260 have an opposite polarity than the distal magnets 250. The electromagnets 270 are disposed about and connected to the hot side heat sink 230, which as noted above is attached to the TEC 220. The cooling system 200 also optionally includes a fan 280 in communication with the hot side heat sink 230 and one or more sealing gaskets 290 disposed between the cold side heat sink 210 and the hot side heat sink 230 and circumferentially about the TEC 220.

As discussed further below, circuitry and one or more batteries are optionally disposed in or on the vessel 120. For example, in one implementation, circuitry, sensors and/or batteries are disposed in a cavity in the distal end 124 of the vessel body 120, such as below the base wall 126B of the vessel 120, and can communicate with electrical contacts on the proximal end 122 of the vessel 120 that can contact corresponding electrical contacts (e.g., pogo pins, contact rings) on the lid L. In another implementation, the lid L can be connected to the proximal end 122 of the vessel 120 via a hinge, and electrical wires can extend through the hinge between the circuitry disposed in the distal end 124 of the vessel 120 and the fan 280 and TEC 220 in the lid L. Further discussion of the electronics in the cooling system 200 is provided further below. In another implementation, the circuitry and one or more batteries can be in a removable pack (e.g., De Walt battery pack) that attaches to the distal end 124 of the vessel 120, where one or more contacts in the removable pack contact one or more contacts on the distal end 124 of the vessel 120. The one or more contacts on the distal end 124 of the vessel 120 are electrically connected (via one or more wires or one or more intermediate components) with the electrical connections on the proximal 122 of the vessel 120, or via the hinge, as discussed above, to provide power to the components of the cooling system 200.

In operation, the one or more electromagnets 270 are operated to have a polarity that is opposite that of the one or more distal magnets 250 and/or the same as the polarity of the one or more proximal magnets 260, causing the electromagnets 270 to move toward and contact the distal magnets 250, thereby causing the TEC 220 to contact the cold side heat sink 210 (see FIG. 1C). The TEC 220 can be operated to draw heat from the chamber 126 via the cold side heat sink 210, which the TEC 220 transfers to the hot side heat sink 230. The fan 280 can optionally be operated to dissipate heat from the hot side heat sink 230, allowing the TEC 220 to draw more heat out of the chamber 126 to thereby cool the chamber 126. Once the desired temperature is achieved in the chamber 126 (e.g., as sensed by one or more sensors in thermal communication with the chamber 126), the fan 280 is turned off and the polarity of the one or more electromagnets 270 can be switched (e.g., switched off) so that the electromagnets 270 are repelled from the distal magnets 250 and/or attracted to the proximal magnets 260, thereby causing the TEC 220 to be spaced apart from (i.e., no longer contact) the cold side heat sink 210 (see FIG. 1D) within the housing 225. The separation between the TEC 220 and the cold side heat sink 210 advantageously

prevents heat in the hot side heat sink or due to ambient temperature from flowing back to the cold side heat sink, which prolongs the cooled state in the chamber 126.

FIGS. 2A-2B schematically illustrate a container system 100B that includes the cooling system 200B. The container system 100B can include the vessel 120 (as described above). Some of the features of the cooling system 200B are similar to features in the cooling system 200 in FIGS. 1A-1D. Thus, references numerals used to designate the various components of the cooling system 200B are identical to those used for identifying the corresponding components of the cooling system 200 in FIGS. 1A-1D, except that a “B” is added to the numerical identifier. Therefore, the structure and description for the various components of the cooling system 200 in FIGS. 1A-1D are understood to also apply to the corresponding components of the cooling system 200B in FIGS. 2A-2B, except as described below.

The TEC 220B can optionally be selectively slid into alignment between the cold side heat sink 210B and the hot side heat sink 230B, such that operation of the TEC 220B draws heat from the chamber 126 via the cold side heat sink 210B and transfers it to the hot side heat sink 230B. The fan 280B is optionally operated to further dissipate heat from the hot side heat sink 230B, allowing it to draw more heat from the chamber 126 via the TEC 220B. Optionally, one or more springs 212B (e.g., coil springs) resiliently couple the cold side heat sink 210B with the insulator 240B to maintain an efficient thermal connection between the cold side heat sink 210B and the TEC 220 when aligned together.

The TEC 220B can optionally be selectively slid out of alignment between the cold side heat sink 210B and the hot side heat sink 230B to thereby disallow heat transfer through the TEC 220B (e.g., once the desired temperature in the chamber 126 has been achieved). Optionally, the TEC 220B is slid into a cavity 242B in the insulator 240B.

The TEC 220B can be slid into and out of alignment between the cold side heat sink 210B and the hot side heat sink 230B with a number of suitable mechanisms. In one implementation, an electric motor can drive a gear in contact with a gear rack (e.g., rack and pinion), where the TEC 220B can be attached to the rack that linearly moved via rotation of the gear by the electric motor. In another implementation, a solenoid motor can be attached to TEC 220B to effect the linear movement of the TEC 220B. In still another implementation a pneumatic or electromechanical system can actuate movement of a piston attached to the TEC 220B to effect the linear movement of the TEC 220B.

FIG. 2C schematically illustrates a portion of a container system 100B' that includes the cooling system 200B'. The container system 100B' can include the vessel 120 (as described above). Some of the features of the cooling system 200B' are similar to features in the cooling system 200B in FIGS. 2A-2B. Thus, references numerals used to designate the various components of the cooling system 200B' are identical to those used for identifying the corresponding components of the cooling system 200B in FIGS. 2A-2B, except that a “'” is added to the numerical identifier. Therefore, the structure and description for the various components of the cooling system 200B in FIGS. 2A-2B are understood to also apply to the corresponding components of the cooling system 200B' in FIG. 2C, except as described below.

The cooling system 200B' differs from the cooling system 200B in that the TEC 220B' is tapered or wedge shaped. An actuator 20A (e.g., electric motor) is coupled to the TEC 220B' via a driver 20B. The actuator 20A is selectively actuatable to move the TEC 220B' into and out of engage-

ment (e.g., into and out of contact) with the hot side heat sink 230B' and the cold side heat sink 210B' to allow for heat transfer therebetween. Optionally, the hot side heat sink 230B' and/or the cold side heat sink 210B' can have a tapered surface that thermally communicates with (e.g., operatively contacts) one or more tapered surfaces (e.g., wedge shaped surfaces) of the TEC 220B' when the TEC 220B' is moved into thermal communication (e.g., into contact) with the hot side heat sink 230B' and the cold side heat sink 210B'.

FIGS. 3A-3C schematically illustrate a container system 100C that includes the cooling system 200C. The container system 100C can include the vessel 120 (as described above). Some of the features of the cooling system 200C are similar to features in the cooling system 200B in FIGS. 2A-2B. Thus, references numerals used to designate the various components of the cooling system 200C are identical to those used for identifying the corresponding components of the cooling system 200B in FIGS. 2A-2B, except that a “C” is used instead of a “B”. Therefore, the structure and description for the various components of the cooling system 200B in FIGS. 2A-2B are understood to also apply to the corresponding components of the cooling system 200C in FIGS. 3A-3C, except as described below.

The cooling system 200C differs from the cooling system 200B in that the TEC 220C is in a fixed position adjacent the hot side heat sink 230C. The insulator member 240C has one or more thermal conductors 244C embedded therein, and the insulator member 240C can be selectively rotated about an axis (e.g., an axis offset from the axis Z of the vessel 120) to align at least one of the thermal conductors 244C with the TEC 220C and the cold side heat sink 210C to allow heat transfer between the chamber 126 and the hot side heat sink 230C. The insulator member 240C can also be selectively rotated to move the one or more thermal conductors 244C out of alignment with the TEC 220C so that instead an insulating portion 246C is interposed between the TEC 220C and the cold side heat sink 210C, thereby inhibiting (e.g., preventing) heat transfer between the TEC 220C and the cold side heat sink 210C to prolong the cooled state in the chamber 126. With reference to FIGS. 3B-3C, in one implementation, the insulator member 240C can be rotated by a motor 248C (e.g., electric motor) via a pulley cable or band 249C.

FIGS. 4A-4C schematically illustrate a container system 100D that includes the cooling system 200D. The container system 100D can include the vessel 120 (as described above). Some of the features of the cooling system 200D are similar to features in the cooling system 200C in FIGS. 3A-3C. Thus, references numerals used to designate the various components of the cooling system 200D are identical to those used for identifying the corresponding components of the cooling system 200C in FIGS. 3A-3C, except that a “D” is used instead of a “C”. Therefore, the structure and description for the various components of the cooling system 200C in FIGS. 3A-3C are understood to also apply to the corresponding components of the cooling system 200D in FIGS. 4A-4C, except as described below.

The cooling system 200D differs from the cooling system 200C in the mechanism for rotating the insulator member 240D. In particular, the insulator member 240D has one or more thermal conductors 244D embedded therein, and the insulator member 240D can be selectively rotated about an axis (e.g., an axis offset from the axis Z of the vessel 120) to align at least one of the thermal conductors 244D with the TEC 220D and the cold side heat sink 210D to allow heat transfer between the chamber 126 and the hot side heat sink 230D. The insulator member 240D can also be selectively

rotated to move the one or more thermal conductors **244D** out of alignment with the TEC **220D** so that instead an insulating portion **246D** is interposed between the TEC **220D** and the cold side heat sink **210D**, thereby inhibiting (e.g., preventing) heat transfer between the TEC **220D** and the cold side heat sink **210D** to prolong the cooled state in the chamber **126**. With reference to FIGS. **4B-4C**, in one implementation, the insulator member **240D** can be rotated by a motor **248D** (e.g., electric motor) via a gear train or geared connection **249D**.

FIGS. **5A-5B** schematically illustrate a container system **100E** that includes the cooling system **200E**. The container system **100E** can include the vessel **120** (as described above). Some of the features of the cooling system **200D** are similar to features in the cooling system **200B** in FIGS. **2A-2B**. Thus, references numerals used to designate the various components of the cooling system **200E** are identical to those used for identifying the corresponding components of the cooling system **200B** in FIGS. **2A-2B**, except that an “E” is used instead of a “B”. Therefore, the structure and description for the various components of the cooling system **200B** in FIGS. **2A-2B** are understood to also apply to the corresponding components of the cooling system **200E** in FIGS. **5A-5B**, except as described below.

An assembly **A** including the hot side heat sink **230E**, fan **280E**, TEC **220E** and an insulator segment **244E** can optionally be selectively slid relative to the vessel **120** to bring the TEC **220E** into alignment (e.g., contact) between the cold side heat sink **210E** and the hot side heat sink **230E**, such that operation of the TEC **220E** draws heat from the chamber **126** via the cold side heat sink **210E** and transfers it to the hot side heat sink **230E**. The fan **280E** is optionally operated to further dissipate heat from the hot side heat sink **230E**, allowing it to draw more heat from the chamber **126** via the TEC **220E**. Optionally, one or more springs **212E** (e.g., coil springs) resiliently couple the cold side heat sink **210E** with the insulator **240E** to maintain an efficient thermal connection between the cold side heat sink **210E** and the TEC **220E** when aligned together.

The assembly **A** can optionally be selectively slid to move the TEC **220E** out of alignment (e.g., contact) between the cold side heat sink **210E** and the hot side heat sink **230E**. This causes the insulator segment **244E** to instead be placed in alignment (e.g., contact) between the cold side heat sink **210E** and the hot side heat sink **230E**, which disallows heat transfer through the TEC **220E** (e.g., once the desired temperature in the chamber **126** has been achieved).

The assembly **A** can be slid with a number of suitable mechanisms. In one implementation, an electric motor can drive a gear in contact with a gear rack (e.g., rack and pinion), where the assembly **A** can be attached to the rack that linearly moves via rotation of the gear by the electric motor. In another implementation, a solenoid motor can be attached to assembly **A** to effect the linear movement of the assembly **A**. In still another implementation a pneumatic or electromechanical system can actuate movement of a piston attached to the assembly **A** to effect the linear movement of the assembly **A**.

FIGS. **6A-6B** schematically illustrate a container system **100F** that includes the cooling system **200F**. The container system **100F** can include the vessel **120** (as described above). Some of the features of the cooling system **200F** are similar to features in the cooling system **200** in FIGS. **1A-1D**. Thus, references numerals used to designate the various components of the cooling system **200F** are identical to those used for identifying the corresponding components of the cooling system **200** in FIGS. **1A-1D**, except that a “G”

is added to the numerical identifiers. Therefore, the structure and description for the various components of the cooling system **200** in FIGS. **1A-1D** are understood to also apply to the corresponding components of the cooling system **200F** in FIGS. **6A-6B**, except as described below.

As shown in FIGS. **6A-6B**, the hot side heat sink **230F** is in contact with the TEC **220F**. One or more springs **212F** (e.g., coil springs) can be disposed between the hot side heat sink **230F** and the insulator member **240F**. The one or more springs **212F** exert a (bias) force on the hot side heat sink **230F** to bias it toward contact with the insulator member **240F**. One or more expandable bladders **250F** are disposed between the insulator member **240F** and the hot side heat sink **230F**.

When the one or more expandable bladders **250F** are in a collapsed state (see FIG. **6A**), the one or more springs **212F** draw the hot side heat sink **230F** toward the insulator member **240F** so that the TEC **220F** contacts the cold side heat sink **210F**. The TEC **220F** can be operated to draw heat out of the chamber **126** via the cold side heat sink **210F**, which is then transferred via the TEC **220F** to the hot side heat sink **230F**. Optionally, the fan **280F** can be operated to dissipate heat from the hot side heat sink **230F**, allowing the hot side heat sink **230F** to draw additional heat from the chamber **126** via the contact between the cold side heat sink **210F**, the TEC **220F** and the hot side heat sink **230F**. Accordingly, with the one or more expandable bladders **250F** in the collapsed state, the cooling system **200F** can be operated to draw heat from the chamber **126** to cool the chamber to a predetermined temperature or temperature range.

When the one or more expandable bladders **250F** are in an expanded state (see FIG. **6B**), they can exert a force on the hot side heat sink **230F** in a direction opposite to the bias force of the one or more springs **212F**, causing the hot side heat sink **230F** to separate from (e.g., lift from) the insulator member **240F**. Such separation between the hot side heat sink **230F** and the insulator member **240F** also causes the TEC **220F** to become spaced apart from the cold side heat sink **210F**, inhibiting (e.g., preventing) heat transfer between the cold side heat sink **210F** and the TEC **220F**. Accordingly, once the predetermined temperature or temperature range has been achieved in the chamber **126**, the one or more expandable bladders **250F** can be transitioned to the expanded state to thermally disconnect the cold side heat sink **210F** from the TEC **220F** to thereby maintain the chamber **126** in a prolonged cooled state.

In one implementation, the one or more expandable bladders **250F** form part of a pneumatic system (e.g., having a pump, one or more valves, and/or a gas reservoir) that selectively fills the bladders **250F** with a gas to move the bladders **250F** to the expanded state and selectively empties the one or more expandable bladders **250F** to move the bladders **250F** to the collapsed state.

In another implementation, the one or more expandable bladders **250F** form part of a hydraulic system (e.g., having a pump, one or more valves, and/or a liquid reservoir) that selectively fills the bladders **250F** with a liquid to move the bladders **250F** to the expanded state and selectively empties the one or more expandable bladders **250F** to move the bladders **250F** to the collapsed state.

FIGS. **7A-7B** schematically illustrate a container system **100G** that includes the cooling system **200G**. The container system **100G** can include the vessel **120** (as described above). Some of the features of the cooling system **200G** are similar to features in the cooling system **200F** in FIGS. **6A-6B**. Thus, references numerals used to designate the

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various components of the cooling system **200G** are identical to those used for identifying the corresponding components of the cooling system **200F** in FIGS. **6A-6B**, except that a “G” is used instead of an “F”. Therefore, the structure and description for the various components of the cooling system **200F** in FIGS. **6A-6B** are understood to also apply to the corresponding components of the cooling system **200G** in FIGS. **7A-7B**, except as described below.

The cooling system **200G** differs from the cooling system **200F** in the position of the one or more springs **212G** and the one or more expandable bladders **250G**. As shown in FIGS. **7A-7B**, the one or more springs **212G** (e.g., coil springs) can be disposed between the cold side heat sink **210G** and the insulator member **240G**. The one or more springs **212G** exert a (bias) force on the cold side heat sink **210G** to bias it toward contact with the insulator member **240G**. The one or more expandable bladders **250G** are disposed between the insulator member **240G** and the cold side heat sink **230G**.

When the one or more expandable bladders **250G** are in a collapsed state (see FIG. **7A**), the one or more springs **212G** draw the cold side heat sink **230G** (up) toward the insulator member **240G** so that the TEC **220G** contacts the cold side heat sink **210G**. The TEC **220G** can be operated to draw heat out of the chamber **126** via the cold side heat sink **210G**, which is then transferred via the TEC **220G** to the hot side heat sink **230G**. Optionally, the fan **280G** can be operated to dissipate heat from the hot side heat sink **230G**, allowing the hot side heat sink **230G** to draw additional heat from the chamber **126** via the contact between the cold side heat sink **210G**, the TEC **220G** and the hot side heat sink **230G**. Accordingly, with the one or more expandable bladders **250G** in the collapsed state, the cooling system **200G** can be operated to draw heat from the chamber **126** to cool the chamber to a predetermined temperature or temperature range.

When the one or more expandable bladders **250G** are in an expanded state (see FIG. **7B**), they can exert a force on the cold side heat sink **210G** in a direction opposite to the bias force of the one or more springs **212G**, causing the cold side heat sink **210G** to separate from (e.g., move down relative to) the insulator member **240G**. Such separation between the cold side heat sink **210G** and the insulator member **240G** also causes the TEC **220G** to become spaced apart from the cold side heat sink **210G**, inhibiting (e.g., preventing) heat transfer between the cold side heat sink **210G** and the TEC **220G**. Accordingly, once the predetermined temperature or temperature range has been achieved in the chamber **126**, the one or more expandable bladders **250G** can be transitioned to the expanded state to thermally disconnect the cold side heat sink **210G** from the TEC **220G** to thereby maintain the chamber **126** in a prolonged cooled state.

In one implementation, the one or more expandable bladders **250G** form part of a pneumatic system (e.g., having a pump, one or more valves, and/or a gas reservoir) that selectively fills the bladders **250G** with a gas to move the bladders **250G** to the expanded state and selectively empties the one or more expandable bladders **250G** to move the bladders **250G** to the collapsed state.

In another implementation, the one or more expandable bladders **250G** form part of a hydraulic system (e.g., having a pump, one or more valves, and/or a liquid reservoir) that selectively fills the bladders **250G** with a liquid to move the bladders **250G** to the expanded state and selectively empties the one or more expandable bladders **250G** to move the bladders **250G** to the collapsed state.

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FIGS. **8A-8B** schematically illustrate a container system **100H** that includes the cooling system **200H**. The container system **100H** can include the vessel **120** (as described above). Some of the features of the cooling system **200H** are similar to features in the cooling system **200F** in FIGS. **6A-6B**. Thus, references numerals used to designate the various components of the cooling system **200H** are identical to those used for identifying the corresponding components of the cooling system **200F** in FIGS. **6A-6B**, except that an “H” is used instead of an “F”. Therefore, the structure and description for the various components of the cooling system **200F** in FIGS. **6A-6B** are understood to also apply to the corresponding components of the cooling system **200H** in FIGS. **8A-8B**, except as described below.

The cooling system **200H** differs from the cooling system **200F** in that one or more expandable bladders **255H** are included instead of the one or more springs **212F** to provide a force in a direction opposite to the force exerted by the one or more expandable bladders **250H**. As shown in FIGS. **8A-8B**, the one or more expandable bladders **255H** are disposed between a housing **225H** and a portion of the hot side heat sink **230H**, and one or more expandable bladders **250H** are disposed between the insulator member **240H** and the hot side heat sink **230H**. Optionally, the one or more expandable bladders **250H** are in fluid communication with the one or more expandable bladders **255H**, and the fluid is moved between the two expandable bladders **250H**, **255H**. That is, when the one or more expandable bladders **250H** are in the expanded state, the one or more expandable bladders **255H** are in the collapsed state, and when the expandable bladders **250H** are in the collapsed state, the expandable bladders **255H** are in the expanded state.

When the one or more expandable bladders **250H** are in a collapsed state (see FIG. **8A**), the one or more expandable bladders **255H** are in the expanded state and exert a force on the hot side heat sink **230H** toward the insulator member **240H** so that the TEC **220H** contacts the cold side heat sink **210H**. The TEC **220H** can be operated to draw heat out of the chamber **126** via the cold side heat sink **210H**, which is then transferred via the TEC **220H** to the hot side heat sink **230H**. Optionally, the fan **280H** can be operated to dissipate heat from the hot side heat sink **230H**, allowing the hot side heat sink **230H** to draw additional heat from the chamber **126** via the contact between the cold side heat sink **210H**, the TEC **220H** and the hot side heat sink **230H**. Accordingly, with the one or more expandable bladders **250H** in the collapsed state, the cooling system **200H** can be operated to draw heat from the chamber **126** to cool the chamber to a predetermined temperature or temperature range.

When the one or more expandable bladders **250H** are in an expanded state (see FIG. **8B**), the one or more expandable bladders **255H** are in a collapsed state. The expanded state of the expandable bladders **250H** exerts a force on the hot side heat sink **230H** that causes the hot side heat sink **230H** to separate from (e.g., lift from) the insulator member **240H**. Such separation between the hot side heat sink **230H** and the insulator member **240H** also causes the TEC **220H** to become spaced apart from (e.g., lift from) the cold side heat sink **210H**, thereby thermally disconnecting (e.g., inhibiting heat transfer between) the cold side heat sink **210H** and the TEC **220H**. Accordingly, once the predetermined temperature or temperature range has been achieved in the chamber **126**, the one or more expandable bladders **250H** can be transitioned to the expanded state (e.g., by transferring the fluid from the expandable bladders **255H** to the expandable bladders **250H**) to thermally disconnect the cold side heat

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sink 210H from the TEC 220H to thereby maintain the chamber 126 in a prolonged cooled state.

In one implementation, the one or more expandable bladders 250H, 255H form part of a pneumatic system (e.g., having a pump, one or more valves, and/or a gas reservoir) that selectively fills and empties the bladders 250H, 255H with a gas to move them between an expanded and a collapsed state.

In one implementation, the one or more expandable bladders 250H, 255H form part of a hydraulic system (e.g., having a pump, one or more valves, and/or a liquid reservoir) that selectively fills and empties the bladders 250H, 255H with a liquid to move them between an expanded and a collapsed state.

FIGS. 9A-9B schematically illustrate a container system 100I that includes the cooling system 200I. The container system 100I can include the vessel 120 (as described above). Some of the features of the cooling system 200I are similar to features in the cooling system 200G in FIGS. 7A-7B. Thus, references numerals used to designate the various components of the cooling system 200I are identical to those used for identifying the corresponding components of the cooling system 200G in FIGS. 7A-7B, except that an "I" is used instead of a "G". Therefore, the structure and description for the various components of the cooling system 200G in FIGS. 7A-7B are understood to also apply to the corresponding components of the cooling system 200I in FIGS. 9A-9B, except as described below.

The cooling system 200I differs from the cooling system 200G in that the one or more rotatable cams 250I are used instead of one or more expandable bladders 250G. As shown in FIGS. 9A-9B, the one or more springs 212I (e.g., coil springs) can be disposed between the cold side heat sink 210I and the insulator member 240I. The one or more springs 212I exert a (bias) force on the cold side heat sink 210I to bias it toward contact with the insulator member 240I. The one or more rotatable cams 250I are rotatably coupled to the insulator member 240I and rotatable to selectively contact a proximal surface of the cold side heat sink 230I.

In a cooling state (see FIG. 9A), the rotatable cams 250I are not in contact with the cold side heat sink 210I, such that the one or more springs 212I bias the cold side heat sink 210I into contact with the TEC 220I, thereby allowing heat transfer therebetween. The TEC 220I can be operated to draw heat out of the chamber 126 via the cold side heat sink 210I, which is then transferred via the TEC 220I to the hot side heat sink 230I. Optionally, the fan 280I can be operated to dissipate heat from the hot side heat sink 230I, allowing the hot side heat sink 230I to draw additional heat from the chamber 126 via the contact between the cold side heat sink 210I, the TEC 220I and the hot side heat sink 230I. Accordingly, with the one or more rotatable cams 250I in a retracted state, the cooling system 200I can be operated to draw heat from the chamber 126 to cool the chamber to a predetermined temperature or temperature range.

When the one or more rotatable cams 250I are moved to the deployed state (see FIG. 9B), the cams 250I bear against the cold side heat sink 210I, overcoming the bias force of the springs 212I. In the deployed state, the one or more cams 250I exert a force on the cold side heat sink 210I that causes the cold side heat sink 210I to separate from (e.g., move down relative to) the insulator member 240I. Such separation between the cold side heat sink 210I and the insulator member 240I also causes the cold side heat sink 210I to become spaced apart from (e.g., move down relative to) the TEC 220I, thereby thermally disconnecting (e.g., inhibiting

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heat transfer between) the cold side heat sink 210I and the TEC 220I. Accordingly, once the predetermined temperature or temperature range has been achieved in the chamber 126, the one or more rotatable cams 250I can be moved to the deployed state to thermally disconnect the cold side heat sink 210I from the TEC 220I to thereby maintain the chamber 126 in a prolonged cooled state.

FIGS. 10A-10B schematically illustrate a container system 100J that includes the cooling system 200J. The container system 100J can include the vessel 120 (as described above). Some of the features of the cooling system 200J are similar to features in the cooling system 200I in FIGS. 9A-9B. Thus, references numerals used to designate the various components of the cooling system 200J are identical to those used for identifying the corresponding components of the cooling system 200I in FIGS. 9A-9B, except that an "J" is used instead of an "I". Therefore, the structure and description for the various components of the cooling system 200I in FIGS. 9A-9B are understood to also apply to the corresponding components of the cooling system 200J in FIGS. 10A-10B, except as described below.

The cooling system 200J differs from the cooling system 200I in the location of the one or more springs 212J and the one or more cams 250J. As shown in FIGS. 10A-10B, the one or more springs 212J are disposed between the insulator member 240J and the hot side heat sink 230J and exert a bias force between the two biasing the hot side heat sink 230J down toward contact with the insulator member 240J. Such bias force also biases the TEC 220J (which is attached to or in contact with the hot side heat sink 230J) into contact with the cold side heat sink 210J.

When the one or more rotatable cams 250J are in a retracted state (see FIG. 10A), the cams 250J allow the TEC 220J to contact the cold side heat sink 210J. The TEC 220J can be operated to draw heat out of the chamber 126 via the cold side heat sink 210J, which is then transferred via the TEC 220J to the hot side heat sink 230J. Optionally, the fan 280J can be operated to dissipate heat from the hot side heat sink 230J, allowing the hot side heat sink 230J to draw additional heat from the chamber 126 via the contact between the cold side heat sink 210J, the TEC 220J and the hot side heat sink 230J. Accordingly, with the one or more rotatable cams 250J in a retracted state, the cooling system 200J can be operated to draw heat from the chamber 126 to cool the chamber to a predetermined temperature or temperature range.

When the one or more rotatable cams 250J are moved to the deployed state (see FIG. 10B), the cams 250J bear against the hot side heat sink 230J, overcoming the bias force of the springs 212J. In the deployed state, the one or more cams 250J exert a force on the hot side heat sink 230J that causes the hot side heat sink 230J to separate from (e.g., lift from) the insulator member 240J. Such separation also causes the TEC 220J (attached to the hot side heat sink 230J) to become spaced apart from (e.g., lift from) the cold side heat sink 210J, thereby thermally disconnecting (e.g., inhibiting heat transfer between) the cold side heat sink 210J and the TEC 220J. Accordingly, once the predetermined temperature or temperature range has been achieved in the chamber 126, the one or more rotatable cams 250J can be moved to the deployed state to thermally disconnect the cold side heat sink 210J from the TEC 220J to thereby maintain the chamber 126 in a prolonged cooled state.

FIG. 11A schematically illustrates a container system 100K that includes the cooling system 200K. The container system 100K can include the vessel 120 (as described above) removably sealed by a lid L'. Some of the features of

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the cooling system **200K** are similar to features in the cooling system **200** in FIGS. 1A-1D. Thus, reference numerals used to designate the various components of the cooling system **200K** are similar to those used for identifying the corresponding components of the cooling system **200** in FIGS. 1A-1D, except that an “K” is used. Therefore, the structure and description for said similar components of the cooling system **200** in FIGS. 1A-1D are understood to also apply to the corresponding components of the cooling system **200K** in FIG. 11A, except as described below.

With reference to FIG. 11A, the vessel **120** optionally has a cavity **128** (e.g., annular cavity or chamber) between the inner wall **126A** and the outer wall **121**. The cavity **128** can be under vacuum, so that the vessel **120** is vacuum sealed. The lid **L'** that removably seals the vessel **120** is optionally also a vacuum sealed lid. The vacuum sealed vessel **120** and/or lid **L'** advantageously inhibits heat transfer there-through, thereby inhibiting a passive change in temperature in the chamber **126** when the lid **L'** is attached to the vessel **120** (e.g., via passive loss of cooling through the wall of the vessel **120** and/or lid **L'**).

The cooling system **200K** includes a hot side heat sink **230K** in thermal communication with the thermoelectric element (TEC) (e.g., Peltier element) **220K**, so that the heat sink **230K** can draw heat away from the TEC **220K**. Optionally, a fan **280K** can be in thermal communication with the hot side heat sink **230K** and be selectively operable to further dissipate heat from the hot side heat sink **230K**, thereby allowing the heat sink **230K** to further draw heat from the TEC **220K**.

The TEC **220K** is in thermal communication with a cold side heat sink **210K**, which is in turn in thermal communication with the chamber **126** in the vessel **120**. The cold side heat sink **210K** optionally includes a flow path **214K** that extends from an opening **132K** in the lid **L'** adjacent the chamber **126** to an opening **134K** in the lid **L'** adjacent the chamber **126**. In one implementation, the opening **132K** is optionally located generally at a center of the lid **L'**, as shown in FIG. 11A. In one implementation, the opening **134K** is optionally located in the lid **L'** at a location proximate the inner wall **126A** of the vessel **120** when the lid **L'** is attached to the vessel **120**. Optionally, the cold side heat sink **210K** includes a fan **216K** disposed along the flow path **214K** between the openings **132K**, **134K**. As shown in FIG. 11A, at least a portion of the flow path **214K** is in thermal communication with the TEC **220K** (e.g., with a cold side of the TEC).

In operation, air in the chamber **126** enters the flow path **214K** via the opening **132K** and flows through the flow path **214K** so that it passes through the portion of the flow path **214K** that is proximate the TEC **220K**, where the TEC **220K** is selectively operated to cool (e.g., reduce the temperature of) the air flow passing therein. The cooled airflow continues to flow through the flow path **214K** and exits the flow path **214K** at opening **134K** where it enters the chamber **126**. Optionally, the fan **216K** is operable to draw (e.g., cause or facilitate) the flow of air through the flow path **214K**.

Though FIG. 11A shows the cooling system **200** disposed on a side of the vessel **120**, one of skill in the art will recognize that the cooling system **200** can be disposed in other suitable locations (e.g., on the bottom of the vessel **120**, on top of the lid **L'**, in a separate module attachable to the top of the lid **L'**, etc.) and that such implementations are contemplated by the invention.

FIG. 11B schematically illustrates a container system **100K'** that includes the cooling system **200K'**. The container system **100K'** can include the vessel **120** (as described

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above). Some of the features of the cooling system **200K'** are similar to features in the cooling system **200K** in FIG. 11A. Thus, reference numerals used to designate the various components of the cooling system **200K'** are similar to those used for identifying the corresponding components of the cooling system **200K** in FIG. 11A, except that an “'” is used. Therefore, the structure and description for said similar components of the cooling system **200K** in FIG. 11A are understood to also apply to the corresponding components of the cooling system **200K'** in FIG. 11B, except as described below.

The container system **100K'** is optionally a self-chilled container (e.g. self-chilled water container, such as a water bottle). The cooling system **200K'** differs from the cooling system **200K** in that a liquid is used as a cooling medium that is circulated through the body of the vessel **120**. A conduit **134K'** can deliver chilled liquid to the body of the vessel **120**, and a conduit **132K'** can remove a warm liquid from the body of the vessel **120**. In the body of the vessel **120**, the chilled liquid can absorb energy from one or more walls of the vessel **120** (e.g., one or more walls that define the chamber **126**) of a liquid in the chamber **126**, and the heated liquid can exit the body of the vessel **120** via conduit **132K'**. In this manner, one or more surfaces of the body of the vessel **120** (e.g., of the chamber **126**) are maintained in the cooled state. Though not shown, the conduits **132K'**, **134K'** connect to a cooling system, such as one having a TEC **220K** in contact with a hot side heat sink **230K**, as described above for container system **100K**.

FIGS. 12A-12B schematically illustrate a container system **100L** that includes the cooling system **200L**. The container system **100L** can include the vessel **120** (as described above). Some of the features of the cooling system **200L**, which optionally serves as part of the lid **L** that selectively seals the vessel **120**, are similar to features in the cooling system **200** in FIGS. 1A-1D. Thus, reference numerals used to designate the various components of the cooling system **200L** are similar to those used for identifying the corresponding components of the cooling system **200** in FIGS. 1A-1D, except that an “L” is used. Therefore, the structure and description for said similar components of the cooling system **200** in FIGS. 1A-1D are understood to also apply to the corresponding components of the cooling system **200L** in FIGS. 12A-12B, except as described below.

With reference to FIGS. 12A-12B, the cooling system **200L** can optionally include a cavity **214L** disposed between the thermoelectric element (TEC) **220L** and the cold side heat sink **210L**. The cooling system **200L** can optionally include a pump **216L** (e.g., a peristaltic pump) in fluid communication with the cavity **214L** and with a reservoir **213L**. The pump **216L** is operable to move a conductive fluid **217L** (e.g., a conductive liquid), such as a volume of conductive fluid **217L**, between the reservoir **213L** and the cavity **214L**. Optionally, the conductive fluid **217L** can be mercury; however, the conductive fluid **217L** can be other suitable liquids.

In operation, when the cooling system **200L** is operated in a cooling stage, the pump **216L** is selectively operable to pump the conductive fluid **217L** into the cavity **214L** (e.g., to fill the cavity **214L**), thereby allowing heat transfer between the cold side heat sink **210L** and the TEC **220L** (e.g., allowing the TEC **220L** to be operated to draw heat from the cold side heat sink **210L** and transfer it to the hot side heat sink **230L**). Optionally, the fan **280L** is selectively operable to dissipate heat from the hot side heat sink **230L**,

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thereby allowing the TEC 220L to draw further heat from the chamber 126 via the cold side heat sink 210L and the conductive fluid 217L.

With reference to FIG. 12A, when the cooling system 200L is operated in an insulating state, the pump 216L is selectively operated to remove (e.g., drain) the conductive fluid 217L from the cavity 214L (e.g., by moving the conductive fluid 217L into the reservoir 213L), thereby leaving the cavity 214L unfilled (e.g., empty). Such removal (e.g., complete removal) of the conductive fluid 217L from the cavity 214L thermally disconnects the cold side heat sink 210L from the TEC 220L, thereby inhibiting (e.g., preventing) heat transfer between the TEC 220L and the chamber 126 via the cold side heat sink 210L, which advantageously prevents heat in the hot side heat sink 230L or due to ambient temperature from flowing back to the cold side heat sink 210L, thereby prolonging the cooled state in the chamber 126.

FIG. 12C schematically illustrate a container system 100L' that includes the cooling system 200L'. The container system 100L' can include the vessel 120 (as described above). Some of the features of the cooling system 200L' are similar to features in the cooling system 200L in FIGS. 12A-12B. Thus, references numerals used to designate the various components of the cooling system 200L' are similar to those used for identifying the corresponding components of the cooling system 200L in FIGS. 12A-12B, except that an "'" is used. Therefore, the structure and description for said similar components of the cooling system 200L in FIGS. 12A-12B are understood to also apply to the corresponding components of the cooling system 200L' in FIG. 12C, except as described below.

The cooling system 200L' differs from the cooling system 200L in that a heat pipe 132L' is used to connect the hot side heat sink 230L' to the cold side heat sink 210L'. The heat pipe 132L' can be selectively turned on and off. Optionally, the heat pipe 132L' can include a phase change material (PCM). Optionally, the heat pipe 132L' can be turned off by removing the working fluid from inside the heat pipe 132L', and turned on by inserting or injecting the working fluid in the heat pipe 132L'. For example, the TEC 220L', when in operation, can freeze the liquid in the heat pipe 132L', to thereby provide a thermal break within the heat pipe 132L', disconnecting the chamber of the vessel 120 from the TEC 220L' that is operated to cool the chamber. When the TEC 220L' is not in operation, the liquid in the heat pipe 132L' can flow along the length of the heat pipe 132L'. For example, the fluid can flow within the heat pipe 132L' into thermal contact with a cold side of the TEC 220L', which can cool the liquid, the liquid can then flow to the hot side of the heat pipe 132L' and draw heat away from the chamber of the vessel 120 which heats such liquid, and the heated liquid can then again flow to the opposite end of the heat pipe 132L' where the TEC 220L' can again remove heat from it to cool the liquid before it again flows back to the other end of the heat pipe 132L' to draw more heat from the chamber.

FIGS. 13A-13B schematically illustrate a container system 100M that includes the cooling system 200M. The container system 100M can include the vessel 120 (as described above). Some of the features of the cooling system 200M, which optionally serves as part of the lid L that selectively seals the vessel 120, are similar to features in the cooling system 200 in FIGS. 1A-1D. Thus, references numerals used to designate the various components of the cooling system 200M are similar to those used for identifying the corresponding components of the cooling system 200 in FIGS. 1A-1D, except that an "M" is used. Therefore,

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the structure and description for said similar components of the cooling system 200 in FIGS. 1A-1D are understood to also apply to the corresponding components of the cooling system 200M in FIGS. 13A-13B, except as described below.

With reference to FIGS. 13A-13B, the cooling system 200M can include a cold side heat sink 210M in thermal communication with a thermoelectric element (TEC) 220M and can selectively be in thermal communication with the chamber 126 of the vessel. Optionally, the cooling system 200M can include a fan 216M selectively operable to draw air from the chamber 126 into contact with the cold side heat sink 210M. Optionally, cooling system 200M can include an insulator member 246M selectively movable (e.g., slidable) between one or more positions. As shown in FIGS. 13A-13B, the insulator member 246M can be disposed adjacent or in communication with the chamber 126.

With reference to FIG. 13A, when the cooling system 200M is operated in a cooling state, the insulator member 246M is disposed at least partially apart (e.g., laterally apart) relative to the cold side heat sink 210M and fan 216M. The TEC 220M is selectively operated to draw heat from the cold side heat sink 210M and transfer it to the hot side heat sink 230M. Optionally, a fan 280M is selectively operable to dissipate heat from the hot side heat sink 230M, thereby allowing the TEC 220M to draw further heat from the chamber 126 via the cold side heat sink 210M.

With reference to FIG. 13B, when the cooling system 200M is operated in an insulating stage, the insulator member 246M is moved (e.g., slid) into a position adjacent to the cold side heat sink 210M so as to be disposed between the cold side heat sink 210M and the chamber 126, thereby blocking air flow to the cold side heat sink 210M (e.g., thermally disconnecting the cold side heat sink 210M from the chamber 126) to thereby inhibit heat transfer to and from the chamber 126 (e.g., to maintain the chamber 126 in an insulated state).

The insulator member 246M can be moved between the position in the cooling state (see FIG. 13A) and the position in the insulating stage (see FIG. 13B) using any suitable mechanism (e.g., electric motor, solenoid motor, a pneumatic or electromechanical system actuating a piston attached to the insulator member 246M, etc.). Though the insulator member 246M is shown in FIGS. 13A-13B as sliding between said positions, in another implementation, the insulator member 246M can rotate between the cooling stage position and the insulating stage position.

FIGS. 14A-14B schematically illustrate a container system 100N that includes the cooling system 200N. The container system 100N can include the vessel 120 (as described above). Some of the features of the cooling system 200N, which optionally serves as part of the lid L that selectively seals the vessel 120, are similar to features in the cooling system 200M in FIGS. 13A-13B. Thus, references numerals used to designate the various components of the cooling system 200N are similar to those used for identifying the corresponding components of the cooling system 200M in FIGS. 13A-13B, except that an "N" is used. Therefore, the structure and description for said similar components of the cooling system 200M in FIGS. 13A-13B are understood to also apply to the corresponding components of the cooling system 200N in FIGS. 14A-14B, except as described below.

With reference to FIGS. 14A-14B, the cooling system 200N can include a cold side heat sink 210N in thermal communication with a thermoelectric element (TEC) 220N and can selectively be in thermal communication with the chamber 126 of the vessel 120. Optionally, the cooling

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system 200N can include a fan 216N selectively operable to draw air from the chamber 126 into contact with the cold side heat sink 210N via openings 132N, 134N and cavities or chambers 213N, 214N. Optionally, cooling system 200N can include insulator members 246N, 247N selectively movable (e.g., pivotable) between one or more positions relative to the openings 134N, 132N, respectively. As shown in FIGS. 14A-14B, the insulator member 246N can be disposed adjacent or in communication with the chamber 126 and be movable to selectively allow and disallow airflow through the opening 134N, and the insulator member 247N can be disposed in the chamber 214N and be movable to selectively allow and disallow airflow through the opening 132N.

With reference to FIG. 14A, when the cooling system 200N is operated in a cooling state, the insulator members 246N, 247N are disposed at least partially apart from the openings 134N, 132N, respectively, allowing air flow from the chamber 126 through the openings 132N, 134N and cavities 213N, 214N. Optionally, the fan 216N can be operated to draw said airflow from the chamber 126, through the opening 132N into the chamber 214N and over the cold side heat sink 210N, then through the chamber 213N and opening 134N and back to the chamber 126. The TEC 220N is selectively operated to draw heat from the cold side heat sink 210N and transfer it to the hot side heat sink 230N. Optionally, a fan 280N is selectively operable to dissipate heat from the hot side heat sink 230N, thereby allowing the TEC 220N to draw further heat from the chamber 126 via the cold side heat sink 210N.

With reference to FIG. 14B, when the cooling system 200N is operated in an insulating stage, the insulator members 246N, 247N are moved (e.g., pivoted) into a position adjacent to the openings 134N, 132N, respectively to close said openings, thereby blocking air flow to the cold side heat sink 210N (e.g., thermally disconnecting the cold side heat sink 210N from the chamber 126) to thereby inhibit heat transfer to and from the chamber 126 (e.g., to maintain the chamber 126 in an insulated state).

The insulator members 246N, 247N can be moved between the position in the cooling state (see FIG. 14A) and the position in the insulating stage (see FIG. 14B) using any suitable mechanism (e.g., electric motor, solenoid motor, etc.). Optionally, the insulator members 246N, 247N are spring loaded into the closed position (e.g., adjacent the openings 134N, 132N), such that the insulator members 246N, 247N are pivoted to the open position (see FIG. 14A) automatically with an increase in air pressure generated by the operation of the fan 216N. Though the insulator members 246N, 247N are shown in FIGS. 14A-14B as pivoting between said positions, in another implementation, the insulator members 246N, 247N can slide or translate between the cooling stage position and the insulating stage position.

FIGS. 15A-15B schematically illustrate a container system 100P that includes the cooling system 200P. The container system 100P can include the vessel 120 (as described above). Some of the features of the cooling system 200P, which optionally serves as part of the lid L that selectively seals the vessel 120, are similar to features in the cooling system 200M in FIGS. 13A-13B. Thus, references numerals used to designate the various components of the cooling system 200P are similar to those used for identifying the corresponding components of the cooling system 200M in FIGS. 13A-13B, except that an "P" is used. Therefore, the structure and description for said similar components of the cooling system 200M in FIGS. 13A-13B are understood to

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also apply to the corresponding components of the cooling system 200P in FIGS. 15A-15B, except as described below.

With reference to FIGS. 15A-15B, the cooling system 200P can include a cold side heat sink 210P in thermal communication with a thermoelectric element (TEC) 220P and can selectively be in thermal communication with the chamber 126 of the vessel 120. Optionally, the cooling system 200P can include a fan 216P selectively operable to draw air from the chamber 126 into contact with the cold side heat sink 210P. Optionally, cooling system 200P can include insulator members 246P, 247P selectively movable (e.g., slidable) between one or more positions relative to the cold side heat sink 210P.

With reference to FIG. 15A, when the cooling system 200P is operated in a cooling state, the insulator members 246P, 247P are disposed at least partially apart from the cold side heat sink 210P, allowing air flow from the chamber 126 to contact (e.g., be cooled by) the cold side heat sink 210P. Optionally, the fan 216P can be operated to draw said airflow from the chamber 126 and over the cold side heat sink 210P. The TEC 220P is selectively operated to draw heat from the cold side heat sink 210P and transfer it to the hot side heat sink 230P. Optionally, a fan 280P is selectively operable to dissipate heat from the hot side heat sink 230P, thereby allowing the TEC 220P to draw further heat from the chamber 126 via the cold side heat sink 210P.

With reference to FIG. 15B, when the cooling system 200P is operated in an insulating stage, the insulator members 246P, 247P are moved (e.g., slid) into a position between the cold side heat sink 210P and the chamber 126, thereby blocking air flow to the cold side heat sink 210P (e.g., thermally disconnecting the cold side heat sink 210P from the chamber 126) to thereby inhibit heat transfer to and from the chamber 126 (e.g., to maintain the chamber 126 in an insulated state).

The insulator members 246P, 247P can be moved between the position in the cooling state (see FIG. 15A) and the position in the insulating stage (see FIG. 15B) using any suitable mechanism (e.g., electric motor, solenoid motor, etc.). Though the insulator members 246P, 247P are shown in FIGS. 15A-15B as sliding between said positions, in another implementation, the insulator members 246P, 247P can pivot between the cooling stage position and the insulating stage position.

FIGS. 16A-16B schematically illustrate a container system 100Q that includes the cooling system 200Q. The container system 100Q can include the vessel 120 (as described above). Some of the features of the cooling system 200Q, which optionally serves as part of the lid L that selectively seals the vessel 120, are similar to features in the cooling system 200M in FIGS. 13A-13B. Thus, references numerals used to designate the various components of the cooling system 200Q are similar to those used for identifying the corresponding components of the cooling system 200M in FIGS. 13A-13B, except that an "Q" is used. Therefore, the structure and description for said similar components of the cooling system 200M in FIGS. 13A-13B are understood to also apply to the corresponding components of the cooling system 200Q in FIGS. 16A-16B, except as described below.

With reference to FIGS. 16A-16B, the cooling system 200Q can include a cold side heat sink 210Q in thermal communication with a thermoelectric element (TEC) 220Q and can selectively be in thermal communication with the chamber 126 of the vessel 120. Optionally, the cooling system 200Q can include a fan 216Q selectively operable to draw air from the chamber 126 into contact with the cold

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side heat sink **210Q**. Optionally, the cooling system **200Q** can include an expandable members **246Q** selectively movable between A deflated state and an expanded state relative to the cold side heat sink **210P**.

With reference to FIG. **16A**, when the cooling system **200Q** is operated in a cooling state, the expandable member **246Q** is in the deflated state, allowing air flow from the chamber **126** to contact (e.g., be cooled by) the cold side heat sink **210Q**. Optionally, the fan **216Q** can be operated to draw said airflow from the chamber **126** and over the cold side heat sink **210Q**. The TEC **220Q** is selectively operated to draw heat from the cold side heat sink **210Q** and transfer it to the hot side heat sink **230Q**. Optionally, a fan **280Q** is selectively operable to dissipate heat from the hot side heat sink **230Q**, thereby allowing the TEC **220Q** to draw further heat from the chamber **126** via the cold side heat sink **210Q**.

With reference to FIG. **16B**, when the cooling system **200Q** is operated in an insulating stage, the expandable member **246Q** is moved into the expanded state so that the expandable member **246Q** is between the cold side heat sink **210Q** and the chamber **126**, thereby blocking air flow to the cold side heat sink **210Q** (e.g., thermally disconnecting the cold side heat sink **210Q** from the chamber **126**) to thereby inhibit heat transfer to and from the chamber **126** (e.g., to maintain the chamber **126** in an insulated state).

The expandable member **246Q** is optionally disposed or housed in a cavity or chamber **242Q** defined in the insulator member **240Q**. Optionally, the expandable member **246Q** is part of a pneumatic system and filled with a gas (e.g., air) to move it into the expanded state. In another implementation, the expandable member **246Q** is part of a hydraulic system and filled with a liquid (e.g., water) to move it into the expanded state.

FIGS. **17A-17B** schematically illustrate a container system **100R** that includes the cooling system **200R**. The container system **100R** can include the vessel **120** (as described above). Some of the features of the cooling system **200R**, which optionally serves as part of the lid **L** that selectively seals the vessel **120**, are similar to features in the cooling system **200M** in FIGS. **13A-13B**. Thus, references numerals used to designate the various components of the cooling system **200R** are similar to those used for identifying the corresponding components of the cooling system **200M** in FIGS. **13A-13B**, except that an "R" is used. Therefore, the structure and description for said similar components of the cooling system **200M** in FIGS. **13A-13B** are understood to also apply to the corresponding components of the cooling system **200R** in FIGS. **17A-17B**, except as described below.

With reference to FIGS. **17A-17B**, the cooling system **200R** can include a cold side heat sink **210R** in thermal communication with a thermoelectric element (TEC) **220R** and can selectively be in thermal communication with the chamber **126** of the vessel. Optionally, the cooling system **200** can include a fan **216R** selectively operable to draw air from the chamber **126** into contact with the cold side heat sink **210R**. Optionally, cooling system **200R** can include an insulator element **246R** selectively movable (e.g., pivotable) between one or more positions. As shown in FIGS. **17A-17B**, the insulator element **246R** can be disposed in a cavity or chamber **242R** defined in the insulator member **240R**.

With reference to FIG. **17A**, when the cooling system **200R** is operated in a cooling state, the insulator element **246R** is disposed relative to the cold side heat sink **210R** so as to allow air flow through the chamber **242R** from the chamber **126** to the cold side heat sink **210R**. Optionally, the fan **216R** is selectively operated to draw air from the

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chamber **126** into contact with the cold side heat sink **210R** (e.g., to cool said air flow and return it to the chamber **126**). The TEC **220R** is selectively operated to draw heat from the cold side heat sink **210R** and transfer it to the hot side heat sink **230R**. Optionally, a fan **280R** is selectively operable to dissipate heat from the hot side heat sink **230R**, thereby allowing the TEC **220R** to draw further heat from the chamber **126** via the cold side heat sink **210R**.

With reference to FIG. **17B**, when the cooling system **200R** is operated in an insulating stage, the insulator element **246R** is moved (e.g., rotated, pivoted) into a position relative to the cold side heat sink **210P** so as to close off the chamber **242R**, thereby blocking air flow from the chamber **126** to the cold side heat sink **210R** (e.g., thermally disconnecting the cold side heat sink **210R** from the chamber **126**) to thereby inhibit heat transfer to and from the chamber **126** (e.g., to maintain the chamber **126** in an insulated state).

The insulator element **246R** can be moved between the position in the cooling state (see FIG. **17A**) and the position in the insulating stage (see FIG. **17B**) using any suitable mechanism (e.g., electric motor, solenoid motor, etc.).

FIG. **18A** is a schematic view of a portion of a cooling system **200S**. The cooling system **200S** is similar to the cooling systems disclosed herein, such as cooling systems **200-200X**, except as described below.

As shown in FIG. **18A**, in the cooling system **200S**, the fan **280S** has air intake **I** that is generally vertical and air exhaust **E** that is generally horizontal, so that the air flows generally horizontally over one or more heat sink surfaces, such as surfaces of the hot side heat sink **230S**.

FIG. **18B** is a schematic view of a portion of a cooling system **200T**. The cooling system **200T** in a cylindrical container **100T** has a fan **280T** that optionally blows air over a heat sink **230T**. Optionally, the cooling system **200T** has a heat pipe **132T** in thermal communication with another portion of the container **100T** via end portion **134T** of heat pipe **132T**, allowing the fan **280T** and heat sink **230T** to remove heat from said portions via the heat pipe **132T**.

FIG. **18C** is a schematic view of a coupling mechanism **30A** for coupling the lid **L** and the vessel **120** for one or more implementations of the container system **100-100X** disclosed herein. In the illustrated embodiment, the lid **L** can be connected to one or more portions of the vessel **120** via a hinge that allows the lid **L** to be selectively moved between an open position (see FIG. **18C**) to allow access to the chamber **126**, and a closed position to disallow access to the chamber **126**.

FIG. **18D** is a schematic view of another embodiment of a coupling mechanism **30B** between the lid **L** and the vessel **120** of the container system **100-100X**. In the illustrated embodiment, the lid **L** can have one or more electrical connectors **31B** that communicate with one or more electrical contacts **32B** on the vessel **120** when the lid **L** is coupled to the vessel **120**, thereby allowing operation of the fan **280**, TEC **220**, etc. that are optionally in the lid **L**. Optionally, one of the electrical connectors **31B** and electrical contacts **32B** can be contact pins (e.g., Pogo pins) and the other of the electrical connectors **31B** and electrical contacts **32B** can be electrical contact pads (e.g., circular contacts) that optionally allows connection of the lid **L** to the vessel **120** irrespective of the angular orientation of the lid **L** relative to the vessel **120**.

FIG. **18E** shows a schematic view of an embodiment of a vessel for the cooler container system, such as the cooler container systems **100-100X** disclosed herein. In the illustrated embodiment, the vessel **120** has electronics (e.g., one or more optional batteries, circuitry, optional transceiver)

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housed in a compartment E on a bottom of the vessel **120**. The electronics can communicate or connect to the fan **280**, TEC **220** or other components in the lid L via electrical connections (such as those shown and described in connection with FIG. **18D**), or via wires that extend through the hinge **30A** (such as that shown in FIG. **18C**).

FIG. **18F** shows a schematic view of an embodiment of a vessel for the cooler container system, such as the cooler container systems **100-100X** disclosed herein. In the illustrated embodiment, the vessel **120** has electronics (e.g., one or more optional batteries, circuitry, optional transceiver) housed in a compartment E on a side of the vessel **120**. The electronics can communicate or connect to the fan **280**, TEC **220** or other components in the lid L via electrical connections (such as those shown and described in connection with FIG. **18D**), or via wires that extend through the hinge **30A** (such as that shown in FIG. **18C**).

FIG. **19** shows another embodiment of a container system **100U** having a cooling system **200U**. The container system **100U** includes a vessel **120** with a chamber **126**. The vessel **120** can be double walled, as shown, with the space between the inner wall and outer wall under vacuum. A TEC **220U** can be in contact with a cold delivery member (e.g., stud) **225U**, which is in contact with the inner wall and can selectively thermally communicate with a hot side heat sink **230U**. The cold delivery member **225U** can be small relative to the size of the vessel **120**, and can extend through an opening **122U** in the vessel **120**. Optionally, the container system **100U** can have a pump P operable to pull a vacuum out from the cavity between the inner and outer walls of the vessel **120**.

FIGS. **20-31** show a container system **100'** that includes a cooling system **200'**. The container system **100'** has a body **120'** that extends from a proximal end **122'** to a distal end **124'** and has an opening **123'** selectively closed by a lid L". The body **120'** can optionally be box shaped. The lid L" can optionally be connected to the proximal end **122'** of the body **120'** by a hinge **130'** on one side of the body **120'**. A groove or handle **106'** can be defined on an opposite side of the body **120'** (e.g., at least partially defined by the lid L" and/or body **120'**), allowing a user to lift the lid L" to access a chamber **126'** in the container **100'**. Optionally, one or both of the lid L" and proximal end **122'** of the body **120'** can have one or more magnets (e.g., electromagnets, permanent magnets) that can apply a magnetic force between the lid L' and body **120'** to maintain the lid L' in a closed state over the body **120'** until a user overcomes said magnetic force to lift the lid L'. However, other suitable fasteners can be used to retain the lid L' in a closed position over the body **120'**.

With reference to FIG. **27**, the body **120'** can include an outer wall **121'** and optionally include an inner wall **126A'** spaced apart from the outer wall **121'** to define a gap (e.g., annular gap, annular chamber) **128'** therebetween. Optionally, the inner wall **126A'** can be suspended relative to the outer wall **121'** in a way that provides the inner wall **126A'** with shock absorption (e.g., energy dissipation). For example, one or more springs can be disposed between the inner wall **126A'** and the outer wall **121'** that provide said shock absorption. Optionally, the container **100'** includes one or more accelerometers (e.g., in communication with the circuitry of the container **100'**) that sense motion (e.g., acceleration) of the container **100'**. Optionally, the one or more accelerometers communicate sensed motion information to the circuitry, and the circuitry optionally operates one or more components to adjust a shock absorption provided by the inner wall **126A'** (e.g., by tuning a shock absorption property of one or more springs, such as magnetorheological

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(MRE) springs) that support the inner surface **126A'**. In one implementation, the container **100'** can include a plastic and/or rubber structure in the gap **128'** between the inner wall **126A'** and the outer wall **121'** to aid in providing such shock absorption.

The gap **128'** can optionally be filled with an insulative material (e.g., foam). In another implementation, the gap **128'** can be under vacuum. In still another implementation, the gap **128'** can be filled with a gas (e.g., air). Optionally, the inner wall **126A'** can be made of metal. Optionally, the outer wall **121'** can be made of plastic. In another implementation, the outer wall **121'** and the inner wall **126A'** are optionally made of the same material.

With continued reference to FIG. **27**, the cooling system **200'** can optionally be housed in a cavity **127'** disposed between a base **125'** of the container body **120'** and the inner wall **126A'**. The cooling system **200'** can optionally include one or more thermoelectric elements (TEC) (e.g., Peltier elements) **220'** in thermal communication with (e.g., in direct contact with) the inner wall **126A'**. In one implementation, the cooling system **200'** has only one TEC **220'**. The one or more TECs **220'** can optionally be in thermal communication with one or more heat sinks **230'**. Optionally, the one or more heat sinks **230'** can be a structure with a plurality of fins. Optionally, one or more fans **280'** can be in thermal communication with (e.g., in fluid communication with) the one or more heat sinks **230'**. The cooling system **200'** can optionally have one or more batteries **277'**, optionally have a converter **279'**, and optionally have a power button **290'**, that communicate with circuitry (e.g., on a printed circuit board **278'**) that controls the operation of the cooling system **200'**.

The optional batteries **277'** provide power to one or more of the circuitry, one of more fans **280'**, one or more TECs **220'**, and one or more sensors (described further below). Optionally, at least a portion of the body **120'** (e.g., a portion of the base **125'**) of the container **100'** is removable to access the one or more optional batteries **277'**. Optionally, the one or more optional batteries **277'** can be provided in a removable battery pack, which can readily be removed and replaced from the container **100'**. Optionally, the container **100'** can include an integrated adaptor and/or retractable cable to allow connection of the container **100'** to a power source (e.g., wall outlet, vehicle power connector) to one or both of power the cooling system **200'** directly and charge the one or more optional batteries **277'**.

With reference to FIGS. **22-23** and **27**, the container system **100'** can have two or more handles **300** on opposite sides of the body **120'** to which a strap **400** can be removably coupled (see FIG. **24**) to facilitate transportation of the container **100'**. For example, the user can carry the container **100'** by placing the strap **400** over their shoulder. Optionally, the strap **400** is adjustable in length. Optionally, the strap **400** can be used to secure the container system **100'** to a vehicle (e.g., moped, bicycle, motorcycle, etc.) for transportation. Optionally, the one or more handles **300** can be movable relative to the outer surface **121'** of the body **120'**. For example, the handles **300** can be selectively movable between a retracted position (see e.g., FIG. **22**) and an extended position (see e.g., FIG. **23**). Optionally, the handles **300** can be mounted within the body **120'** in a spring-loaded manner and be actuated in a push-to-open and push-to-close manner.

With reference to FIGS. **26-27**, the body **120'** can include one or more sets of vents on a surface thereof to allow air flow into and out of the body **120'**. For example, the body **120'** can have one or more vents **203'** defined on the bottom

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portion of the base 125' of the body 120' and can optionally have one or more vents 205' at one or both ends of the base 125'. Optionally, the vents 203' can be air intake vents, and the vents 205' can be air exhaust vents.

With reference to FIG. 25A, the chamber 126 is optionally sized to receive and hold one or more trays 500 therein (e.g., hold a plurality of trays in a stacked configuration). Each tray 500 optionally has a plurality of receptacles 510, where each receptacle 510 is sized to receive a container (e.g., a vial) 520 therein. The container 520 can optionally hold a liquid (e.g., a medication, such as insulin or a vaccine). Optionally, the tray 500 (e.g., the receptacle 510) can releasably lock the containers 520 therein (e.g., lock the containers 520 in the receptacles 510) to inhibit movement, dislodgement and/or damage to the containers 520 during transit of the container system 100'. Optionally, the tray 500 can have one or more handles 530 to facilitate carrying of the tray 500 and/or pulling the tray 500 out of the chamber 126 or placing the tray 500 in the chamber 126. Optionally, the one or more handles 530 are movable between a retracted position (see FIG. 28) and an extended position (see FIG. 26). Optionally, the one or more handles 530 can be mounted within the tray 500 in a spring-loaded manner and be actuated in a push-to-extend and push-to-retract manner. In another implementation, the one or more handles 530 are fixed (e.g., not movable between a retracted and an extended position).

With reference to FIGS. 25B-25D, the tray 500 can include an outer tray 502 that removably receives one or more inner trays 504, 504', where different inner trays 504, 504' can have a different number and/or arrangement of the plurality of receptacles 510 that receive the one or more containers (e.g., vials) 520 therein, thereby advantageously allowing the container 100' to accommodate different number of containers 520 (e.g., for different medications, etc.). In one implementation, shown in FIG. 25C, the inner tray 504 can have a relatively smaller number of receptacles 510 (e.g., sixteen), for example to accommodate relatively larger sized containers 520 (e.g., vials of medicine, such as vaccines and insulin, biological fluid, such as blood, etc.), and in another implementation, shown in FIG. 25D, the inner tray 504' can have a relatively larger number of receptacles 510 (e.g., thirty-eight), for example to accommodate relatively smaller sized containers 520 (e.g., vials of medicine, biological fluid, such as blood, etc.).

With reference to FIG. 28, the container system 100' can have one or more lighting elements 550 that can advantageously facilitate users to readily see the contents in the chamber 126' when in a dark environment (e.g., outdoors at night, in a rural or remote environment, such as mountainous, desert or rainforest region). In one implementation, the one or more lighting elements can be one or more light strips (e.g., LED strips) disposed at least partially on one or more surfaces of the chamber 126' (e.g., embedded in a surface of the chamber 126', such as near the proximal opening of the chamber 126'). Optionally, the one or more lighting elements 550 can automatically illuminate when the lid L" is opened. Once illuminated, the one or more lighting elements 550 can optionally automatically shut off when the lid L" is closed over the chamber 126'. Optionally, the one or more lighting elements 550 can communicate with circuitry of the container 100', which can also communicate with a light sensor of the container 100' (e.g., a light sensor disposed on an outer surface of the container 100'). The light sensor can generate a signal when the sensed light is below a predetermined level (e.g., when container 100' in a building without power or is in the dark, etc.) and communicate said

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signal to the circuitry, and the circuitry can operate the one or more lighting elements 550 upon receipt of such signal (e.g., and upon receipt of the signal indicating the lid L" is open).

The container system 100' can have a housing with one of a plurality of colors. Such different color housings can optionally be used with different types of contents (e.g., medicines, biological fluids), allowing a user to readily identify the contents of the container 100' by its housing color. Optionally, such different colors can aid users in distinguishing different containers 100' in their possession/use without having to open the containers 100' to check their contents.

With reference to FIGS. 29A-29C, the container 100' can optionally communicate (e.g., one-way communication, two-way communication) with one or more remote electronic device (e.g., mobile phone, tablet computer, desktop computer, remote server) 600, via one or both of a wired or wireless connection (e.g., 802.11b, 802.11a, 802.11g, 802.11n standards, etc.). Optionally, the container 100' can communicate with the remote electronic device 600 via an app (mobile application software) that is optionally downloaded (e.g., from the cloud) onto the remote electronic device 600. The app can provide one or more graphical user interface screens 610A, 610B, 610C via which the remote electronic device 600 can display one or more data received from the container 100'. Optionally, a user can provide instructions to the container 100' via one or more of the graphical user interface screens 610A, 610B, 610C on the remote electronic device 600.

In one implementation, the graphical user interface (GUI) screen 610A can provide one or more temperature presets corresponding to one or more particular medications (e.g., epinephrine/adrenaline for allergic reactions, insulin, vaccines, etc.). The GUI screen 610A can optionally allow the turning on and off of the cooling system 200'. The GUI screen 610A can optionally allow the setting of the control temperature to which the chamber 126' in the container 100' is cooled by the cooling system 200'.

In another implementation, the graphical user interface (GUI) screen 610B can provide a dashboard display of one or more parameters of the container 100' (e.g., ambient temperature, internal temperature in the chamber 126', temperature of the heat sink 230', temperature of the battery 277, etc.). The GUI screen 610B can optionally provide an indication (e.g., display) of power supply left in the one or more batteries 277 (e.g., % of life left, time remaining before battery power drains completely). Optionally, the GUI screen 610B can also include information (e.g., a display) of how many of the receptacles 510 in the tray 500 are occupied (e.g., by containers 520). Optionally, the GUI screen 610B can also include information on the contents of the container 100' (e.g., medication type or disease medication is meant to treat), information on the destination for the container 100' and/or information (e.g., name, identification no.) for the individual assigned to the container 100'.

In another implementation, the GUI screen 610C can include a list of notifications provided to the user of the container 100', including alerts on battery power available, alerts on ambient temperature effect on operation of container 100', alerts on a temperature of a heat sink of the container 100', alert on temperature of the chamber 126, 126', 126V, alert on low air flow through the intake vent 203', 203", 203V and/or exhaust vent 205', 205", 205V indicating they may be blocked/clogged, etc. One of skill in the art will recognize that the app can provide the plurality

of GUI screens 610A, 610B, 610C to the user, allowing the user to swipe between the different screens.

Optionally, as discussed further below, the container 100' can communicate information, such as temperature history of the chamber 126' and/or first heat sink 210 that generally corresponds to a temperature of the containers 520, 520V (e.g., medicine containers, vials, cartridges, injectors), power level history of the batteries 277, ambient temperature history, etc. to the cloud (e.g., on a periodic basis, such as every hour; on a continuous basis in real time, etc.) to one or more of a) an RFID tag on the container system 100, 100', 100", 100B-100V that can later be read (e.g., at the delivery location), b) to a remote electronic device (e.g., a mobile electronic device such as a smartphone or tablet computer or laptop computer or desktop computer), including wirelessly (e.g., via WiFi 802.11, BLUETOOTH®, or other RF communication), and c) to the cloud (e.g., to a cloud-based data storage system or server) including wirelessly (e.g., via WiFi 802.11, BLUETOOTH®, or other RF communication). Such communication can occur on a periodic basis (e.g., every hour; on a continuous basis in real time, etc.). Once stored on the RFID tag or remote electronic device or cloud, such information can be accessed via one or more remote electronic devices (e.g., via a dashboard on a smart phone, tablet computer, laptop computer, desktop computer, etc.). Additionally, or alternatively, the container system 100, 100', 100", 100B-100V can store in a memory (e.g., part of the electronics in the container system 100, 100', 100", 100B-100V) information, such as temperature history of the chamber 126, 126', 126V, temperature history of the first heat sink 210, 210B-210V, power level history of the batteries 277, ambient temperature history, etc., which can be accessed from the container system 100, 100', 100", 100B-100V by the user via a wired or wireless connection (e.g., via the remote electronic device 600).

With reference to FIG. 30, the body 120' of the container 100' can have a visual display 140 on an outer surface 121' of the body 120'. The visual display 140' can optionally display one or more of the temperature in the chamber 126', the ambient temperature, a charge level or percentage for the one or more batteries 277, and amount of time left before recharging of the batteries 277 is needed. The visual display 140' can include a user interface (e.g., pressure sensitive buttons, capacitance touch buttons, etc.) to adjust (up or down) the temperature preset at which the cooling system 200' is to cool the chamber 126' to. Accordingly, the operation of the container 100' (e.g., of the cooling system 200') can be selected via the visual display and user interface 140' on a surface of the container 100'. Optionally, the visual display 140' can include one or more hidden-til-lit LEDs. Optionally, the visual display 140' can include an electronic ink (e-ink) display. In one implementation, the container 100' can optionally include a hidden-til-lit LED 142' (see FIG. 34) that can selectively illuminate (e.g., to indicate one or more operating functions of the container 100', such as to indicate that the cooling system 200' is in operation). The LED 142' can optionally be a multi-color LED selectively operable to indicate one or more operating conditions of the container 100' (e.g., green if normal operation, red if abnormal operation, such as low battery charge or inadequate cooling for sensed ambient temperature, etc.).

With reference to FIG. 31, the container 100' can include one or more security features that allow opening of the container 100' only when the security feature(s) are met. In one implementation, the container 100' can include a keypad 150 via which an access code can be entered to unlock the lid L" to allow access to the chamber 126' when it matches

the access code key programmed to the container 100'. In another implementation, the container 100' can additionally or alternatively have a biometric sensor 150', via which the user can provide a biometric identification (e.g., fingerprint) that will unlock the lid L" and allow access to the chamber 126' when it matches the biometric key programmed to the container 100'. Optionally, the container 100' remains locked until it reaches its destination, at which point the access code and/or biometric identification can be utilized to unlock the container 100' to access the contents (e.g., medication) in the chamber 126'.

The container 100' can optionally be powered in a variety of ways. In one implementation, the container system 100' is powered using 12 VDC power (e.g., from one or more batteries 277'). In another implementation, the container system 100' is powered using 120 VAC or 240 VAC power. In another implementation, the cooling system 200' can be powered via solar power. For example, the container 100' can be removably connected to one or more solar panels so that electricity generated by the solar panels is transferred to the container 100', where circuitry of the container 100' optionally charges the one or more batteries 277 with the solar power. In another implementation, the solar power from said one or more solar panels directly operates the cooling system 200' (e.g., where batteries 277 are excluded from the container 100'). The circuitry in the container 100' can include a surge protector to inhibit damage to the electronics in the container 100' from a power surge.

In operation, the cooling system 200' can optionally be actuated by pressing the power button 290. Optionally, the cooling system 200' can additionally (or alternatively) be actuated remotely (e.g., wirelessly) via a remote electronic device, such as a mobile phone, tablet computer, laptop computer, etc. that wirelessly communicates with the cooling system 200' (e.g., with a receiver or transceiver of the circuitry). The chamber 126' can be cooled to a predetermined and/or a user selected temperature or temperature range. The user selected temperature or temperature range can be selected via a user interface on the container 100' and/or via the remote electronic device.

The circuitry optionally operates the one or more TECs 220' so that the side of the one or more TECs 220' adjacent the inner wall 126A' is cooled and so that the side of the one or more TECs 220' adjacent the one or more heat sinks 230' is heated. The TECs 220' thereby cool the inner wall 126A' and thereby cools the chamber 126' and the contents (e.g., tray 500 with containers (e.g., vials) 520 therein). Though not shown in the drawings, one or more sensors (e.g., temperature sensors) are in thermal communication with the inner wall 126A' and/or the chamber 126' and communicate information to the circuitry indicative of the sensed temperature. The circuitry operates one or more of the TECs 220' and one or more fans 280' based at least in part on the sensed temperature information to cool the chamber 126' to the predetermined temperature and/or user selected temperature. The circuitry operates the one or more fans 280' to flow air (e.g., received via the intake vents 203') over the one or more heat sinks 230' to dissipate heat therefrom, thereby allowing the one or more heat sinks 230' to draw more heat from the one or more TECs 220', which in turn allows the one or more TECs 220' to draw more heat from (i.e., cool) the inner wall 126A' to thereby further cool the chamber 126'. Said air flow, once it passes over the one or more heat sinks 230', is exhausted from the body 120' via the exhaust vents 205'.

FIGS. 32-34 schematically illustrate a container 100" that includes a cooling system 200". The container system 100"

can include a vessel body 120 removably sealed by a lid L". Some of the features of the container 100" and cooling system 200" are similar to the features of the container 100' and cooling system 200' in FIGS. 20-31. Thus, reference numerals used to designate the various components of the container 100" and cooling system 200" are similar to those used for identifying the corresponding components of the cooling system 200' in FIGS. 20-31, except that an "" is used. Therefore, the structure and description for said components of the cooling system 200' of FIGS. 20-31 are understood to also apply to the corresponding components of the container 100" and cooling system 200" in FIGS. 32-34, except as described below. FIG. 33A is a front view of the container 100" in FIG. 32. FIG. 33B is a smaller version of the container 100" and optionally has the same internal components as shown for the container in FIG. 33A (e.g., as shown in FIGS. 37-39).

With reference to FIGS. 32-34, the container 100" differs from the container 100' in that the container 100" has a generally cylindrical or tube-like body 120" with a generally cylindrical outer surface 121". The container 100" can have similar internal components as the container 100', such as a chamber 126" defined by an inner wall 126A", TEC 220", heat sink 230", one or more fans 280", one or more optional batteries 277", converter 279" and power button 290". The lid L" can have one or more vents 203", 205" defined therein, and operate in a similar manner as the vents 203', 205' described above. The container 100" can have a variety of sizes (see FIGS. 33A-33B) that can accommodate a different number and/or size of containers 520". The container 100" and cooling system 200" operate in a similar manner described above for the container 100' and cooling system 200'.

The container 100" can optionally include a display similar to the display 140' described above for the container 100' (e.g., that displays one or more of the temperature in the chamber 126", the ambient temperature, a charge level or percentage for the one or more batteries 277", and amount of time left before recharging of the batteries 277" is needed). The container 100" can optionally include a hidden-til-lit LED 142" (see FIG. 36) that can selectively illuminate (e.g., to indicate one or more operating functions of the container 100", such as to indicate that the cooling system 200" is in operation). The LED 142" can optionally be a multi-color LED selectively operable to indicate one or more operating conditions of the container 100" (e.g., green if normal operation, red if abnormal operation, such as low battery charge or inadequate cooling for sensed ambient temperature, etc.).

With reference to FIG. 34, the container 100" can be removably placed on a base 700", which can connect to a power source (e.g., wall outlet) via a cable 702". In one implementation, the base 700" directly powers the cooling system 200" of the container 100" (e.g., to cool the contents in the container 100") to the desired temperature (e.g., the temperature required by the medication, such as insulin, in the chamber 126" of the container 100"). In another implementation, the base 700" can additionally or alternatively charge the one or more optional batteries 277", so that the batteries 277" take over powering of the cooling system 200" when the container 100" is removed from the base 700". Optionally, the vessel 120" of the container system 100" can have one or more electrical contacts EC1 (e.g., contact rings) that communicate with one or more electrical contacts EC2 (e.g., pogo pins) of the base 700" when the vessel 120" is placed on the base 700". In another implementation, the base 700" can transfer power to the vessel

120" of the container system 100" via inductive coupling (e.g., electromagnetic induction).

With reference to FIGS. 35A-35C, the container 100" can optionally communicate (e.g., one-way communication, two-way communication) with one or more remote electronic device (e.g., mobile phone, tablet computer, desktop computer) 600, via one or both of a wired or wireless connection. Optionally, the container 100" can communicate with the remote electronic device 600 via an app (mobile application software) that is optionally downloaded (e.g., from the cloud) onto the remote electronic device 600. The app can provide one or more graphical user interface screens 610A", 610B", 610C" via which the remote electronic device 600 can display one or more data received from the container 100". Optionally, a user can provide instructions to the container 100" via one or more of the graphical user interface screens 610A", 610B", 610C" on the remote electronic device 600.

In one implementation, the graphical user interface (GUI) screen 610A" can provide one or more temperature presets corresponding to one or more particular medications (e.g., insulin). The GUI 610A" can optionally allow the turning on and off of the cooling system 200". The GUI 610A" can optionally allow the setting of the control temperature to which the chamber 126" in the container 100" is cooled by the cooling system 200".

In another implementation, the graphical user interface (GUI) screen 610B" can provide a dashboard display of one or more parameters of the container 100" (e.g., ambient temperature, internal temperature in the chamber 126", etc.). The GUI screen 610B" can optionally provide an indication (e.g., display) of power supply left in the one or more batteries 277" (e.g., % of life left, time remaining before battery power drains completely). Optionally, the GUI screen 610B" can also include information (e.g., a display) of how many of the receptacles 510" in the tray 500" are occupied (e.g., by containers 520"). Optionally, the GUI screen 610B" can also include information on the contents of the container 100" (e.g., medication type or disease medication is meant to treat), information on the physician (e.g., name of doctor and contact phone no) and/or information (e.g., name, date of birth, medical record no.) for the individual assigned to the container 100".

In another implementation, the GUI screen 610C" can include a list of notifications provided to the user of the container 100", including alerts on battery power available, alerts on ambient temperature effect on operation of container 100", etc. One of skill in the art will recognize that the app can provide the plurality of GUI screens 610A", 610B", 610C" to the user, allowing the user to swipe between the different screens. Optionally, as discussed further below, the container 100" can communicate information, such as temperature history of the chamber 126", power level history of the batteries 277", ambient temperature history, etc. to the cloud (e.g., on a periodic basis, such as every hour; on a continuous basis in real time, etc.).

In some implementations, the container system 100, 100', 100", 100B-100X can include one or both of a radiofrequency identification (RFID) reader and a barcode reader. For example, the RFID reader and/or barcode reader can be disposed proximate (e.g., around) a rim of the chamber 126, 126', 126" to that it can read content units (e.g., vials, containers) placed into or removed from the chamber 126, 126', 126". The RFID reader or barcode reader can communicate data to the circuitry in the container system, which as discussed above, can optionally store such data in a memory or the container system and/or communicate such data to a

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separate or remote computing system, such as a remote computer server (e.g., accessible by a doctor treating the patient with the medication in the container), a mobile electronic device, such as a mobile phone or tablet computer. Such communication can optionally be in one or both of a wired manner (via a connector on the container body) or wireless manner (via a transmitter or transceiver of the container in communication with the circuitry of the container). Each of the contents placed in the chamber of the container (e.g., each medicine unit, such as each vial or container) optionally has an RFID tag or barcode that is read by the RFID reader or barcode reader as it is placed in and/or removed from the chamber of the container, thereby allowing the tracking of the contents of the container system **100**, **100'**, **100"**, **100B-100X**. Optionally, the container system (e.g., the RFID reader, barcode reader and/or circuitry) of the container system, send a notification (e.g., to a remote computer server, to one or more computing systems, to a mobile electronic device such as a smartphone or tablet computer or laptop computer or desktop computer) every time a medicine unit (e.g., vial, container) is placed into and/or removed from the chamber of the container system **100**, **100'**, **100"**, **100B-100X**.

In some implementations, the container system **100**, **100'**, **100"**, **100B-100X** can additionally or alternatively (to the RFID reader and/or barcode reader) include a proximity sensor, for example in the chamber **126**, **126'**, **126"** to advantageously track one or both of the insertion of and removal of content units (e.g., medicine units such as vials, containers, pills, etc.) from the container system. Such a proximity sensor can communicate with the circuitry of the container and advantageously facilitate tracking, for example, of the user taking medication in the container, or the frequency with which the user takes the medication. Optionally, operation of the proximity sensor can be triggered by a signal indicating the lid **L**, **L'**, **L"** has been opened. The proximity sensor can communicate data to the circuitry in the container system, which as discussed above, can optionally store such data in a memory or the container system and/or communicate such data to a separate or remote computing system, such as a remote computer server (e.g., accessible by a doctor treating the patient with the medication in the container), a mobile electronic device, such as a mobile phone or tablet computer. Such communication can optionally be in one or both of a wired manner (via a connector on the container body) or wireless manner (via a transmitter or transceiver of the container in communication with the circuitry of the container).

In some implementations, the container system **100**, **100'**, **100"**, **100B-100X** can additionally or alternatively (to the RFID reader and/or barcode reader) include a weight sensor, for example in the chamber **126**, **126'**, **126"** to advantageously track the removal of content units (e.g. medicine units such as vials, containers, pills, etc.) from the container system. Such a weight sensor can communicate with the circuitry of the container and advantageously facilitate tracking, for example, of the user taking medication in the container, or the frequency with which the user takes the medication. Optionally, operation of the weight sensor can be triggered by a signal indicating the lid **L**, **L'**, **L"** has been opened. The weight sensor can communicate data to the circuitry in the container system, which as discussed above, can optionally store such data in a memory or the container system and/or communicate such data to a separate or remote computing system, such as a remote computer server (e.g., accessible by a doctor treating the patient with the medication in the container), a mobile electronic device,

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such as a mobile phone or tablet computer. Such communication can optionally be in one or both of a wired manner (via a connector on the container body) or wireless manner (via a transmitter or transceiver of the container in communication with the circuitry of the container).

FIG. **36** shows a container system, such as the container systems **100**, **100'**, **100"**, **100A-100X** described herein, removably connectable to a battery pack **B** (e.g., a Dewalt battery pack), which can provide power to one or more electrical components (e.g., TEC, fan, circuitry, etc.) of the container systems or the cooling systems **200**, **200'**, **200"**, **200A-200T**. Optionally, the vessel **120** of the container system can have one or more electrical contacts **EC1** (e.g., contact rings) that communicate with one or more electrical contacts **EC2** (e.g., pogo pins) when the vessel **120** is placed on the battery pack **B**. In another implementation, the battery pack **B** can transfer power to the vessel **120** of the container system via inductive coupling (e.g., electromagnetic induction).

FIGS. **37-39** show a schematic cross-sectional view of a container system **100V** that includes a cooling system **200V**. Optionally, the container system **100V** has a container vessel **120V** that is optionally cylindrical and symmetrical about a longitudinal axis, and one of ordinary skill in the art will recognize that at least some of the features shown in cross-section in FIGS. **37-39** are defined by rotating them about the axis to define the features of the container **100V** and cooling system **200V**. Some of the features of the cooling system **200V**, which optionally serves as part of the lid **L"** that selectively seals the vessel **120V**, are similar to features in the cooling system **200M** in FIGS. **13A-13B**. Thus, reference numerals used to designate the various components of the cooling system **200V** are similar to those used for identifying the corresponding components of the cooling system **200M** in FIGS. **13A-13B**, except that an "V" is used. Therefore, the structure and description for said similar components of the cooling system **200M** in FIGS. **13A-13B** are understood to also apply to the corresponding components of the cooling system **200V** in FIGS. **37-39**, except as described below.

With reference to FIGS. **37-39**, the cooling system **200V** can include a heat sink (cold side heat sink) **210V** in thermal communication with a thermoelectric element (TEC) **220V** and can be in thermal communication with the chamber **126V** of the vessel **120V**. Optionally, the cooling system **200V** can include a fan **216V** selectively operable to draw air from the chamber **126V** into contact with the cold side heat sink **210V**. Optionally, cooling system **200V** can include an insulator member **270V** disposed between the heat sink **210V** and an optional lid top plate **202V**, where the lid top plate **202V** is disposed between the heat sink (hot side heat sink) **230V** and the insulator **270V**, the insulator **270V** disposed about the TEC **220V**. As shown in FIG. **39**, air flow **Fr** is drawn by the fan **216V** from the chamber **126V** and into contact with the heat sink (cold side heat sink) **210V** (e.g., to cool the air flow **Fr**), and then returned to the chamber **126V**. Optionally, the air flow **Fr** is returned via one or more openings **218V** in a cover plate **217V** located distally of the heat sink **210V** and fan **216V**.

With continued reference to FIGS. **37-39**, the TEC **220V** is selectively operated to draw heat from the heat sink (e.g., cold-side heat sink) **210V** and transfer it to the heat sink (hot-side heat sink) **230V**. A fan **280V** is selectively operable to dissipate heat from the heat sink **230V**, thereby allowing the TEC **220V** to draw further heat from the chamber **126V** via the heat sink **210V**. As shown in FIG. **40**, during operation of the fan **280V**, intake air flow **Fi** is drawn through one or

more openings **203V** in the lid cover **L'''** and over the heat sink **230V** (where the air flow removes heat from the heat sink **230V**), after which the exhaust air flow **Fe** flows out of one or more openings **205V** in the lid cover **L'''**. Optionally, both the fan **280V** and the fan **216V** are operated simultaneously. In another implementation, the fan **280V** and the fan **216V** are operated at different times (e.g., so that operation of the fan **216V** does not overlap with operation of the fan **280V**).

As shown in FIGS. **37-39**, the chamber **126V** optionally receives and holds one or more (e.g., a plurality of) trays **500V**, each tray **500V** supporting one or more (e.g., a plurality of) liquid containers **520V** (e.g., vials, such as vaccines, medications, etc.). The lid **L'''** can have a handle **400V** used to remove the lid **L'''** from the vessel **120V** to remove contents from the chamber **126V** or place contents in the chamber **126V** (e.g., remove the trays **500** via handle **530V**). The lid **L'''** can have a sealing gasket **G**, such as disposed circumferentially about the insulator **270V** to seal the lid **L'''** against the chamber **126V**. The inner wall **136V** of the vessel **120V** is spaced from the outer wall **121V** to define a gap (e.g., an annular gap) **128V** therebetween. Optionally, the gap **128V** can be under vacuum. Optionally, the inner wall **136V** defines at least a portion of an inner vessel **130V**. Optionally, the inner vessel **130V** is disposed on a bottom plate **272V**.

The bottom plate **272V** can be spaced from a bottom **275V** of the vessel **120V** to define a cavity **127V** therebetween. The cavity **127V** can optionally house one or more batteries **277V**, a printed circuit board (PCBA) **278V** and at least partially house a power button or switch **290V**. Optionally, the bottom **275V** defines at least a portion of an end cap **279V** attached to the outer wall **121V**. Optionally, the end cap **279V** is removable to access the electronics in the cavity **127V** (e.g., to replace the one or more batteries **277V**, perform maintenance on the electronics, such as the PCBA **278V**, etc.). The power button or switch **290V** is accessible by a user (e.g., can be pressed to turn on the cooling system **200V**, pressed to turn off the cooling system **200V**, pressed to pair the cooling system **200V** with a mobile electronic device, etc.). As shown in FIG. **37**, the power switch **290V** can be located generally at the center of the end cap **279V** (e.g., so that it aligns/extends along the longitudinal axis of the vessel **120V**).

The electronics (e.g., PCBA **278V**, batteries **277V**) can electrically communicate with the fans **280V**, **216V** and TEC **220V** in the lid **L'''** via one or more electrical contacts (e.g., electrical contact pads, Pogo pins) in the lid **L'''** that contact one or more electrical contacts (e.g., Pogo pins, electrical contact pads) in the portion of the vessel **120V** that engages the lid **L'''**, such as in a similar manner to that described above for FIG. **18D**.

FIG. **40** shows a block diagram of a communication system for (e.g., incorporated into) the devices described herein (e.g., the one or more container systems **100**, **100'**, **100''**, **100A-100X**). In the illustrated embodiment, circuitry EM can receive sensed information from one or more sensors **S1-Sn** (e.g., level sensors, volume sensors, temperature sensors, battery charge sensors, biometric sensors, load sensors, Global Positioning System or GPS sensors, radio-frequency identification or RFID reader, etc.). The circuitry EM can be housed in the container, such as in the vessel **120** (e.g., bottom of vessel **120**, side of vessel **120**, as discussed above) or in a lid **L** of the container. The circuitry EM can receive information from and/or transmit information (e.g., instructions) to one or more heating or cooling elements HC, such as the TEC **220**, **220'**, **220A-220X** (e.g., to operate each

of the heating or cooling elements in a heating mode and/or in a cooling mode, turn off, turn on, vary power output of, etc.) and optionally to one or more power storage devices PS (e.g., batteries, such as to charge the batteries or manage the power provided by the batteries to the one or more heating or cooling elements).

Optionally, the circuitry EM can include a wireless transmitter, receiver and/or transceiver to communicate with (e.g., transmit information, such as sensed temperature and/or position data, to and receive information, such as user instructions, from one or more of: a) a user interface UI1 on the unit (e.g., on the body of the vessel **120**), b) an electronic device ED (e.g., a mobile electronic device such as a mobile phone, PDA, tablet computer, laptop computer, electronic watch, a desktop computer, remote server), c) via the cloud CL, or d) via a wireless communication system such as WiFi and/or Bluetooth BT. The electronic device ED can have a user interface UI2, that can display information associated with the operation of the container system (such as the interfaces disclosed above, see FIGS. **29A-29C**, **35A-35C**), and that can receive information (e.g., instructions) from a user and communicate said information to the container system **100**, **100'**, **100''**, **100A-100X** (e.g., to adjust an operation of the cooling system **200**, **200'**, **200''**, **200A-200X**).

In operation, the container system can operate to maintain the chamber **126** of the vessel **120** at a preselected temperature or a user selected temperature. The cooling system can operate the one or more TECs to cool the chamber **126** (e.g., if the temperature of the chamber is above the preselected temperature, such as when the ambient temperature is above the preselected temperature) or to heat the chamber **126** (e.g., if the temperature of the chamber **126** is below the preselected temperature, such as when the ambient temperature is below the preselected temperature). The preselected temperature may be tailored to the contents of the container (e.g., a specific medication, a specific vaccine), and can be stored in a memory of the container, and the cooling system or heating system, depending on how the temperature control system is operated, can operate the TEC to approach the preselected or set point temperature.

Optionally, the circuitry EM can communicate (e.g., wirelessly) information to a remote location (e.g., cloud-based data storage system, remote computer, remote server, mobile electronic device such as a smartphone or tablet computer or laptop or desktop computer) and/or to the individual carrying the container (e.g., via their mobile phone, via a visual interface on the container, etc.), such as a temperature history of the chamber **126** to provide a record that can be used to evaluate the efficacy of the medication in the container and/or alerts on the status of the medication in the container. Optionally, the temperature control system (e.g., cooling system, heating system) automatically operates the TEC to heat or cool the chamber **126** of the vessel **120** to approach the preselected temperature. In one implementation, the cooling system **200**, **200'**, **200''**, **200B-200X** can cool and maintain one or both of the chamber **126**, **126'**, **126V** and the containers **520**, **520V** at or below 15 degrees Celsius, such as at or below 10 degrees Celsius, in some examples at approximately 5 degrees Celsius.

In one implementation, the one or more sensors **S1-Sn** can include one more air flow sensors in the lid **L** that can monitor airflow through one or both of the intake vent **203'**, **203''**, **203V** and exhaust vent **205'**, **205''**, **205V**. If said one or more flow sensors senses that the intake vent **203'**, **203''**, **203V** is becoming clogged (e.g., with dust) due to a decrease in air flow, the circuitry EM (e.g., on the PCBA **278V**) can

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optionally reverse the operation of the fan **280**, **280'**, **280B-280P**, **280V** for one or more predetermined periods of time to draw air through the exhaust vent **205'**, **205"**, **205V** and exhaust air through the intake vent **203'**, **203"**, **203V** to clear (e.g., unclog, remove the dust from) the intake vent **203'**, **203"**, **203V**. In another implementation, the circuitry EM can additionally or alternatively send an alert to the user (e.g., via a user interface on the container **100**, **100'**, **100"**, **100B-100X**, wirelessly to a remote electronic device such as the user's mobile phone via GUI **610A-610C**, **610A'-610C'**) to inform the user of the potential clogging of the intake vent **203'**, **203"**, **203V**, so that the user can inspect the container **100**, **100'**, **100"**, **100B-100X** and can instruct the circuitry EM (e.g., via an app on the user's mobile phone) to run an "cleaning" operation, for example, by running the fan **280**, **280'**, **280B-280P**, **280V** in reverse to exhaust air through the intake vent **203'**, **203"**, **203V**.

In one implementation, the one or more sensors **S1-Sn** can include one more Global Positioning System (GPS) sensors for tracking the location of the container system **100**, **100'**, **100"**, **100B-100X**. The location information can be communicated, as discussed above, by a transmitter and/or transceiver associated with the circuitry EM to a remote location (e.g., a mobile electronic device, a cloud-based data storage system, etc.).

FIG. **41A** shows a container system **100X** (e.g., a medicine cooler container) that includes a cooling system **200X**. Though the container system **100X** has a generally box shape, in other implementations it can have a generally cylindrical or tube shape, similar to the container system **100**, **100"**, **100B**, **100C**, **100D**, **100E**, **100F**, **100G**, **100H**, **100I**, **100J**, **100K**, **100K'**, **100L**, **100L'**, **100M**, **100N**, **100P**, **100Q**, **100R**, **100T**, **100U**, **100V**, or the features disclosed below for container system **100X** can be incorporated into the generally cylindrical or tube shaped containers noted above. In other implementations, the features disclosed below for container system **100X** can be incorporated into containers **100'** disclosed above. In one implementation, the cooling system **200X** can be in the lid **L** of the container system **100X** and can be similar to (e.g., have the same or similar components as) the cooling system **200**, **200"**, **200B**, **200B'**, **200C**, **200D**, **200E**, **200F**, **200G**, **200H**, **200I**, **200J**, **200K**, **200K'**, **200L**, **200L'**, **200M**, **200N**, **200P**, **200Q**, **200R**, **200S**, **200T**, **200V** described above. In another implementation, the cooling system can be disposed in a portion of the container vessel **120X** (e.g. a bottom portion of the container vessel **120X**, similar to cooling system **200'** in vessel **120'** described above).

As shown in FIG. **41A**, the container system **100X** can include a display screen **188X**. Though FIG. **41A** shows the display screen **188X** on the lid **L**, it can alternatively (or additionally) be incorporated into a side surface **122X** of the container vessel **120X**. The display screen **188X** can optionally be an electronic ink or E-ink display (e.g., electrophoretic ink display). In another implementation, the display screen **188X** can be a digital display (e.g., liquid crystal display or LCD, light emitting diode or LED, etc.). Optionally, the display screen **188X** can display a label **189X** (e.g., a shipping label with one or more of an address of sender, an address of recipient, a Maxi Code machine readable symbol, a QR code, a routing code, a barcode, and a tracking number), but can optionally additionally or alternatively display other information (e.g., temperature history information, information on the contents of the container system **100X**). The container system **100X** can optionally also include a user interface **184X**. In FIG. **41A**, the user interface **184X** is a button on the lid **L**. In another implementa-

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tion, the user interface **184X** is disposed on the side surface **122X** of the container vessel **120X**. In one implementation, the user interface **184X** is a depressible button. In another implementation, the user interface **184X** is a capacitive sensor (e.g., touch sensitive sensor). In another implementation, the user interface **184X** is a sliding switch (e.g., sliding lever). In another implementation, the user interface **184X** is a rotatable dial. In still another implementation, the user interface **184X** can be a touch screen portion (e.g., separate from or incorporated as part of the display screen **188X**). Advantageously, actuation of the user interface **184X** can alter the information shown on the display **188X**, such as the form of a shipping label shown on an E-ink display **188X**. For example, actuation of the user interface **184X**, can switch the text associated with the sender and receiver, allowing the container system **100X** to be shipped back to the sender once the receiving party is done with it.

FIG. **41B** shows a block diagram of electronics **180** of the container system **100X**. The electronics **180** can include circuitry EM' (e.g., including one or more processors on a printed circuit board). The circuitry EM' communicate with one or more batteries PS', with the display screen **188X**, and with the user interface **184X**. Optionally, a memory module **185X** is in communication with the circuitry EM'. In one implementation, the memory module **185X** can optionally be disposed on the same printed circuit board as other components of the circuitry EM'. The circuitry EM' optionally controls the information displayed on the display screen **188X**. Information (e.g., sender address, recipient address, etc.) can be communicated to the circuitry EM' via an input module **186X**. The input module **186X** can receive such information wirelessly (e.g., via radiofrequency or RF communication, via infrared or IR communication, via WiFi 802.11, via BLUETOOTH®, etc.), such as using a wand (e.g., a radiofrequency or RF wand that is waved over the container system **100X**, such as over the display screen **188X**, where the wand is connected to a computer system where the shipping information is contained). Once received by the input module **186X**, the information (e.g., shipping information for a shipping label to be displayed on the display screen **188X** can be electronically saved in the memory module **185X**). Advantageously, the one or more batteries PS' can power the electronics **180**, and therefore the display screen **188X** for a plurality of uses of the container **100X** (e.g., during shipping of the container system **100X** up to one-thousand times).

FIG. **42A** shows a block diagram of one method **800A** for shipping the container system **100X**. At step **810**, one or more containers, such as containers **520** (e.g., medicine containers, such as vials, cartridges (such as for injector pens), injector pens, vaccines, medicine such as insulin, epinephrine, etc.) are placed in the container vessel **120X** of the container system **100X**, such as at a distribution facility for the containers **520**. At step **820**, the lid **L** is closed over the container vessel **120X** once finished loading all containers **520** into the container vessel **120X**. Optionally, the lid **L** is locked to the container vessel **120X** (e.g., via a magnetically actuated lock, including an electromagnet actuated when the lid is closed that can be turned off with a code, such as a digital code). At step **830**, information (e.g., shipping label information) is communicated to the container system **100X**. For example, as discussed above, a radiofrequency (RF) wand can be waved over the container system **100X** (e.g., over the lid **L**) to transfer the shipping information to the input module **186X** of the electronics **80** of the container system **100X**. At step **840**, the container system **100X** is

shipped to the recipient (e.g., displayed on the shipping label **189X** on the display screen **188X**).

FIG. **42B** shows a block diagram of a method **800B** for returning the container **100X**. At step **850**, after receiving the container system **100X**, the lid **L** can be opened relative to the container vessel **120X**. Optionally, prior to opening the lid **L**, the lid **L** is unlocked relative to the container vessel **100X** (e.g., using a code, such as a digital code, provided to the recipient from the shipper) via keypad and/or biometric identification (e.g., fingerprint on the container vessel, as discussed above with respect to FIG. **31**). At step **860**, the one or more containers **520** are removed from the container vessel **120X**. At step **870**, the lid **L** is closed over the container vessel **120X**. At step **880**, the user interface **184X** (e.g., button) is actuated to switch the information of the sender and recipient in the display screen **188X** with each other, advantageously allowing the return of the container system **100X** to the original sender to be used again without having to reenter shipping information on the display screen **188X**. The display screen **188X** and label **189X** advantageously facilitate the shipping of the container system **100X** without having to print any separate labels for the container system **100X**. Further, the display screen **188X** and user interface **184X** advantageously facilitate return of the container system **100X** to the sender (e.g. without having to reenter shipping information, without having to print any labels), where the container system **100X** can be reused to ship containers **520** (e.g., medicine containers, such as vials, cartridges (such as for injector pens), injector pens, vaccines, medicine such as insulin, epinephrine, etc.) again, such as to the same or a different recipient. The reuse of the container system **100K** for delivery of perishable material (e.g., medicine) advantageously reduces the cost of shipping by allowing the reuse of the container vessel **120X** (e.g., as compared to commonly used cardboard containers, which are disposed of after one use).

ADDITIONAL EMBODIMENTS

In embodiments of the present invention, a portable cooler container with active temperature control, may be in accordance with any of the following clauses:

Clause 1. A portable cooler container with active temperature control, comprising:

- a container body having a chamber configured to receive and hold one or more containers of medicine;
- a lid removably coupleable to the container body to access the chamber; and
- a temperature control system comprising
 - one or more thermoelectric elements configured to actively heat or cool at least a portion of the chamber, one or more batteries,
 - circuitry configured to control an operation of the one or more thermoelectric elements to heat or cool at least a portion of the chamber to a predetermined temperature or temperature range; and
 - a display screen disposed on one or both of the container body and the lid, the display screen configured to selectively display shipping information for the portable cooler container using electronic ink.

Clause 2. The portable cooler container of any preceding clause, further comprising a button or touch screen actuable by a user to automatically switch sender and recipient information on the display screen to facilitate return of the portable cooler container to a sender.

Clause 3. The portable cooler container of any preceding clause, wherein the body comprises an outer peripheral wall

and a bottom portion attached to the outer peripheral wall, the inner peripheral wall being spaced relative to the outer peripheral wall to define a gap between the inner peripheral wall and the outer peripheral wall, the base spaced apart from the bottom portion to define a cavity between the base and the bottom portion, the one or more batteries and circuitry at least partially disposed in the cavity.

Clause 4. The portable cooler container of any preceding clause, wherein the one or more thermoelectric elements are housed in the lid, the temperature control system further comprising a first heat sink unit in thermal communication with one side of the one or more thermoelectric elements, a second heat sink unit in thermal communication with an opposite side of the one or more thermoelectric elements, and one or more fans, wherein the one or more fans, first heat sink unit and second heat sink unit are at least partially housed in the lid, the first heat sink configured to heat or cool at least a portion of the chamber.

Clause 5. The portable cooler container of any preceding clause, further comprising one or more sensors configured to sense the one or more parameters of the chamber or temperature control system and to communicate the sensed information to the circuitry.

Clause 6. The portable cooler container of any preceding clause, wherein at least one of the one or more sensors is a temperature sensor configured to sense a temperature in the chamber and to communicate the sensed temperature to the circuitry, the circuitry configured to communicate the sensed temperature data to the cloud-based data storage system or remote electronic device.

Clause 7. The portable cooler container of any preceding clause, further comprising one or more electrical contacts on a rim of the container body configured to contact one or more electrical contacts on the lid when the lid is coupled to the container body so that the circuitry controls the operation of the one or more thermoelectric elements and one or more fans when the lid is coupled to the container body.

Clause 8. The portable cooler container of any preceding clause, wherein the gap is under vacuum.

Clause 9. The portable cooler container of any preceding clause, further comprising a removable tray configured to removably receive the containers of medicine therein and to releasably lock the containers in the tray to inhibit dislodgement of the medicine containers from the tray during shipping of the portable cooler container.

Clause 10. The portable cooler container of any preceding clause, further comprising means for thermally disconnecting the one or more thermoelectric elements from the chamber to inhibit heat transfer between the one or more thermoelectric elements and the chamber.

Clause 11. A portable cooler container with active temperature control, comprising:

- a container body having a chamber configured to receive and hold one or more medicine containers, the chamber defined by a base and an inner peripheral wall of the container body;
- a lid removably coupleable to the container body to access the chamber; and
- a temperature control system comprising
 - one or more thermoelectric elements and one or more fans, one or both of the thermoelectric elements and fans configured to actively heat or cool at least a portion of the chamber,
 - one or more batteries, and
 - circuitry configured to control an operation of the one or more thermoelectric elements to heat or cool at

least a portion of the chamber to a predetermined temperature or temperature range.

Clause 12. The portable cooler container of clause 11, wherein the body comprises an outer peripheral wall and a bottom portion attached to the outer peripheral wall, the inner peripheral wall being spaced relative to the outer peripheral wall to define a gap between the inner peripheral wall and the outer peripheral wall, the base spaced apart from the bottom portion to define a cavity between the base and the bottom portion, the one or more batteries and circuitry at least partially disposed in the cavity.

Clause 13. The portable cooler container of any of clauses 11-12, wherein the one or more thermoelectric elements are housed in the lid, the temperature control system further comprising a first heat sink unit in thermal communication with one side of the one or more thermoelectric elements, a second heat sink unit in thermal communication with an opposite side of the one or more thermoelectric elements, wherein the one or more fans, first heat sink unit and second heat sink unit are at least partially housed in the lid, the first heat sink unit configured to heat or cool at least a portion of the chamber.

Clause 14. The portable cooler container of any of clauses 11-13, further comprising one or more sensors, at least one of the one or more sensors is a temperature sensor configured to sense a temperature in the chamber and to communicate the sensed temperature to the circuitry.

Clause 15. The portable cooler container of any of clauses 11-14, wherein the circuitry further comprises a transmitter configured to transmit one or both of temperature and position information for the portable cooler container to one or more of a memory of the portable cooler container, a radiofrequency identification tag of the portable cooler containers, a cloud-based data storage system, and a remote electronic device.

Clause 16. The portable cooler container of any of clauses 11-15, further comprising a display on one or both of the container body and the lid, the display configured to display information indicative of a temperature of the chamber.

Clause 17. The portable cooler container of any of clauses 11-16, further comprising one or more electrical contacts on a rim of the container body configured to contact one or more electrical contacts on the lid when the lid is coupled to the container body, the circuitry being housed in the container body and the one or more thermoelectric elements being housed in the lid, the electrical contacts facilitating control of the operation of the one or more thermoelectric elements and one or more fans by the circuitry when the lid is coupled to the container body.

Clause 18. The portable cooler container of any of clauses 11-17, wherein the gap is under vacuum.

Clause 19. The portable cooler container of any of clauses 11-18, further comprising means for thermally disconnecting the one or more thermoelectric elements from the chamber to inhibit heat transfer between the one or more thermoelectric elements and the chamber.

Clause 20. A portable cooler container with active temperature control, comprising:

- a container body having a chamber configured to receive and hold one or more volumes of perishable liquid, the chamber defined by a base and an inner peripheral wall of the container body;
- a lid movably coupled to the container body by one or more hinges; and
- a temperature control system, comprising
 - one or more thermoelectric elements configured to actively heat or cool at least a portion of the chamber,

one or more power storage elements, circuitry configured to control an operation of the one or more thermoelectric elements to heat or cool at least a portion of the chamber to a predetermined temperature or temperature range, the circuitry further configured to wirelessly communicate with a cloud-based data storage system or a remote electronic device; and

an electronic display screen disposed on one or both of the container body and the lid, the display screen configured to selectively display shipping information for the portable cooler container.

Clause 21. The portable cooler container of clause 20, wherein the electronic display screen is an electrophoretic display screen.

Clause 22. The portable cooler container of any of clauses 20-21, further comprising a button or touch screen actuable by a user to automatically switch sender and recipient information on the display screen to facilitate return of the portable cooler container to a sender.

Clause 23. The portable cooler container of any of clauses 20-22, further comprising means for thermally disconnecting the one or more thermoelectric elements from the chamber to inhibit heat transfer between the one or more thermoelectric elements and the chamber.

While certain embodiments of the inventions have been described, these embodiments have been presented by way of example only, and are not intended to limit the scope of the disclosure. Indeed, the novel methods and systems described herein may be embodied in a variety of other forms. For example, though the features disclosed herein are in described for medicine containers, the features are applicable to containers that are not medicine containers (e.g., portable coolers for food, etc.) and the invention is understood to extend to such other containers. Furthermore, various omissions, substitutions and changes in the systems and methods described herein may be made without departing from the spirit of the disclosure. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of the disclosure. Accordingly, the scope of the present inventions is defined only by reference to the appended claims.

Features, materials, characteristics, or groups described in conjunction with a particular aspect, embodiment, or example are to be understood to be applicable to any other aspect, embodiment or example described in this section or elsewhere in this specification unless incompatible therewith. All of the features disclosed in this specification (including any accompanying claims, abstract and drawings), and/or all of the steps of any method or process so disclosed, may be combined in any combination, except combinations where at least some of such features and/or steps are mutually exclusive. The protection is not restricted to the details of any foregoing embodiments. The protection extends to any novel one, or any novel combination, of the features disclosed in this specification (including any accompanying claims, abstract and drawings), or to any novel one, or any novel combination, of the steps of any method or process so disclosed.

Furthermore, certain features that are described in this disclosure in the context of separate implementations can also be implemented in combination in a single implementation. Conversely, various features that are described in the context of a single implementation can also be implemented in multiple implementations separately or in any suitable subcombination. Moreover, although features may be

described above as acting in certain combinations, one or more features from a claimed combination can, in some cases, be excised from the combination, and the combination may be claimed as a subcombination or variation of a subcombination.

Moreover, while operations may be depicted in the drawings or described in the specification in a particular order, such operations need not be performed in the particular order shown or in sequential order, or that all operations be performed, to achieve desirable results. Other operations that are not depicted or described can be incorporated in the example methods and processes. For example, one or more additional operations can be performed before, after, simultaneously, or between any of the described operations. Further, the operations may be rearranged or reordered in other implementations. Those skilled in the art will appreciate that in some embodiments, the actual steps taken in the processes illustrated and/or disclosed may differ from those shown in the figures. Depending on the embodiment, certain of the steps described above may be removed, others may be added. Furthermore, the features and attributes of the specific embodiments disclosed above may be combined in different ways to form additional embodiments, all of which fall within the scope of the present disclosure. Also, the separation of various system components in the implementations described above should not be understood as requiring such separation in all implementations, and it should be understood that the described components and systems can generally be integrated together in a single product or packaged into multiple products.

For purposes of this disclosure, certain aspects, advantages, and novel features are described herein. Not necessarily all such advantages may be achieved in accordance with any particular embodiment. Thus, for example, those skilled in the art will recognize that the disclosure may be embodied or carried out in a manner that achieves one advantage or a group of advantages as taught herein without necessarily achieving other advantages as may be taught or suggested herein.

Conditional language, such as “can,” “could,” “might,” or “may,” unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain embodiments include, while other embodiments do not include, certain features, elements, and/or steps. Thus, such conditional language is not generally intended to imply that features, elements, and/or steps are in any way required for one or more embodiments or that one or more embodiments necessarily include logic for deciding, with or without user input or prompting, whether these features, elements, and/or steps are included or are to be performed in any particular embodiment.

Conjunctive language such as the phrase “at least one of X, Y, and Z,” unless specifically stated otherwise, is otherwise understood with the context as used in general to convey that an item, term, etc. may be either X, Y, or Z. Thus, such conjunctive language is not generally intended to imply that certain embodiments require the presence of at least one of X, at least one of Y, and at least one of Z.

Language of degree used herein, such as the terms “approximately,” “about,” “generally,” and “substantially” as used herein represent a value, amount, or characteristic close to the stated value, amount, or characteristic that still performs a desired function or achieves a desired result. For example, the terms “approximately,” “about,” “generally,” and “substantially” may refer to an amount that is within less than 10% of, within less than 5% of, within less than 1% of, within less than 0.1% of, and within less than 0.01% of the

stated amount. As another example, in certain embodiments, the terms “generally parallel” and “substantially parallel” refer to a value, amount, or characteristic that departs from exactly parallel by less than or equal to 15 degrees, 10 degrees, 5 degrees, 3 degrees, 1 degree, or 0.1 degree.

The scope of the present disclosure is not intended to be limited by the specific disclosures of preferred embodiments in this section or elsewhere in this specification, and may be defined by claims as presented in this section or elsewhere in this specification or as presented in the future. The language of the claims is to be interpreted broadly based on the language employed in the claims and not limited to the examples described in the present specification or during the prosecution of the application, which examples are to be construed as non-exclusive.

What is claimed is:

1. A portable cooler container with active temperature control for medicines, vaccines, biological samples or other medical goods, comprising:

a container body having an insulated chamber configured to receive and hold one or more medicines, vaccines, biological samples or other medical goods, the container body comprising an outer peripheral wall and a bottom portion attached to the outer peripheral wall, the insulated chamber suspended relative to the outer peripheral wall, a base spaced apart from the bottom portion to define a cavity between the base and the bottom portion;

a lid removably coupleable or hingedly coupleable to the container body to access the chamber, the chamber being sealed when the lid is coupled to the container body to close the chamber;

a temperature control system at least partially disposed in the container body between an outer container wall and the insulated chamber, the temperature control system comprising:

one or more thermoelectric elements in thermal communication with an inner wall surface of the insulated chamber and operable to actively heat or cool at least a portion of the chamber, and

circuitry configured to control an operation of the one or more thermoelectric elements to heat or cool at least a portion of the insulated chamber to a predetermined temperature or temperature range;

the container body having a gap between the insulated chamber and the outer container wall that is under vacuum to provide vacuum insulation about the insulated chamber; and

an electronic display screen configured to selectively display shipping information for the portable cooler container.

2. The portable cooler container of claim 1, wherein the circuitry is further configured to automatically switch sender information and recipient information on the display screen to facilitate return of the portable cooler container to a sender.

3. The portable cooler container of claim 2, further comprising one or more temperature sensors configured to sense a temperature in the insulated chamber and one or more GPS sensors configured to sense a location of the container, the one or more temperature sensors and one or more GPS sensors configured to communicate the sensed temperature and sensed location to the circuitry, the circuitry configured to store the sensed temperature and the sensed location in a memory of the container.

4. The portable cooler container of claim 3, further comprising one or more motion sensors configured to sense

a motion of the container body and to communicate sensed motion data to the circuitry, the circuitry configured to store the sensed motion data in the memory of the container.

5. The portable cooler container of claim 4, wherein the circuitry stores the sensed temperature, the sensed location and the sensed motion in the memory of the container.

6. The portable cooler container of claim 1, wherein the one or more thermoelectric elements are in thermal communication with the inner wall surface of the insulated chamber through a wall of the insulated chamber and are operable to heat or cool the insulated chamber via conduction heat transfer through the wall of the insulated chamber.

7. The portable cooler container of claim 1, wherein the temperature control system further comprises a heat sink unit in thermal communication with one side of the one or more thermoelectric elements and one or more fans, the one or more fans operable to draw air through one or more air intake openings and flow air past the heat sink unit to remove heat from the heat sink unit and flow said air out through one or more exhaust openings, the circuitry configured to operate the one or more thermoelectric elements and the one or more fans to heat or cool at least a portion of the insulated chamber to the predetermined temperature or temperature range.

8. The portable cooler container of claim 7, further comprising one or more electrical connectors connectable to a power source to power one or both of the temperature control system and charge one or more batteries of the container.

9. The portable cooler container of claim 7, wherein the container body further comprises one or more electrical contacts configured to contact one or more electrical contacts of a power base when the container body is placed on the power base so that the circuitry controls the operation of the one or more thermoelectric elements and one or more fans when the container body is on the power base.

10. A portable cooler container with active temperature control for medicines, vaccines, biological samples or other medical goods, comprising:

a container body having an insulated chamber configured to receive and hold one or more medicines, vaccines, biological samples or other medical goods, the container body comprising an outer peripheral wall and a bottom portion attached to the outer peripheral wall, the insulated chamber suspended relative to the outer peripheral wall, a base of the insulated chamber spaced apart from the bottom portion to define a cavity between the base and the bottom portion;

a lid hingedly coupleable to the container body to access the chamber, the chamber being sealed when the lid is coupled to the container body to close the chamber;

a temperature control system at least partially disposed in the container body between an outer container wall and the insulated chamber, the temperature control system comprising:

one or more thermoelectric elements in thermal communication with an inner wall surface of the insulated chamber and operable to actively heat or cool at least a portion of the chamber, and

circuitry configured to control an operation of the one or more thermoelectric elements to heat or cool at least a portion of the insulated chamber to a predetermined temperature or temperature range;

the container body having a gap between the insulated chamber and the outer container wall that is under vacuum to provide vacuum insulation about the insulated chamber; and

an electronic display screen actuatable to display shipping address information for the portable cooler container.

11. The portable cooler container of claim 10, wherein the circuitry is further configured to automatically switch sender information and recipient information on the display screen to facilitate return of the portable cooler container to a sender.

12. The portable cooler container of claim 11, further comprising one or more temperature sensors configured to sense a temperature in the insulated chamber and one or more GPS sensors configured to sense a location of the container, the one or more temperature sensors and one or more GPS sensors configured to communicate the sensed temperature and the sensed location to the circuitry, the circuitry configured to store the sensed temperature and the sensed location in a memory of the container.

13. The portable cooler container of claim 12, further comprising one or more motion sensors configured to sense a motion of the container body and to communicate sensed motion data to the circuitry, the circuitry configured to store the sensed motion data in the memory of the container.

14. The portable cooler container of claim 13, wherein the circuitry stores the sensed temperature, the sensed location and the sensed motion in the memory of the container.

15. The portable cooler container of claim 10, wherein the temperature control system further comprises a heat sink unit in thermal communication with one side of the one or more thermoelectric elements and one or more fans, the one or more fans operable to draw air through one or more air intake openings and flow air past the heat sink unit to remove heat from the heat sink unit and flow said air out through one or more exhaust openings, the circuitry configured to operate the one or more thermoelectric elements and the one or more fans to heat or cool at least a portion of the insulated chamber to the predetermined temperature or temperature range.

16. The portable cooler container of claim 15, further comprising one or more electrical connectors connectable to a power source to power one or both of the temperature control system and charge one or more batteries of the container.

17. The portable cooler container of claim 10, wherein the one or more thermoelectric elements are in thermal communication with the inner wall surface of the insulated chamber through a wall of the insulated chamber and are operable to heat or cool the insulated chamber via conduction heat transfer through the wall of the insulated chamber.

18. A portable cooler container with active temperature control for medicines, vaccines, biological samples or other medical goods, comprising:

a container body having a vacuum insulated chamber configured to receive and hold one or more medicines, vaccines, biological samples or other medical goods, the container body comprising an outer peripheral wall and a bottom portion attached to the outer peripheral wall, the insulated chamber suspended relative to the outer peripheral wall, a base spaced apart from the bottom portion to define a cavity between the base and the bottom portion;

a lid removably coupleable or hingedly coupleable to the container body to access the chamber, the chamber being sealed when the lid is coupled to the container body to close the chamber;

a temperature control system at least partially disposed in the container body between an outer container wall and the vacuum insulated chamber, the temperature control system comprising:

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one or more thermoelectric elements in thermal communication with an inner wall surface of the vacuum insulated chamber and operable to actively heat or cool at least a portion of the chamber; and
circuitry configured to control an operation of the one or more thermoelectric elements to heat or cool at least a portion of the vacuum insulated chamber to a predetermined temperature or temperature range;
the container body having a gap between the insulated chamber and the outer container wall that is under vacuum to provide vacuum insulation about the insulated chamber; and
an electronic display screen configured to selectively display information.

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